Curvature-Calibrated Exploration: Matrix-Free GGN Widths and Conditional Regret Bounds

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Abstract

Current-parameter generalized Gauss–Newton (GGN) exploration must reconcile relinearized gradients with the predictable frozen features required by self-normalized confidence. We first isolate a structured resolution: for a fixed-feature scalar-link model, each replayed current gradient is a scalar multiple of its collection-time gradient. This gives relative feature-transfer and centering certificates. More generally, we prove a conditional Gaussian/squared-loss regret theorem for predictable positive-definite curvature operators with explicit linearization, centering, transfer, conjugate-gradient (CG), and realized-width terms. For exact current GGN under certified small drift, the bound transfers to the frozen potential; a rank-r spectral split then exposes the residual tail mass without assuming a known tangent subspace, and a positive-gap corollary follows. Full-GGN widths are computed matrix-free with curvature–vector products and residual-checked CG. These are conditional results, not guarantees for unrestricted neural training. The predeclared scaled-tanh full grid does not support an empirical theorem-instantiation claim: 148 of 4,000 theorem-labeled trajectories fail the required premise conjunction. A separate linear gap study passes its recorded exact/full-CG premises but its bounds remain vacuous; broader operator studies show no uniform fidelity–regret ordering.

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1 Introduction

Contextual bandits often use a differentiable reward model or a pretrained representation with a learned reward head (Zhou et al., 2020; Riquelme et al., 2018). In a white-box model, the natural exploration primitive is the current GGN/Fisher-linearized variance

$$s_t^2(a) \;=\; \mathbf{g}_t(a)^\top \mathbf{C}_t^{-1}\, \mathbf{g}_t(a), \tag{1}$$

where $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and $\mathbf{C}_t$ relinearizes all past examples at the current parameter. That relinearization breaks the rank-one update used by predictable self-normalized analysis. The collection-time frozen Gram retains predictability but answers a different geometric question. The central problem is therefore transfer: when does a confidence statement in the frozen metric remain optimistic in the current metric?

Structured scalar links make this transfer explicit. If $\mu_{\boldsymbol{\theta}}(z) = h(\phi(z)^\top \boldsymbol{\theta})$ with fixed ϕ, then a replayed current gradient is the corresponding frozen gradient multiplied by a scalar derivative ratio. The maximum relative ratio error directly controls a two-sided current/frozen curvature sandwich and yields a structured centering identity (Lemma 20). This mechanism gives an analytic scaled-tanh specialization (Corollary 18), but that corollary remains conditional: the completed full experiment fails its required premise conjunction and is reported as a stop, not as empirical validation.

The general theorem separates the remaining obligations. Under conditionally sub-Gaussian noise and Gaussian squared loss, Theorem 1 handles any predictable SPD algorithmic curvature through a one-sided transfer factor, a residual-checked CG factor, explicit centering and linearization errors, and a realized width complexity. Exact current GGN under certified small drift transfers further to the monotone frozen potential. A rank-r head/tail split closes its information gain without requiring a known subspace, and a positive-gap argument gives a separate conditional bound (Corollaries 2 to 4). The formal presentation follows this dependency order: master reduction, exact-current transfer, spectral and gap closures, then operational specializations. None controls unrestricted fine-tuning.

The operator is accessible without a dense curvature matrix. Curvature–vector products and CG solve $\mathbf{C}_t\mathbf{u} = \mathbf{g}_t(a)$ with a constant number of $O(d)$ Krylov vectors. Exact full-history products still replay raw examples; model state, activations, replay storage, and any historical parameter checkpoints are not included in this additional curvature-state count. Related implicit predictive-variance methods appear outside bandits (Maddox et al., 2021); NeuralUCB, neural-linear, and structured Laplace methods motivate the operator comparisons (Zhou et al., 2020; Riquelme et al., 2018; Ritter et al., 2018; Daxberger et al., 2021). We use full curvature as a reference, not as a presumed winner: independently executed policy contrasts are descriptive because methods induce different histories.

Contributions. **(1)** Relative transfer and centering certificates for fixed-feature scalar-link models, including an analytic scaled-tanh conditional specialization (Lemma 20 and Corollary 18). **(2)** A modular realized-width regret theorem with explicit linearization, centering, one-sided operator transfer, and CG terms (Theorem 1 and Lemma 19). **(3)** Exact-current closures through a sharp two-sided drift sandwich, a full-dimensional rank-plus-spectral-tail information bound, and a positive-gap corollary (Corollaries 2 to 4 and Lemma 8). **(4)** Matrix-free residual-checked GGN widths plus predictable Welford path certificates, a supplied-subspace shortcut, and a growing-window accounting bound (Corollaries 5 and 16 and Theorem 6). **(5)** Predeclared failure-mode studies that retain negative results: the scaled-tanh premise stop, premise-clean but vacuous gap validation, mixed operator comparisons, and isolated real-autodiff systems measurements (Section 5 and Proposition 36).

2 Problem Setup

We present the problem setup for the *Gaussian/squared-loss* instantiation, which is the setting analyzed in Theorem 1. An extension to exponential-family likelihoods is discussed in Appendix J.

Protocol. At each round $t = 1, \dots, T$ the learner observes a context $\mathbf{x}_t \in \mathcal{X}$ and a finite action set $\mathcal{A}_t$. It selects $a_t \in \mathcal{A}_t$ and observes a stochastic reward

$$r_t \;=\; \mu^*(\mathbf{x}_t, a_t) + \eta_t, \qquad \eta_t \text{ is } \sigma\text{-sub-Gaussian}, \quad (2)$$

where $\mu^* : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$ is the unknown mean reward. The goal is to minimize cumulative regret $R_T = \sum_{t=1}^{T}\big[\mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)\big]$, where $a_t^* = \arg\max_{a\in\mathcal{A}_t} \mu^*(\mathbf{x}_t, a)$. We assume $|\mathcal{A}_t| \le A_{\max}$ deterministically for all t.

Reward model and parameterization convention. We fix a pretrained initialization $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ and parameterize the model by the *displacement* $\boldsymbol{\theta} \in \mathbb{R}^d$ from it: the deployed predictor is $\mu_{\boldsymbol{\theta}_0+\boldsymbol{\theta}}(\mathbf{x}, a)$, and we abbreviate $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) := \mu_{\boldsymbol{\theta}_0+\boldsymbol{\theta}}(\mathbf{x}, a)$ throughout. The initial displacement is $\boldsymbol{\theta}_1 = \mathbf{0}$ (the deployed model starts at the pretrained weights $\boldsymbol{\theta}_0$). With this convention the ridge penalty below is a penalty on the *displacement from the pretrained weights*, not a zero-centered prior on the raw weights, and the realizability radius S bounds $\|\boldsymbol{\theta}^\star\| = \|\boldsymbol{\theta}^\star_{\mathrm{orig}} - \boldsymbol{\theta}_0\|$, the distance of the true parameter from the pretrained initialization. Define the regularized loss at round t:

$$\mathcal{L}_t(\boldsymbol{\theta}) \;:=\; \frac{1}{2\sigma^2}\sum_{s=1}^{t-1}\big(r_s - \mu_{\boldsymbol{\theta}}(\mathbf{x}_s, a_s)\big)^2 \;+\; \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2. \quad (3)$$

We write $\boldsymbol{\theta}_t$ for the parameter returned by the warm-started optimizer on $\mathcal{L}_t$. Let $\mathcal{D}_{t-1} := \{(\mathbf{x}_s, a_s, r_s) : s < t\}$ and $N_t := |\mathcal{D}_{t-1}| = t - 1$. Equation (3) is a *full-history* objective. The curvature replay set $\mathcal{S}_t \subseteq \{1, \dots, t-1\}$ and its size $m_t := |\mathcal{S}_t|$ affect $\mathcal{C}_t$, but do not replace $\mathcal{D}_{t-1}$ in (3). Thus a generic full-batch evaluation of $\nabla\mathcal{L}_t$ replays N_t observations even when $m_t \ll N_t$. In convex settings (e.g. a frozen backbone with a linear head), under exact optimization this coincides with the global minimizer; in nonconvex settings $\boldsymbol{\theta}_t$ may be a local minimizer or approximate stationary point. The gap between $\boldsymbol{\theta}_t$ and an idealized global minimizer is absorbed by the action-scaled centering certificate (Assumption 4).

Gradient features and curvature matrices. Define the *gradient feature* (Jacobian of the mean reward) at the current parameter estimate:

$$\mathbf{g}_t(\mathbf{x}, a) \;:=\; \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \;\in\; \mathbb{R}^d. \quad (4)$$

The algorithm uses the *relinearized curvature matrix*

$$\mathbf{C}_t \;=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1}\mathbf{g}_t(\mathbf{x}_s, a_s)\,\mathbf{g}_t(\mathbf{x}_s, a_s)^\top, \quad (5)$$

the *damped GGN curvature* for the Gaussian/squared-loss model at the current parameter $\boldsymbol{\theta}_t$. (For the Gaussian working model underlying this squared-loss quasi-likelihood, the GGN coincides with the model Fisher information; see the remark below.) While $\mathbf{C}_t$ is the natural object for computing predictive variances at $\boldsymbol{\theta}_t$, it re-evaluates all past Jacobians at $\boldsymbol{\theta}_t$ and therefore does *not* admit a sequential rank-one update—all summands change when $\boldsymbol{\theta}_t$ changes.

For the regret analysis we introduce the *frozen-feature Gram matrix*:

$$\bar{\mathbf{C}}_t \;:=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1}\mathbf{g}_s(\mathbf{x}_s, a_s)\,\mathbf{g}_s(\mathbf{x}_s, a_s)^\top, \quad (6)$$

where each $\mathbf{g}_s(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$ is evaluated at the parameter $\boldsymbol{\theta}_s$ that was current when the sample was collected. This matrix satisfies the recursive update

$$\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top, \qquad (7)$$

and, critically, each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is measurable with respect to the pre-reward sigma-algebra $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ of (105), which is the filtration used in the self-normalized bound. Both the additive update (7) and this measurability are structural requirements of the self-normalized martingale bound and the elliptic potential lemma invoked in Section 4. The algorithm computes UCB widths using the algorithmic curvature $\mathcal{C}_t$ (via CG; the exact case is $\mathcal{C}_t = \mathbf{C}_t$), but the confidence analysis is conducted in terms of $\bar{\mathbf{C}}_t$; the one-sided optimism-transfer factor of Assumption 5, certified by Lemma 19, provides the bridge.

3 Method: Curvature-Calibrated Exploration

3.1 CC-UCB Algorithm

For a predictable SPD oracle $\mathcal{C}_t \succeq \lambda\mathbf{I}$, set $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ for the original nonlinear center. Algorithm 1 is the analyzed full-enumeration policy. The same fixed curvature operator is reused across separate per-action CG solves; a solution for one action is not reused as the solution for another action.

Algorithm 1 Curvature-Calibrated UCB (CC-UCB)

1: Observe context $\mathbf{x}_t$ and finite action set $\mathcal{A}_t$.
2: Construct a fixed predictable SPD oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$ with $\mathcal{C}_t \succeq \lambda\mathbf{I}$.
3: **for** each $a \in \mathcal{A}_t$ **do**
4: Compute $\widehat{\mu}_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.
5: If $\mathbf{g}_t(a) \neq \mathbf{0}$, solve $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$ separately by CG until the certified tolerance holds; otherwise set $\tilde{\mathbf{u}} = \mathbf{0}$.
6: Set $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}$.
7: **end for**
8: Select and play $a_t \in \arg\max_{a \in \mathcal{A}_t}\{\widehat{\mu}_t(a) + \omega_t\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}\sqrt{\tilde{s}_t^2(a)}\}$.
9: Observe r_t, append $(\mathbf{x}_t, a_t, r_t)$, and update $\boldsymbol{\theta}_{t+1}$.

The oracle is accessed only through matrix–vector products. At each round the agent computes, for every candidate action a, the gradient feature $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and the *curvature-calibrated variance proxy*, the generic algorithmic width,

$$s_{\mathcal{C},t}^2(a) \;=\; \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1}\, \mathbf{g}_t(a). \qquad (8)$$

Under the GGN/Fisher-linearized Gaussian approximation with covariance $\mathcal{C}_t^{-1}$, the quantity $s_{\mathcal{C},t}^2(a)$ is the metric-dependent predictive variance of the linearized reward at $(\mathbf{x}_t, a)$. The *exact* full-history instantiation of CC-UCB takes $\mathcal{C}_t = \mathbf{C}_t$, the relinearized curvature (5); weighted replay, subsampling, sliding windows, periodic refreshes, and randomized sketches are special implementations of $\mathcal{C}_t$ (§C.3), covered by the theory only when the corresponding one-sided transfer factor is certified.

Because $\mathcal{C}_t \succ 0$ by construction, conjugate gradients (CG) is guaranteed to converge on the system $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$, provided the operator is held *fixed* across all iterations of the solve (the randomized sketch/subsample is drawn into Ω_t before action selection; §2). For each action with $\mathbf{g}_t(a) \neq \mathbf{0}$ we run CG to the per-action relative energy-norm error

$$\varepsilon_{t,a} \;:=\; \frac{\|\mathbf{u}_{t,a} - \tilde{\mathbf{u}}_{t,a}\|_{\mathcal{C}_t}}{\|\mathbf{u}_{t,a}\|_{\mathcal{C}_t}}, \qquad \mathbf{u}_{t,a} := \mathcal{C}_t^{-1}\mathbf{g}_t(a), \quad (9)$$

yielding the computable proxy $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_{t,a}$; if $\mathbf{g}_t(a) = \mathbf{0}$ we set $\tilde{s}_t^2(a) = 0$ and skip the solve (leaving the residual ratio $\|r_I\|/\|\mathbf{g}\|$ undefined), and by convention put $\varepsilon_{t,a} := 0$ in this case, so that $\varepsilon_{t,a}$ is defined for every $a \in \mathcal{A}_t$ and (90) holds trivially there. Lemma 26 shows $\tilde{s}_t^2(a)$ approximates $s_{\mathcal{C},t}^2(a)$ within a multiplicative $(1 \pm \varepsilon_{t,a})$ factor. To ensure the UCB bonus *upper-bounds* the confidence width $\omega_t\, \bar{s}_t(a)$ despite the approximation, the algorithm inflates the width by $\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}$ (line 8), where $\bar{\varepsilon}_t \in (0,1)$ is a *per-round* known tolerance enforced uniformly over all actions,

$$\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \;\leq\; \bar{\varepsilon}_t \;<\; 1, \qquad (10)$$

and $u_t(a) \geq 1$ is the one-sided transfer factor (Assumption 5).

4 Theoretical Analysis

The motivating scalar-link structure is formalized in Lemma 20, but the proof dependencies run in the opposite direction: we first state a master transfer theorem, then its exact-current, spectral-tail, and gap-dependent closures, and finally the operational and scalar-link specializations. The master theorem analyzes CC-UCB (Algorithm 1, with full action enumeration $K{=}|\mathcal{A}_t|$) under the Gaussian/squared-loss setup and Assumptions 1 to 5. It is not a full exponential-family or generalized-linear-bandit proof; the extension in Appendix J defines curvature only.

For a general predictable SPD sequence $\mathcal{C}_t$, confidence is proved in the frozen Gram $\bar{\mathbf{C}}_t$, whose update and measurability support self-normalized concentration. A one-sided factor $u_t(a)$ transfers optimism to $\mathcal{C}_t$, CG contributes α_t, and nonmonotonicity remains in the realized width complexity. Appendix C.5 gives exact determinant bookkeeping, not a primitive sublinear bound. Later specializations make the nonlinear inputs observable, but their smallness still requires a restricted parameter path.

Core conditions and links. Assume conditionally σ-sub-Gaussian noise, $\|\mathbf{g}_t(\mathbf{x}_s, a)\| \leq G$, realizability by $\|\boldsymbol{\theta}^*\| \leq S$, and a local expansion with round-t remainder at most $\varepsilon_{\text{lin}}(t)$. For the original center assume the action-scaled discrepancy is at most $\bar{\psi}_t \bar{s}_t(a)$. The operator $\mathcal{C}_t$ is predictable, SPD, satisfies $\mathcal{C}_t \succeq \lambda \mathbf{I}$ and $\mathcal{C}_1 = \lambda \mathbf{I}$, and remains fixed throughout every CG solve. Define

$$\bar{s}_t^2(a) = \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a), \quad s_{\mathcal{C},t}^2(a) = \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a), \\ \tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_t(a). \tag{11}$$

The analysis requires only the one-sided transfer and uniform CG relations

$$\begin{gathered} \bar{s}_t^2(a) \leq u_t(a) s_{\mathcal{C},t}^2(a), \\ (1-\bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a) \leq \tilde{s}_t^2(a) \leq (1+\bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a), \end{gathered} \tag{12}$$

for every candidate action, where $u_t(a) \geq 1$, $\bar{\varepsilon}_t < 1$, and $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$. Sufficient feature-drift and approximate-operator constructions are given in Lemma 19 and Theorem 34; no lower spectral transfer is needed.

Dynamic width complexity. The realized quantity used below is

$$\Lambda_T^{\mathcal{C}} := \sum_{t=1}^{T} \log\bigl(1 + \sigma^{-2} s_{\mathcal{C},t}^2(a_t)\bigr). \tag{13}$$

The bounded-width cap gives

$$\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda) \Lambda_T^{\mathcal{C}}. \tag{14}$$

For a general predictable nonmonotone SPD sequence, we do not derive a primitive sublinear bound on $\Lambda_T^{\mathcal{C}}$. Appendix Appendix C.5 gives an exact determinant bookkeeping identity, but its upper bound is informative only when both the terminal log determinant and determinant-decreasing refresh charge are independently controlled, for example under the rank and spectral-floor hypotheses of Lemma 24.

4.1 Regret Bound

Let $\mathcal{E}_{\text{conf}}$ denote the simultaneous self-normalized confidence event and let $\mathcal{E}_{\text{cert}}$ collect the predictable linearization, centering, transfer, condition-number, CG, and randomized-operator certificates. The appendix defines each component and its failure allocation; write their total failure probability as δ_{cert}.

Theorem 1 (Dynamic transfer regret bound for CC-UCB (Gaussian/squared-loss))**.** *Under the conditions above and Assumptions 1 to 5, suppose $|\mathcal{A}_t| \leq A_{\max}$, the optimal action is present, and Algorithm 1 enumerates all actions using a predictable schedule $\bar{\beta}_t \geq \beta_t$ and uniform per-round CG tolerance. Let $E_T = \sum_{t=1}^T \varepsilon_{\text{lin}}(t)$, $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, and $S_T = \sum_{t=1}^T \alpha_t^2 u_t(a_t) \omega_t^2$. On $\mathcal{E}_{\text{conf}} \cap \mathcal{E}_{\text{cert}}$,*

$$R_T \;\leq\; 2\sqrt{(\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}}\, S_T} \;+\; 2\, E_T, \tag{15}$$

with probability at least $1 - \delta - \delta_{\text{cert}}$. Deterministic or almost-sure certificates contribute zero to δ_{cert}. For the corrected center, the same statement holds without Assumption 4 after replacing ω_t by $\bar{\beta}_t$.

In particular, the bound is sublinear whenever $\Lambda_T^{\mathcal{C}} S_T = o(T^2)$ and $E_T = o(T)$. The exact-curvature instantiation is $\mathcal{C}_t = \mathbf{C}_t$ with $u_t = (1+\bar{\chi}_t)^2$ from Lemma 19; approximate operators are covered through Theorem 34. The proof is in Appendix F.

Proof sketch. Self-normalized confidence plus the centering and linearization terms gives $|\mu^*(a) - \mu_{\boldsymbol{\theta}_t}(a)| \leq \omega_t \bar{s}_t(a) + \varepsilon_{\text{lin}}(t)$. The one-sided relation and CG lower bound make the computed bonus optimistic, so instantaneous regret is at most $2w_t(a_t) + 2\varepsilon_{\text{lin}}(t)$. The CG upper bound gives $w_t(a_t) \leq \alpha_t \omega_t \sqrt{u_t(a_t)} s_{\mathcal{C},t}(a_t)$. Summing, applying Cauchy–Schwarz only once, and using (14) yields (15); hence time-varying factors remain inside S_T rather than being replaced by horizon maxima. The corrected center replaces ω_t by $\bar{\beta}_t$ and removes the centering certificate, but requires exact or certified frozen-feature access (Corollary 10).

Corollary 2 (Exact-current regret through the frozen potential)**.** *Adopt Theorem 1 with exact current curvature $\mathcal{C}_t = \mathbf{C}_t$. Suppose a predictable certificate satisfies $\chi_t \leq \bar{\chi}_t < 1$ for every t, and use the valid transfer factor $u_t = (1+\bar{\chi}_t)^2$. Then, on the event of Theorem 1,*

$$R_T \leq 2\left\{ (\sigma^2 + G^2/\lambda)\, \gamma_T \times \sum_{t=1}^{T} \alpha_t^2 \omega_t^2 \left(\frac{1+\bar{\chi}_t}{1-\bar{\chi}_t}\right)^2 \right\}^{1/2} + 2E_T. \tag{16}$$

For the corrected center of Corollary 10, the same statement holds without Assumption 4 after replacing ω_t by $\bar{\beta}_t$. This removes the nonmonotone dynamic-width term only in the small-drift, exact-current regime.

Proof. The sharpened sandwich in Corollary 21 and Loewner inversion give

$$s_t(a) \le \frac{\bar{s}_t(a)}{1-\bar{\chi}_t}.$$

The per-round proof of Theorem 1, together with $\sqrt{u_t} = 1 + \bar{\chi}_t$, therefore gives

$$\text{regret}_t \le 2\alpha_t\omega_t \frac{1+\bar{\chi}_t}{1-\bar{\chi}_t}\bar{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t).$$

Sum and apply Cauchy–Schwarz once. The frozen rank-one update and the usual elliptic-potential inequality give $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\gamma_T$, proving the display. The corrected-center claim follows from the prediction identity in Corollary 10. □

For an integer $0 \le r \le d$, define

$$\begin{aligned} \Delta_{T,r} &:= \sum_{i>r} \nu_i, \qquad r_T := \min\{r, T\}, \\ \Gamma^{\text{tail}}_{T,r} &:= r_T \log\left(1 + \frac{TG^2}{r_T\lambda\sigma^2}\right) + \frac{\Delta_{T,r}}{\lambda}, \end{aligned} \tag{17}$$

where $\nu_1 \ge \cdots \ge \nu_d \ge 0$ are the eigenvalues of $\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}$ and the first term is defined to be zero when $r_T = 0$. Lemma 8 proves $\gamma_T \le \Gamma^{\text{tail}}_{T,r}$. Writing $\tau_T = \sum_i \nu_i$ and $q_{T,r} = \min\{d-r, \max\{T-r, 0\}\}$, the same lemma proves the sharper realized split bound

$$\begin{aligned} \Gamma^{\text{split}}_{T,r} := r_T \log\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T\lambda}\right) \\ + q_{T,r} \log\left(1 + \frac{\Delta_{T,r}}{q_{T,r}\lambda}\right), \end{aligned} \tag{18}$$

where a term with zero multiplier is defined as zero.

Corollary 3 (Approximate-rank exact-current GGN)**.** *Under Corollary 2, on the same confidence event the following holds simultaneously for every $r \in \{0, \ldots, d\}$:*

$$\begin{aligned} R_T \le 2\Bigg\{ &(\sigma^2 + G^2/\lambda)\,\Gamma^{\text{tail}}_{T,r} \\ &\times \sum_{t=1}^{T} \alpha_t^2\omega_t^2 \left(\frac{1+\bar{\chi}_t}{1-\bar{\chi}_t}\right)^2 \Bigg\}^{1/2} + 2E_T. \end{aligned} \tag{19}$$

The same display holds with $\Gamma^{\text{tail}}_{T,r}$ replaced by the smaller $\Gamma^{\text{split}}_{T,r}$ from (18)*. This is a realized-complexity statement: the policy may retain the existing observable $\hat{\gamma}_{t-1}$ schedule, but a terminal realized $\Delta_{T,r}$ is not used in any action-time radius. Because the deterministic spectral inequality holds for every r, the terminal report may minimize $\Gamma^{\text{tail}}_{T,r}$ over r a posteriori, provided that choice does not alter any past action, radius, or other action-time quantity.*

Operationally, if before round t a supplied deterministic or $\mathcal{H}_t$-measurable envelope $\bar{\Delta}_{t-1,r}$ upper-bounds the spectral tail of $\bar{\mathbf{C}}_t - \lambda\mathbf{I}$, the old predictable envelope below remains valid:

$$\begin{aligned} r_{t-1} &:= \min\{r, t-1\}, \\ \bar{\Gamma}^{\text{tail}}_{t-1,r} &:= r_{t-1} \log\left(1 + \frac{(t-1)G^2}{r_{t-1}\lambda\sigma^2}\right) + \frac{\bar{\Delta}_{t-1,r}}{\lambda}. \end{aligned}$$

again taking the first term as zero when $r_{t-1} = 0$. Replacing γ_{t-1} in the constructive confidence radius by this predictable upper bound is valid. In that version, (19) *holds with $\Gamma^{\text{tail}}_{T,r}$ replaced by any supplied terminal envelope $\bar{\Gamma}^{\text{tail}}_{T,r}$ and with ω_t formed from the resulting radius.*

A refined predictable alternative uses the observable frozen trace $\tau_{t-1} = \operatorname{tr}(\bar{\mathbf{C}}_t - \lambda\mathbf{I}) = \sigma^{-2}\sum_{s<t}\|\mathbf{g}_s\|^2$. With $q_{t-1,r} = \min\{d-r, \max\{t-1-r, 0\}\}$, set

$$\begin{aligned} \bar{\Gamma}^{\text{split}}_{t-1,r} := r_{t-1} \log\left(1 + \frac{\tau_{t-1}}{r_{t-1}\lambda}\right) \\ + q_{t-1,r} \log\left(1 + \frac{\bar{\Delta}_{t-1,r}}{q_{t-1,r}\lambda}\right), \end{aligned} \tag{20}$$

again omitting zero-multiplier terms. The head term deliberately uses the full trace: subtracting an upper *tail envelope could reduce the head term and invalidate an upper bound. Thus $\bar{\Gamma}^{\text{split}}_{t-1,r} \ge \gamma_{t-1}$ is a valid action-time schedule, although it can be looser than the realized split expression.*

No exact tangent subspace is assumed. A small spectral tail can contain decision-relevant directions and does not supply their basis, so full-dimensional CG may still be necessary. Exact rank is recovered when $\Delta_{T,r} = 0$; the supplied-subspace result Corollary 16 is then an additional computational special case. The corrected-center version again replaces ω_t by $\bar{\beta}_t$.

Proof. Apply Lemma 8 to substitute either $\gamma_T \le \Gamma^{\text{split}}_{T,r} \le \Gamma^{\text{tail}}_{T,r}$ in (16). The prefix version of the same lemma makes either $\bar{\Gamma}^{\text{tail}}_{t-1,r}$ or $\bar{\Gamma}^{\text{split}}_{t-1,r}$ a predictable upper bound on γ_{t-1}, so Lemma 29 and Corollary 30 justify the operational radius. No eigenspace is used by either argument. □

Corollary 4 (Gap-dependent exact-current regret)**.** *Adopt Corollary 2. Define the realized gap $\Delta_t := \mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)$ and suppose every suboptimal candidate action at every round has gap at least a deterministic (or almost-sure) $\Delta_{\min} > 0$. This gap is an analysis condition and is not used to tune the policy; if it holds only on a high-probability event, that event must be added to the certificate failure allocation. Suppose also $\varepsilon_{\text{lin}}(t) \le \Delta_{\min}/4$ for all t, and set*

$$H^{\text{gap}}_t := \alpha_t\omega_t \frac{1+\bar{\chi}_t}{1-\bar{\chi}_t}, \qquad H^{\text{gap}}_{\max} := \max_{t\le T} H^{\text{gap}}_t.$$

Then, on the event of Theorem 1,

$$R_T \le \frac{16(H_{\max}^{\mathrm{gap}})^2}{\Delta_{\min}} \left(\sigma^2 + \frac{G^2}{\lambda}\right) \gamma_T. \tag{21}$$

Consequently γ_T in this display may be replaced by the exact-rank bound $\Gamma_{T,r}$ of (27), the simple tail bound $\Gamma_{T,r}^{\mathrm{tail}}$ of (17), or the refined $\Gamma_{T,r}^{\mathrm{split}}$ of (18), under the corresponding premises. The corrected-center version replaces ω_t by $\bar{\beta}_t$ as usual. This result is not a minimax-optimality claim: unlike the gap-free result, it requires a uniform positive gap and a per-round linearization error smaller than one quarter of that gap.

Proof. The per-round argument in Corollary 2 gives

$$\Delta_t \le 2H_t^{\mathrm{gap}} \bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

On a suboptimal round, $\Delta_t \ge \Delta_{\min}$ and hence $2\varepsilon_{\mathrm{lin}}(t) \le \Delta_t/2$. Rearranging and using $H_t^{\mathrm{gap}} \le H_{\max}^{\mathrm{gap}}$ gives $\Delta_t \le 4H_{\max}^{\mathrm{gap}} \bar{s}_t(a_t)$. Squaring this inequality and using $\Delta_t \ge \Delta_{\min}$ yields

$$\Delta_t \le \frac{\Delta_t^2}{\Delta_{\min}} \le \frac{16(H_{\max}^{\mathrm{gap}})^2}{\Delta_{\min}} \bar{s}_t^2(a_t).$$

Optimal rounds contribute zero. Sum over t and apply $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\gamma_T$ to prove the display; the three substitutions follow from Lemmas 7 and 8. ☐

5 Experiments

All regret rows are independently executed policies. Common random numbers are used only where pairing is valid. Exact teacher and dense replay quantities are post-action audits; common-trajectory operator comparisons never report causal regret. Tuning and evaluation seeds are disjoint, intervals use whole-seed trajectories, and stopped or failed studies remain visible. Derived full-grid aggregates retain hash-indexed provenance; release scope is stated in Appendix L.

Scaled-tanh stop. The predeclared finite-support full grid evaluates five horizons, four width ratios, eight methods, and 50 untouched seeds: 8,000 trajectories in total. The four theorem-labeled methods contribute 4,000 trajectories. Their failed-trajectory counts are 43 (exact relative), 51 (full-CG relative), 40 (Welford), and 14 (corrected center), totaling 148; the failed fields are optimizer residual, relative transfer, and, in eight rounds, CG convergence. Every method–horizon–width theorem cell therefore fails the joint premise criterion. Exact and CG policies nevertheless have identical actions and widths agree to at worst 1.3×10^{-12} relatively, which checks implementation but does not rescue the theorem premise. We make no empirical theorem-instantiation or rate claim from this study.

Gap validation. A separate linear study executes 1,000 trajectories over gaps $0.05, 0.1, 0.2, 0.4$. All 400 exact/full-CG trajectories satisfy the recorded premises of Corollary 4; maximum CG energy error is 8.4×10^{-7}. Exact-full mean terminal regret ranges from 5.05 to 7.99, while the gap-free RHS is about 2.2×10^4 and the gap-dependent RHS ranges from 1.28×10^6 to 1.00×10^7. Thus the experiment validates the recorded implication but also shows that this gap-dependent constant is vacuous and looser than the gap-free bound here.

The spectral-tail reversal, premise-clean Krylov grid, mixed coverage-matched operator comparisons, failed small-model baselines, and real JVP/VJP primitive measurements are deferred to Appendix B. They do not support a full-curvature performance advantage. The full Wheel benchmark did not complete; smoke runs are engineering checks only and no Wheel result enters the paper.

6 Conclusion

Fixed-feature scalar links expose a useful structure: current replay gradients are relative rescalings of predictable frozen gradients, so feature transfer and centering can be certified in the same coordinates. The master CC-UCB theorem then separates those obligations from linearization, operator approximation, and CG error. Under exact current curvature and certified small drift, the frozen potential yields full-dimensional spectral-tail and positive-gap closures; matrix-free CVP/CG solves avoid a dense curvature matrix but still require replay.

The experiments delimit rather than enlarge these claims. The scaled-tanh full grid fails the required premise conjunction in 148 of 4,000 theorem-labeled trajectories, so it provides no empirical theorem-instantiation or rate result. All 400 exact/full-CG trajectories in the separate gap study pass their recorded premises, but both displayed bounds are vacuous and the gap-dependent constant is larger. Spectral-tail and operator studies reject a uniform fidelity–regret ordering; real-autodiff results concern isolated primitives; small nonlinear and real-data baselines fail their sanity checks. The full Wheel study remains unrun and contributes no paper evidence.

Limitations. Theory assumes Gaussian squared loss, action enumeration, realizability, smoothness, and, for the original center, an optimizer certificate. Structured scalar-link replay stores per-sample scalar metadata; the corrected center needs frozen-feature access; approximate operators need additional transfer certificates. Controlling drift and spectral tails during unrestricted training remains open. The empirical checks are float64 rather than verified enclosures, the measured autodiff models stop near 10^7 parameters on one accelerator, and independently executed adaptive-policy contrasts are descriptive, not causal.

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A Additional conditional rates

A.1 Online nonlinear certificates

At the start of round t, let $n_t = t-1$, and let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ be maintained by a vector Welford update. With collection residuals $c_s = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) - r_s$, define

$$Q_t = J_{<t} + n_t \|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2, \qquad R_t^{\mathrm{col}} = \sum_{s<t} c_s^2. \tag{22}$$

Both are pre-action quantities and $Q_t = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$ exactly. If prediction gradients are L_g-Lipschitz on the certified region, set

$$\bar{\chi}_t = \frac{L_g \sqrt{Q_t}}{\sigma\sqrt{\lambda}}, \qquad u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2, \tag{23}$$

where $\kappa_{+,t} = 1$ for exact full curvature and for an unrescaled current-parameter subset, and otherwise is a supplied one-sided operator certificate. If $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\| \le \zeta_t$, an observable centering schedule is

$$\begin{aligned} \bar{\psi}_t &= \frac{\zeta_t + \bar{M}_t}{\sqrt{\lambda}}, \\ \bar{M}_t &= \sigma^{-2}\Big[L_g \sqrt{R_t^{\mathrm{col}} Q_t} + \frac{3}{2} G L_g Q_t + \frac{1}{2} L_g^2 Q_t^{3/2}\Big]. \end{aligned} \tag{24}$$

For a known radius-R trust region, the final numerator term improves to $L_g^2 R Q_t$. Finally, for an L_μ-Lipschitz prediction gradient define $\bar{\epsilon}_t^{\mathrm{lin}} = \frac{L_\mu}{2}(S + \|\boldsymbol{\theta}_t\|)^2$, $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\mathrm{lin}})^2$, and

$$\begin{aligned} \bar{\beta}_t &= \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} + \sqrt{\lambda} S + \sigma^{-1}\sqrt{\bar{F}_t}, \\ \hat{\gamma}_{t-1} &= \sum_{s<t} \log(1+q_s), \qquad q_s = \frac{u_s(a_s)\tilde{s}_s^2(a_s)}{\sigma^2(1-\bar{\varepsilon}_s)}. \end{aligned} \tag{25}$$

Corollary 5 (Operational CC-UCB)**.** *Under the structural hypotheses of Theorem 1, with its certificate inputs instantiated by valid smoothness constants, the optimizer residual bound ζ_t, uniform CG certificates, and a valid $\kappa_{+,t}$,* (22)–(25) *are predictable and satisfy every nonlinear input of the theorem. Thus Theorem 1 holds with E_T replaced by $\bar{E}_T = \sum_{t\le T} \bar{\epsilon}_t^{\mathrm{lin}}$ and the observable CG upper bound on $\Lambda_T^{\mathcal{C}}$ from Corollary 25.*

Proof. Lemma 13 proves $\bar{\chi}_t \ge \chi_t$, $\bar{\psi}_t \ge \psi_t$, $\bar{F}_t \ge F_t$, and $\bar{E}_T \ge E_T$. The one-sided operator relation and Corollary 31 give u_t and $\hat{\gamma}_{t-1} \ge \gamma_{t-1}$, hence $\bar{\beta}_t \ge \beta_t$. All quantities use the summary through round $t-1$ and the already formed $\boldsymbol{\theta}_t$; uniform candidate-action CG accuracy is checked before selection. Substitution in Theorem 1 and Corollary 25 proves the claim. □

The proof, filtration-order pseudocode, and a scalar-tanh specialization with $G \le B_\phi$ and $L_\mu = L_g \le 4B_\phi^2/(3\sqrt{3})$ are in Lemma 13 and Corollary 17.

For an abstract near-linearity scale W, suppose $\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R$, $L_\mu \le c_\mu/\sqrt{W}$, $L_g \le c_g/\sqrt{W}$, $\zeta_t \le \zeta_0/\sqrt{W}$, and $R_t^{\mathrm{col}} \le C_R t P_T$ for a supplied $P_T \ge 1$. Then

$$\begin{aligned} \bar{E}_T &\le \frac{2c_\mu R^2 T}{\sqrt{W}}, \qquad \bar{F}_{T+1} \le \frac{4c_\mu^2 R^4 T}{W}, \\ \bar{\chi}_t &\le \frac{2c_g R\sqrt{t-1}}{\sigma\sqrt{\lambda W}}. \end{aligned} \tag{26}$$

and Corollary 14 gives an explicit trust-region bound on $\bar{\psi}_t$. If a supplied fixed active subspace gives rank bound $r \ge 1$, define

$$\Gamma_{T,r} = r_T \log\left(1 + \frac{TG^2}{r_T \lambda\sigma^2}\right), \qquad r_T = \min\{r, T\}. \tag{27}$$

and set $\Gamma_{0,r} := 0$ for the round-one confidence schedule. The bound r must be known before selection; a terminal realized rank is only post-hoc. Then $\gamma_T \le \Gamma_{T,r}$ and Corollary 16 gives the fully explicit conditional projected-Gram bound

$$R_T \le 2\alpha_I \sqrt{\rho_+/\rho_-}\,\Omega_{T,r}\sqrt{(\sigma^2 + G^2/\lambda)T\Gamma_{T,r}} + 2\bar{E}_T, \tag{28}$$

where $x_T = 2c_g R\sqrt{T-1}/(\sigma\sqrt{\lambda W})$, $\rho_- = (1-x_T)^2$, $\rho_+ = (1+x_T)^2$, and $\Omega_{T,r}$ is displayed with every $T, W, r, P_T, \lambda, \sigma$ dependence in the appendix. At fixed r and problem constants, $W = \Omega(T^2 P_T)$ makes the path inflation bounded and gives $R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W})$. This is a low-rank tangent-space or parameter-efficient adaptation regime, not generic full-network training.

Theorem 6 (Conditional growing-window bound under fixed-subspace excitation)**.** *Adopt the hypotheses, events, and valid predictable schedules of Theorem 1, let $m_t = \min\{t-1, \lceil t^q \rceil\}$ for $q \in (0,1]$, and let $\mathcal{C}_t$ be the unrescaled current-parameter Gram over the last m_t observations. Thus Proposition 33 gives $\kappa_{+,t} = 1$ and $u_t = (1+\bar{\chi}_t)^2$. Suppose candidate gradients lie in a fixed subspace U, and let P_U be its orthogonal projector. For a supplied burn-in index $\tau_w \in \{1, \dots, T+1\}$, assume the predictable window-excitation condition*

$$\mathcal{C}_t \succeq \lambda \mathbf{I} + \kappa_w m_t P_U$$

holds for a known $\kappa_w > 0$ and every $t \ge \tau_w$. Then, on the same event of probability at least $1 - \delta - \delta_{\mathrm{cert}}$ as Theorem 1,

$$\begin{aligned}
&\Lambda_T^{\mathcal{C}} \le L_q(T;\tau_w),\\
&L_q(T;\tau_w) := (\tau_w - 1)\log\left(1 + \frac{G^2}{\lambda\sigma^2}\right)\\
&\qquad\qquad + \frac{G^2}{\sigma^2}\sum_{t=\tau_w}^{T} \frac{1}{\lambda + \kappa_w m_t},\\
&L_q(T;\tau_w) = \begin{cases} O(\tau_w + T^{1-q}), & q < 1,\\ O(\tau_w + \log T), & q = 1.\end{cases}
\end{aligned} \tag{29}$$

For $\tau_w = 1$, more explicitly, $L_q(T;1) \le (G^2/\sigma^2)[\lambda^{-1} + \kappa_w^{-1} h_q(T)]$, where, for $T \ge 2$, $h_q(T) = 1 + ((T-1)^{1-q} - 1)/(1-q)$ for $q < 1$ and $h_1(T) = 1 + \log(T-1)$, while $h_q(1) := 0$ in both cases. Writing $r_U = \dim U$, the excitation premise can first hold nontrivially only once $m_t \ge r_U$, because $\operatorname{rank}(\mathcal{C}_t - \lambda I) \le m_t$*; thus τ_w includes the required rank-r_U burn-in. If $H_T = \max_{t \le T} \alpha_t \sqrt{u_t(a_t)}\,\omega_t$, then*

$$R_T \le 2H_T\sqrt{(\sigma^2 + G^2/\lambda)T L_q(T;\tau_w)} + 2E_T. \tag{30}$$

If $H_T = \widetilde{O}(1)$, $E_T = \widetilde{O}(T^{1-q/2})$, and the burn-in obeys $\tau_w = \widetilde{O}(T^{1-q})$ for $q < 1$ (or $\tau_w = \widetilde{O}(1)$ for $q = 1$), then (30) *simplifies to $R_T = \widetilde{O}(T^{1-q/2})$; otherwise its displayed L_q and E_T terms remain. This is conditional on the fixed subspace, excitation, and all confidence, transfer, optimizer, and CG certificates above. Certified worst-case* width-solve *work is $O(KT^{1+3q/2}\log(1/\bar{\varepsilon}))$ sample-CVPs when a fixed uniform target $\bar{\varepsilon} \in (0,1)$ is enforced; this count excludes model optimization, candidate-gradient computation, and certificate construction.*

Proof. Before τ_w, $\mathcal{C}_t \succeq \lambda \mathbf{I}$ bounds each logarithm in $\Lambda_T^{\mathcal{C}}$ by $\log(1 + G^2/(\lambda\sigma^2))$. Thereafter, for $g_t(a) \in U$, inversion of the excitation inequality gives $s^2_{\mathcal{C},t}(a) \le G^2/(\lambda + \kappa_w m_t)$; summing $\log(1+x) \le x$ proves (29). Since $S_T \le TH_T^2$, substitution in Theorem 1 proves (30). The condition-number bound $\kappa(\mathcal{C}_t) \le 1 + m_t G^2/(\lambda\sigma^2)$ makes CG use $O(\sqrt{m_t}\log(1/\bar{\varepsilon}))$ iterations, each replaying m_t samples; summing $Km_t^{3/2}$ proves the cost claim. □

Under the hypotheses of Corollary 16, if the window policy uses the predictable rank envelope $\Gamma_{t-1,r}$ instead of the realized $\hat{\gamma}_{t-1}$ in its confidence radius, then $H_T \le \alpha_I(1 + x_T)\Omega^{\mathrm{win}}_{T,r}$, where $\Omega^{\mathrm{win}}_{T,r}$ is the explicit $\Omega_{T,r}$ in Corollary 16 with $A_{\inf} = 1$. Hence (30) contains no uncontrolled information-gain placeholder. The excitation premise is restrictive and standard in persistently excited identification; contextual diversity alone does not imply it. For $\dim U = r > 1$ one must choose τ_w after a rank-r burn-in, as the formal early charge records. The predictable rank schedule in Corollary 16 makes H_T polylogarithmic under the same bounded-path conditions. Illustrative achieved upper-bound exponents are reported in Appendix C.4; they are not a Pareto frontier or a lower bound. When U is supplied, the projected Gram in (47) can instead be solved directly.

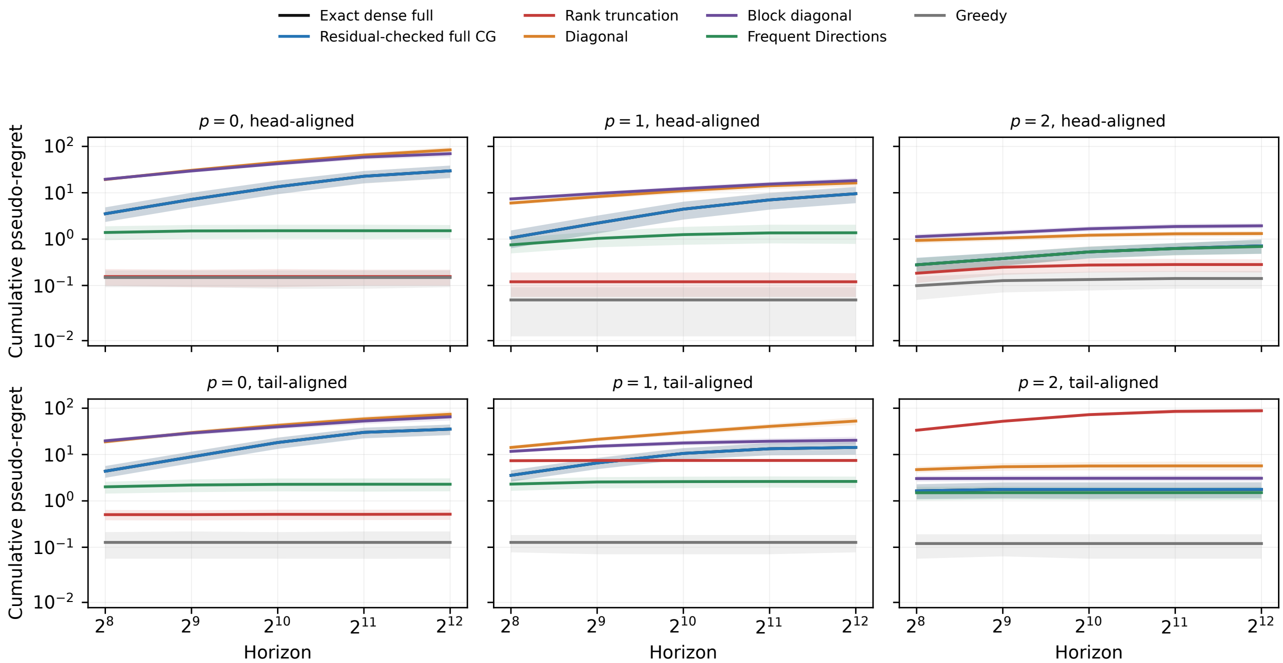

Figure 1: Full spectral-tail evaluation at $d = 256$, $K = 8$, and displayed rank $r = 8$. Curves are means over untouched evaluation seeds 1000–1049; bands are 95% whole-seed bootstrap intervals at five predeclared horizons. Bonuses are selected on tuning seeds 0–9 over the complete predeclared cell set. Head- and tail-aligned regimes include cells unfavorable to full curvature.

B Additional experimental evidence

The spectral-tail run contains 7,440 tuning and 8,400 evaluation trajectories. At $r = 8, p = 2$, head-aligned rank truncation has mean terminal regret 0.283 versus 0.714 for exact full; moving the decision gap to the discarded tail reverses the comparison to 88.176 versus 1.771. In the head-aligned cell, $\gamma_T = 87.87$ and the tail bound is 330.75, below the ambient-rank bound 1888.87 but still conservative. Dense full and full CG agree throughout, but the construction needs only one CG iteration and is an equivalence check, not a Krylov stress test.

B.1 Scaled-tanh premise stop

B.2 Positive-gap validation details

B.3 Premise-clean scaling and operator studies

The complete run contains 10,800 trajectories over three dimensions, three active ranks, three condition numbers, eight methods, and 50 fixed evaluation seeds. Feature drift and linearization error are identically zero, the model update is a closed-form ridge solve, and cyclic excitation has a prespecified analytic lower bound. Every full-CG and window solve converges from zero; more than 90% of rounds use multiple iterations, and the largest observed relative energy error is below 2.4×10^{-4} versus the 10^{-3} target. At $d = 128, r = 8, \kappa = 1000$, mean work slopes are 1.498, 1.674, and 2.027 for $q = 1/2, 2/3, 1$, respectively. They are below the worst-case CG upper-bound exponents because iteration counts saturate in the fixed-rank active space. All methods have the same regret in this sign-symmetric construction, so the experiment validates premises and work accounting, not policy superiority. The implementation uses dense active-coordinate matrices for simulation and exact energy audits; reported work is replay-equivalent sample-CVP accounting rather than controlled wall-clock timing. All fitted slopes are in the generated appendix fit tables.

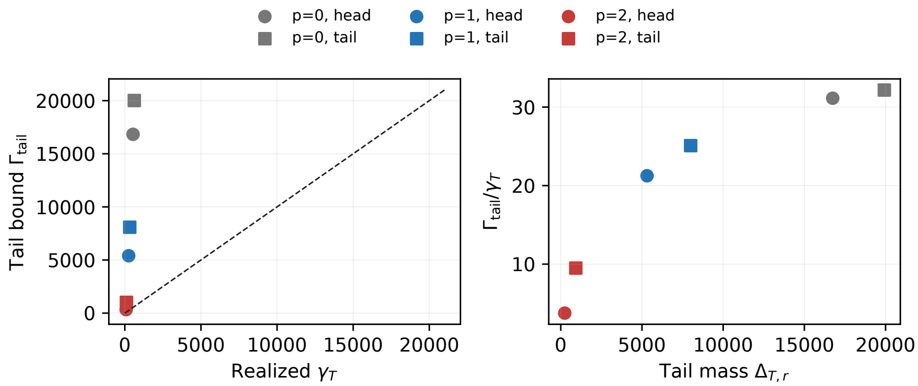

Figure 2: Realized frozen information gain and the spectral-tail upper bound for the exact-full $r = 8$ trajectories. The bound is informative for decaying spectra but is not uniformly tighter than the ambient trace/rank bound; it is an upper bound rather than an estimator of γ_T.

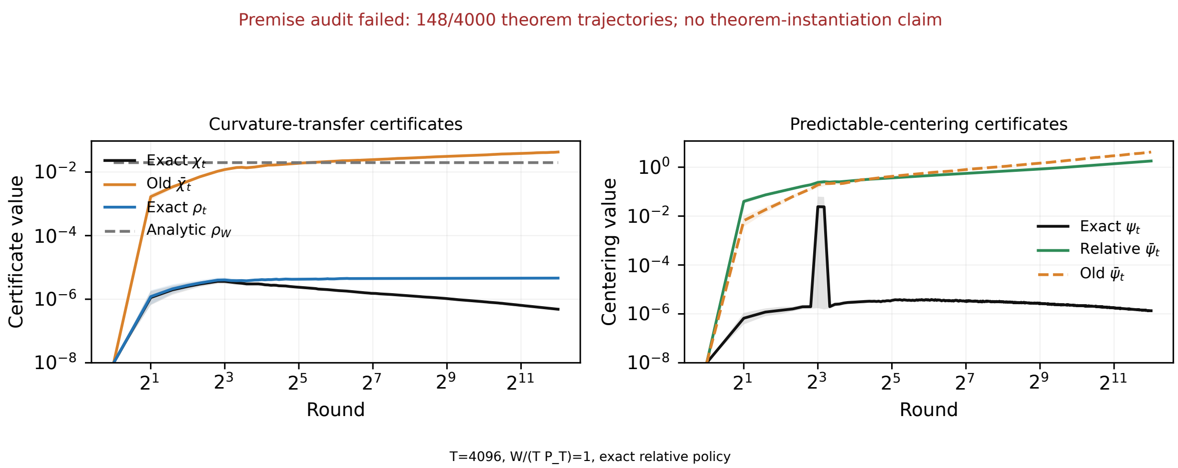

Figure 3: Full scaled-tanh certificate audit over five horizons, four width ratios, and 50 evaluation seeds. The displayed certificates are computed from complete trajectories, but every theorem-labeled method–cell fails the required joint premise criterion. This is a failure audit, not an empirical instantiation of Corollary 18.

B.4 Earlier bounded-tanh diagnostic

This smaller predecessor to the scaled-tanh grid uses the nonlinear Gaussian model $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $d = 6$, four bounded actions, $T = 100$, $\|\theta^*\| \le 0.35$, and a radius-0.5 projected optimizer. The analytic constants in Corollary 17, exact full-objective gradient norm, the path statistics in (22), and the analytic condition bound $1 + (t - 1)G^2/(\lambda\sigma^2)$ are computed before action selection. No teacher, future reward, exact χ_t, exact ψ_t, or exact Taylor error enters the policy. A fixed protocol is evaluated on 50 seeds (160–209); an eight-cell one-factor sensitivity grid is confined to ten tuning seeds.

Every original- and corrected-center trajectory passes every recorded transfer, centering, linearization, information, CG, confidence, optimism, and regret event: zero observed failures over 5,000 rounds per center. Mean regret is 7.86 $[7.66, 8.07]$ for the original center and 5.69 $[5.46, 5.92]$ for the corrected center. The schedules are deliberately conservative: at the final round, original χ_t averages 0.0216 versus $\bar{\chi}_t = 4.89$, and ψ_t averages 0.0520 versus $\bar{\psi}_t = 128.1$. The mean observable RHS/regret ratios are 6,956.7 and 1,328.9, respectively, so the bounds are numerically vacuous. Because the residual and dense audit checks are ordinary float64 point computations, these executions are classified *post-hoc theorem-event verified with fully predictable schedules*, not verified-enclosure theorem certified. They do not override the failed conjunction in the larger predeclared scaled-tanh grid.

T	$W/(TP_T)$	ρ_W	Pass	Fail E/C/W/R	Regret	RHS/T	CG agr.	Width err.	CVPs
256	0.5	0.12	FAIL	0/0/1/0	14.3	3.79	1	8.12e-13	2.35e+06
256	1	0.0834	FAIL	1/1/1/0	11.7	3.05	1	1e-12	2.38e+06
256	2	0.0583	FAIL	1/1/2/0	9.63	2.6	1	1.1e-12	2.42e+06
256	4	0.0409	FAIL	2/2/2/1	8.3	2.3	1	1.15e-12	2.45e+06
512	0.5	0.0829	FAIL	1/2/1/0	13.8	2.63	1	1.01e-12	1.01e+07
512	1	0.058	FAIL	1/1/2/0	10.9	2.2	1	1.12e-12	1.02e+07
512	2	0.0406	FAIL	2/2/2/1	8.88	1.92	1	1.2e-12	1.03e+07
512	4	0.0286	FAIL	3/3/2/1	7.78	1.74	1	1.18e-12	1.03e+07
1024	0.5	0.0576	FAIL	1/2/4/0	11.3	1.87	1	1.13e-12	4.16e+07
1024	1	0.0404	FAIL	2/2/2/1	9.11	1.6	1	1.18e-12	4.17e+07
1024	2	0.0284	FAIL	3/4/2/1	7.86	1.43	1	1.21e-12	4.17e+07
1024	4	0.02	FAIL	2/2/2/0	7.41	1.31	1	1.28e-12	4.17e+07
2048	0.5	0.0402	FAIL	2/2/3/1	9.08	1.35	1	1.2e-12	1.67e+08
2048	1	0.0282	FAIL	3/3/2/1	7.86	1.18	1	1.23e-12	1.67e+08
2048	2	0.0199	FAIL	2/3/2/0	7.41	1.07	1	1.22e-12	1.68e+08
2048	4	0.014	FAIL	4/5/2/2	7.06	0.988	1	1.28e-12	1.68e+08
4096	0.5	0.0281	FAIL	3/4/2/1	7.91	0.981	1	1.18e-12	6.71e+08
4096	1	0.0198	FAIL	2/3/2/0	7.42	0.871	1	1.22e-12	6.71e+08
4096	2	0.0139	FAIL	5/6/2/3	7.07	0.796	1	1.24e-12	6.71e+08
4096	4	0.00984	FAIL	3/3/2/1	6.67	0.744	1	1.28e-12	6.71e+08

Method	Failed trajectories	Failed rounds	Required event-field failures
Exact relative	43	72	optimizer: 58; relative transfer: 14
Full CG relative	51	80	CG convergence: 8; optimizer: 58; relative transfer: 14
Current Welford	40	77	optimizer: 77
Corrected current	14	14	relative transfer: 14

Table 1: Scaled-tanh full-grid audit. "Premises" is the conjunction of all required fields for the corresponding theorem trajectory. Failed-trajectory counts are reported rather than discarding those runs; CG agreement and width error are implementation checks only.

B.5 Reference-operator falsification and matched baselines

An eight-cell phase map runs seven operators on 30 independent policies per cell and replays the exact-full path for detailed width/rank diagnostics. Full CG matches dense full regret; diagonal has lower regret in all eight cells, block in six, while full beats window and stale refresh in all eight and rank-3 Lanczos in seven. Thus fidelity does not order regret. The separate linear-Gram witness changes the diagonal policy's action, but is existential rather than a dominance result (Figure 12 and table 20).

Scalar calibration changes conclusions but does not induce a uniform ordering. For each comparison we use a two-sided paired Student-t test on the 50 seed-level terminal-regret differences; an identically zero difference receives $p = 1$, while a constant nonzero difference receives the limiting value $p = 0$. Holm's step-down procedure controls familywise error at 0.05 over all 168 prespecified tests. Across the 56 surrogate–cell comparisons per protocol, the adjustment leaves 35 significant under the identical coefficient (13 favor a surrogate, 22 favor full), 20 under coverage matching (8/12), and 27 under mean-bonus matching (11/16). Evaluation coverage under the tuning-selected 95% protocol averages 94.6% across method–cell means and ranges from 85.6% to 99.2%. Current and historical full Grams coincide in this fixed-feature linear environment; their separate labels audit the protocol rather than representation drift. The added low-rank-plus-diagonal method is a batch-refit LO-FI-style surrogate, not an official LO-FI implementation; none of its 24 comparisons is significant after Holm adjustment. The frozen coefficients are in Table 22, all mean paired differences and raw/adjusted p-values are in the generated appendix comparison tables, and full coverage/decision heatmaps are in Figure 13.

The balanced nonlinear benchmark uses separate tuning and 30 evaluation seeds; local NeuralUCB is the historical-time full-gradient-Gram comparison, alongside NeuralTS, NeuralLinear, last-layer, diagonal, linear, greedy, and context-free baselines. These are local matched implementations, not verified reproductions. At $T = 200$, CC-UCB full GGN-CG has regret 21.94 $[20.64, 23.23]$; diagonal has 16.11 $[14.62, 17.61]$, LinUCB 10.58 $[9.82, 11.33]$, NeuralLinear 10.01 $[9.33, 10.68]$, and frozen last-layer UCB 7.44 $[6.90, 7.99]$. Context-free controls have regret 90.35 and 89.40; the sanity check passes, but the experiment is negative evidence for full-curvature superiority.

A separate balanced-MNIST pipeline fixes disjoint supervised-pretraining, tuning, and online-evaluation splits. Both online pools have exactly 10% of each digit; 480 tuning policies select among four configurations per method before 240 untouched evaluation policies (12 methods, 20 seeds, $T = 5000$). The deliberately compute-

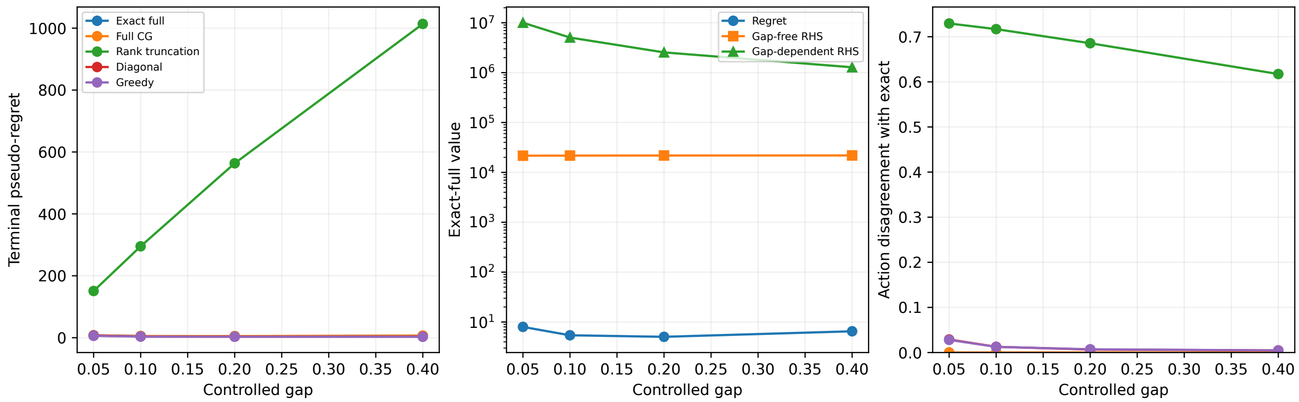

Figure 4: Predeclared positive-gap validation at $T = 4096$ with 50 untouched evaluation seeds per method–gap cell. Exact full and residual-checked full CG satisfy the recorded gap, linearization, confidence, and operator premises in all 400 theorem-eligible trajectories. The gap-dependent RHS dominates regret but is much larger than the gap-free RHS; approximate-operator and greedy rows are decision diagnostics outside this corollary's premise.

bounded nonlinear model uses 64 fixed pixels, one tanh hidden unit, and 85 trainable parameters. Current and historical full Grams use exact dense solves here, not the matrix-free primitive benchmarked in Figure 8; their maximum original-system residual is below 7×10^{-12}. All implementations are local matched baselines, not official reproductions.

The result is a failed algorithmic benchmark (Figure 7 and table 4): mean accuracy ranges only from 9.885% to 10.068%, and mean regret from 4496.6 to 4505.75. Current full GGN has regret 4501.75 $[4490.97, 4512.53]$ and accuracy 9.965%; none of its 11 paired method comparisons is significant after Holm adjustment. We retain the full result as negative evidence rather than selecting a favorable method or seed subset. Multi-process wall times are resource accounting under host contention, not a controlled speed comparison.

B.6 Audits and systems scope

All 280 bounded linear traces pass float64 checks, but lack verified enclosures; their bounds are vacuous. Synthetic CPU operators reach $d = 8192$, but use unmatched stored-feature primitives. Small-model policies likewise materialize Jacobians for dense audits, validating algebra rather than on-the-fly $O(d)$ Krylov state. The separate real-autodiff study executes 360 predeclared cells using `torch.func` JVP/VJP operators without retaining an explicit sample-by-parameter Jacobian for the matrix-free methods (Figure 8). It covers ten evaluation instances for 131,841- and 9,972,737-parameter MLPs, buffers 32/128/512, 5/10 actions, and residual targets $10^{-1}/10^{-2}/10^{-3}$; all cells finish within 0.167 accelerator-hours and none is dropped. These single-host primitive timings do not include model training, distributed communication, or a bandit loop and do not support a deployment or foundation-model claim.

Real-data baseline check. Covertype fails its baseline check: context-free controls beat every contextual method at every horizon (Appendix E.1).

Gap	Method	Regret	Gap-free RHS	Gap RHS	Disagree.	Premise
0.05	Exact full	7.985	21625.0	10021964.2	0.000	PASS
0.05	Full CG	7.571	21625.0	10021982.1	0.000	PASS
0.05	Rank truncation	150.885	21626.6	10023433.6	0.729	N/A
0.05	Diagonal	6.820	21624.9	10021847.9	0.029	N/A
0.05	Greedy	6.093	21624.9	10021853.0	0.028	N/A
0.10	Exact full	5.422	21717.0	5053717.4	0.000	PASS
0.10	Full CG	5.482	21717.0	5053728.0	0.000	PASS
0.10	Rank truncation	295.236	21718.7	5054499.5	0.717	N/A
0.10	Diagonal	3.956	21717.0	5053703.5	0.013	N/A
0.10	Greedy	3.654	21717.0	5053703.7	0.013	N/A
0.20	Exact full	5.052	21803.1	2546937.6	0.000	PASS
0.20	Full CG	5.312	21803.1	2546942.4	0.000	PASS
0.20	Rank truncation	563.424	21804.8	2547335.4	0.686	N/A
0.20	Diagonal	3.264	21803.1	2546933.4	0.007	N/A
0.20	Greedy	3.340	21803.1	2546933.3	0.007	N/A
0.40	Exact full	6.520	21865.5	1280764.2	0.000	PASS
0.40	Full CG	6.624	21865.5	1280766.3	0.000	PASS
0.40	Rank truncation	1013.672	21867.3	1280973.4	0.617	N/A
0.40	Diagonal	3.624	21865.5	1280764.1	0.005	N/A
0.40	Greedy	3.104	21865.5	1280763.4	0.005	N/A

Table 2: Predeclared gap-validation terminal results over 50 evaluation seeds per method–gap cell. PASS applies only to exact full and full CG. The approximate and greedy rows are decision diagnostics outside the premises of Corollary 4.

d	r	κ	Exact current	Full CG	Window $q = 1/2$	Window $q = 2/3$	Window $q = 1$	Frozen	Diagonal	Greedy
128	4	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	4	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	4	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	8	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	8	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	8	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	16	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	16	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
128	16	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	4	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	4	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	4	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	8	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	8	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	8	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	16	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	16	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
512	16	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	4	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	4	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	4	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	8	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	8	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	8	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	16	10	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	16	100	PASS	PASS	PASS	PASS	PASS	PASS	control	control
2048	16	1000	PASS	PASS	PASS	PASS	PASS	PASS	control	control

Table 3: Premise status for all 27 rotated scaling cells. PASS is the conjunction over 50 evaluation trajectories and all rounds. Diagonal and greedy are performance controls rather than theorem methods. These are analytic construction checks plus float64 residual/energy audits, not directed-rounding enclosures.

Table 4: Balanced-MNIST terminal results. Times are contention-affected resource accounting. "Max res." is the maximum original-system relative residual over evaluation seeds for the three dense full-Gram methods.

Method	Regret	Accuracy	Time (s)	Peak MB	Max res.
All-layer diagonal	4496.60	0.101	1.50	1003.5	–
Block Laplace	4501.40	0.100	2.48	1003.5	–
Context-free TS	4502.95	0.099	0.38	1004.6	–
Context-free UCB	4500.35	0.100	0.26	1004.5	–
Current full GGN (dense)	4501.75	0.100	67.18	1003.5	5.2e-12
Frozen last-layer UCB	4502.90	0.099	1.28	1003.3	–
Greedy	4503.80	0.099	0.81	1004.7	–
Historical NeuralUCB	4502.95	0.099	41.58	1003.0	4.1e-12
LinUCB (frozen)	4502.90	0.099	1.34	1004.5	–
LO-FI style	4505.75	0.099	2.74	1004.5	–
NeuralLinear	4497.50	0.101	1.42	1003.8	–
NeuralTS	4500.55	0.100	41.65	1003.1	6.9e-12

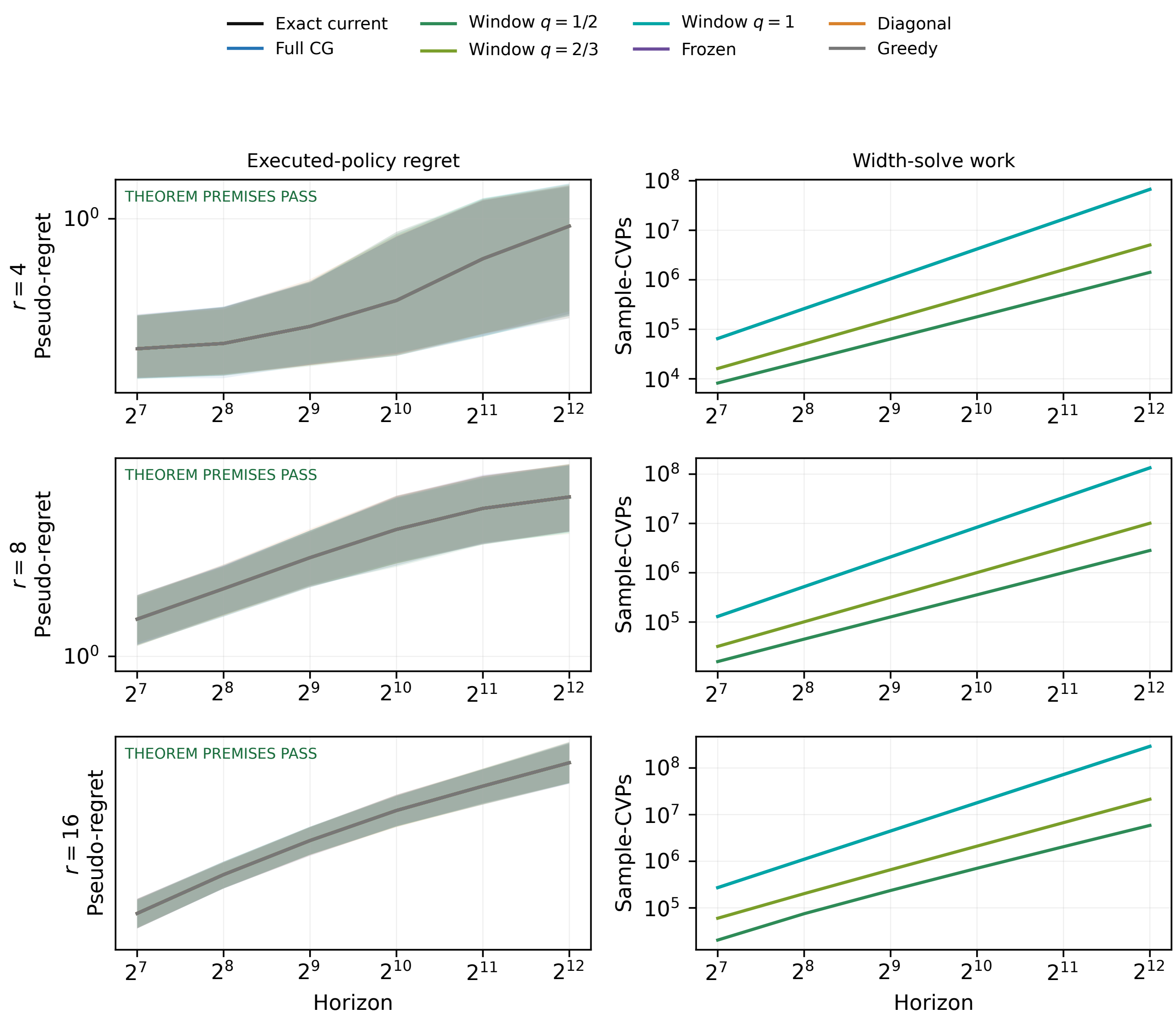

Figure 5: Premise-clean rotated scaling test. The displayed $d = 128$ panels vary active rank and condition number; the full grid also contains $d \in \{512, 2048\}$. Curves are 50-seed means with pointwise whole-trajectory bootstrap 95% intervals over six predeclared horizons. Every theorem method in all 27 cells passes the optimizer, rank-information, excitation, CG convergence, and energy-error fields. Slopes remain finite-grid diagnostics, not asymptotic-rate estimates.

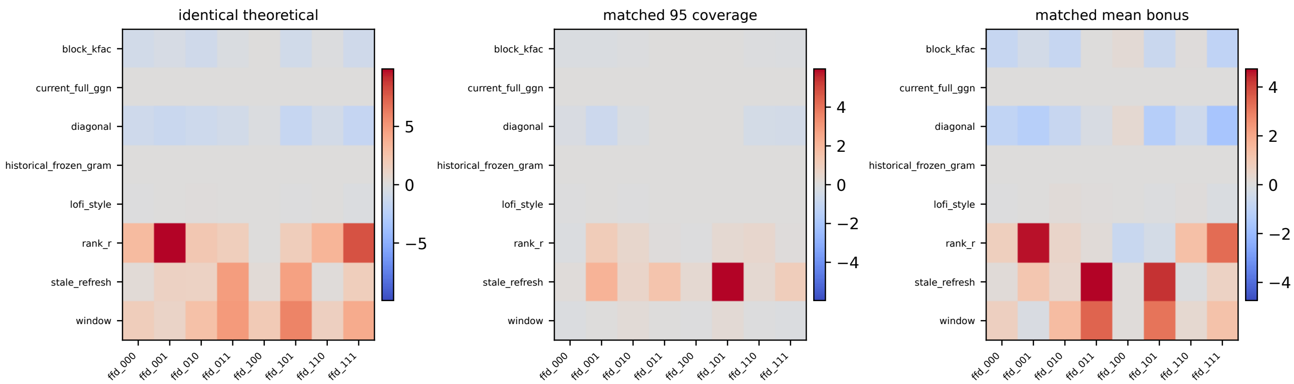

Figure 6: Coverage-matched operator mechanism map. Entries are mean method-minus-current-full-GGN cumulative pseudo-regret in all eight predeclared condition/rotation/gap cells (50 evaluation seeds). Scalar coefficients are either identical, selected on tuning seeds 2000–2019 for 95% one-step coverage, or selected there to match mean bonus magnitude; evaluation seeds 2100–2149 are untouched during selection. Policies execute independently, so differences are descriptive rather than causal operator effects.

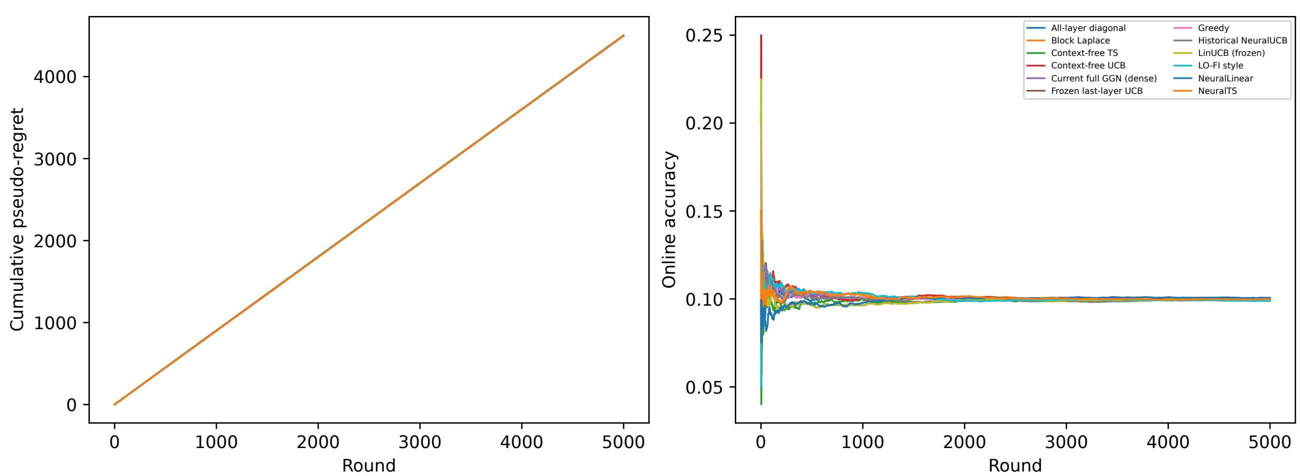

Figure 7: Balanced-MNIST evaluation over 20 untouched seeds. Every method remains near the 10% context-free baseline; the plot is a failed sanity check for this small local model, not evidence of algorithmic parity.

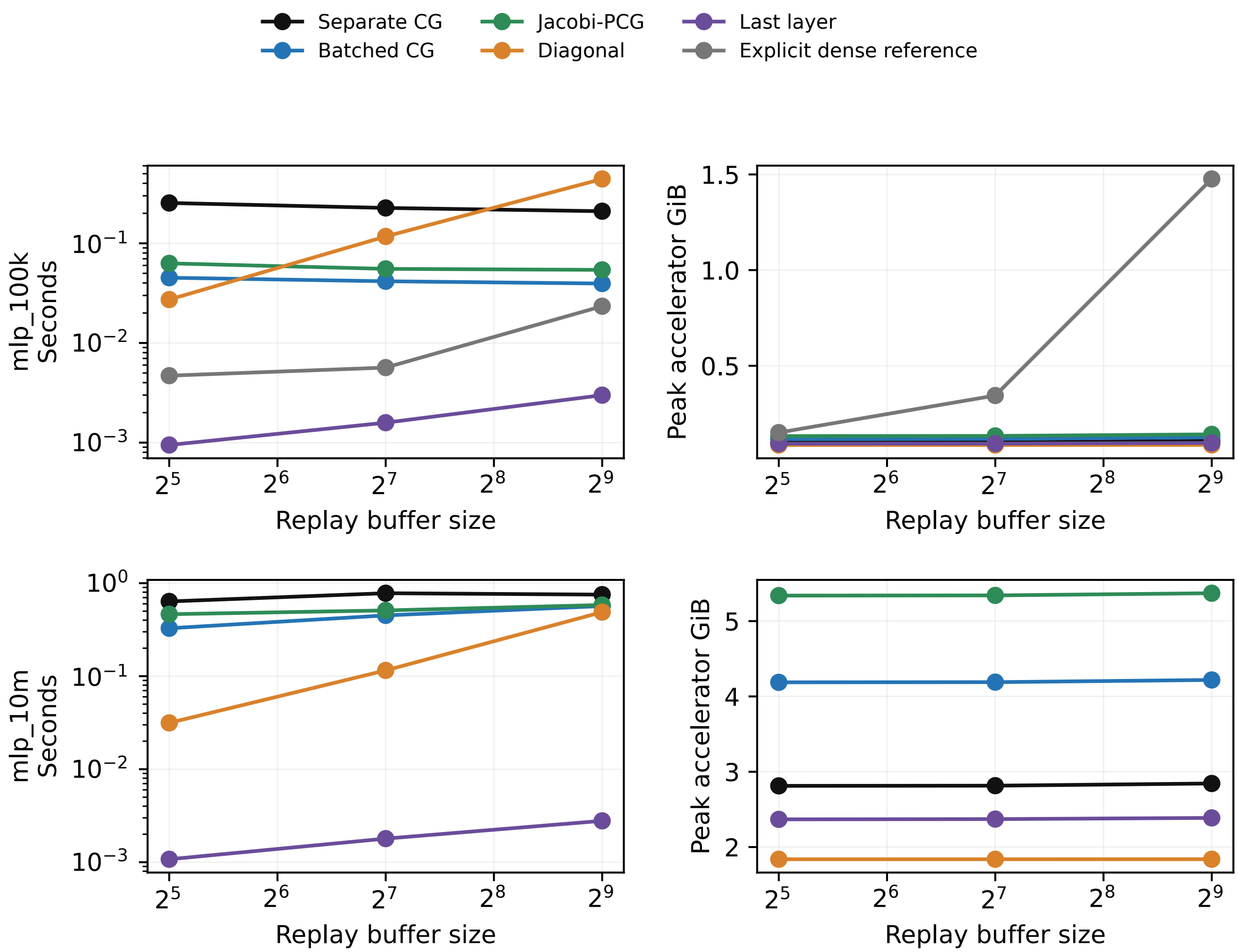

Figure 8: Real neural JVP/VJP systems benchmark (ten evaluation instances per cell; medians are aggregated with 95% seed intervals). Matrix-free methods use `torch.func.jvp/vjp`; the explicit Jacobian is an audit available only for the 131,841-parameter model. The study measures isolated uncertainty primitives after warm-up on one NVIDIA accelerator, not training or end-to-end policy throughput.

C Deferred main-text derivations

This appendix collects the formal statements, proofs, and derivations deferred from the compact main text; all cross-referenced results are stated here.

For the sequential frozen-feature Gram, the usual information gain is

$$\gamma_T := \log \frac{\det(\bar{\mathbf{C}}_{T+1})}{\det(\lambda \mathbf{I})} = \sum_{t=1}^{T} \log\bigl(1 + \sigma^{-2} \bar{s}_t^2(a_t)\bigr) . \tag{31}$$

Lemma 7 (Rank and statistical-dimension information closure)**.** *For $A \succeq 0$, $r \geq \operatorname{rank}(A)$, and $q_A = \|A\|_{\mathrm{op}}/\lambda$,*

$$\log\det(\mathbf{I} + A/\lambda) \leq \min\left\{ r \log\left(1 + \frac{\operatorname{tr} A}{r\lambda}\right), k(q_A) d_{\mathrm{eff},\lambda}(A) \right\}, \tag{32}$$

where $d_{\mathrm{eff},\lambda}(A) = \operatorname{tr}[A(A+\lambda\mathbf{I})^{-1}]$, $k(0) = 1$, and $k(q) = (1+q)\log(1+q)/q$ for $q > 0$; the rank term is defined as zero when $r = 0$ (which forces $A = 0$). Consequently, if $\operatorname{rank}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq r$, then $\gamma_T \leq \Gamma_{T,r}$ from (27).

Proof. Writing $x_i = \lambda_i(A)/\lambda$, concavity of $\log(1+x)$ after padding the spectrum with zeros gives the rank–trace term. Also

$$\log(1+x) = \frac{(1+x)\log(1+x)}{x} \frac{x}{1+x}.$$

The first factor is increasing because its derivative is $(x - \log(1+x))/x^2 \geq 0$; bounding it at q_A and summing proves the statistical-dimension term. For the frozen Gram, $\operatorname{tr}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq TG^2/\sigma^2$, which gives (27) (with $r_T = \min\{r, T\}$). $\square$

The second term is useful only with a predictable or deterministic envelope on $d_{\mathrm{eff},\lambda}$; a realized value is post-hoc. It is not the empirical trace effective rank $\operatorname{tr}(A)/\|A\|_{\mathrm{op}}$, which does not in general control log determinant.

Lemma 8 (Spectral-tail information closure)**.** *Let*

$$A_T := \bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I} = \sigma^{-2} \sum_{t=1}^{T} \mathbf{g}_t \mathbf{g}_t^\top \succeq 0$$

have eigenvalues $\nu_1 \geq \cdots \geq \nu_d \geq 0$. For an integer $r \in \{0, \ldots, d\}$, let $\Delta_{T,r} := \sum_{i>r} \nu_i$, $\tau_T := \operatorname{tr}(A_T)$, $r_T := \min\{r, T\}$, and $q_{T,r} := \min\{d - r, \max\{T - r, 0\}\}$. Then

$$\begin{aligned} \gamma_T &\leq r_T \log\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T \lambda}\right) + q_{T,r} \log\left(1 + \frac{\Delta_{T,r}}{q_{T,r}\lambda}\right) \\ &=: \Gamma_{T,r}^{\mathrm{split}} , \end{aligned} \tag{33}$$

where either term is defined as zero when its multiplier is zero. Moreover,

$$\Gamma_{T,r}^{\mathrm{split}} \leq r_T \log\left(1 + \frac{TG^2}{r_T \lambda \sigma^2}\right) + \frac{\Delta_{T,r}}{\lambda}, \tag{34}$$

where the first term is defined as zero when $r_T = 0$. Thus $r = 0$ gives $\gamma_T \leq \operatorname{tr}(A_T)/\lambda$, while $r \geq \operatorname{rank}(A_T)$ gives $\Delta_{T,r} = 0$ and recovers the rank closure. The statement also covers $T = 0$, when both sides vanish.

Proof. Because A_T is a sum of T rank-one matrices, $\operatorname{rank}(A_T) \leq T$, and Assumption 2 gives $\operatorname{tr}(A_T) \leq TG^2/\sigma^2$. If $r_T > 0$, concavity of $x \mapsto \log(1 + x/\lambda)$ on the leading eigenvalues gives

$$\sum_{i=1}^{r} \log(1 + \nu_i/\lambda) = \sum_{i=1}^{r_T} \log(1 + \nu_i/\lambda) \leq r_T \log\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T \lambda}\right).$$

The equality uses zero eigenvalues when $r > T$. For the tail, at most $q_{T,r}$ eigenvalues can be nonzero: there are at most $d-r$ tail coordinates and at most $T-r$ nonzero eigenvalues after the first r. Concavity, with zero padding when necessary, therefore gives

$$\sum_{i>r} \log(1+\nu_i/\lambda) \le q_{T,r} \log\left(1+\frac{\Delta_{T,r}}{q_{T,r}\lambda}\right).$$

If $q_{T,r}=0$, the rank bound forces $\Delta_{T,r}=0$ and the tail sum is empty. This proves (33); the $r_T=0$ case similarly uses only the tail term.

For the second inequality, use $\tau_T - \Delta_{T,r} \le \tau_T \le TG^2/\sigma^2$ in the head term and $q\log(1+x/q) \le x$ with $x = \Delta_{T,r}/\lambda$ in the tail term. This proves (34) without evaluating a zero denominator. The stated edge cases follow immediately. □

Proposition 9 (Tightness of the split spectral-tail expression)**.** *Fix integers $r, q \ge 1$, masses $H, \Delta \ge 0$, ambient dimension $d = r+q$, and $T \ge d$. Let a PSD $d\times d$ matrix have r head eigenvalues equal to H/r, q tail eigenvalues equal to Δ/q, and all remaining eigenvalues zero, with the head values ordered no smaller than the tail values. Splitting after r makes* (33) *an equality:*

$$\log\det(\mathbf{I}_d + A/\lambda) = r\log\left(1+\frac{H}{r\lambda}\right) + q\log\left(1+\frac{\Delta}{q\lambda}\right).$$

If $\Delta/(q\lambda) \le 1$, then the tail contribution obeys

$$q\log\left(1+\frac{\Delta}{q\lambda}\right) \ge \frac{\Delta}{2\lambda}.$$

Thus dependence linear in the tail mass cannot in general be removed without additional information about the tail rank or spectrum. This is a tightness statement for the information inequality, not a regret lower bound for the nonlinear class. The example is algebraic; to embed it in the bounded-feature bandit class one additionally chooses factors satisfying $H+\Delta \le TG^2/\sigma^2$.

Proof. The equality follows by substituting the two constant spectral blocks. For $x\in[0,1]$, $\log(1+x) \ge x/2$; apply this with $x = \Delta/(q\lambda)$ and multiply by q. □

A deterministic or history-measurable prefix envelope $\bar{\Delta}_{t,r} \ge \Delta_{t,r}$ may enter an online confidence schedule through the prefix form of (34). A second valid schedule is (20), which uses the observable full trace for the head and the tail envelope only for the tail. It does not subtract an upper tail envelope from the trace. In contrast, a realized terminal $\Delta_{T,r}$ is an a posteriori path-complexity bound only. Since the inequality is pathwise and simultaneous over the finite set of ranks, the terminal report may minimize over r. Neither the terminal tail nor that post-hoc rank may choose an eigenspace, a radius, an action, or any other action-time quantity. If valid predictable envelopes are available for several ranks, their pointwise minimum is itself a valid predictable information bound. The existing observable $\hat{\gamma}_t$ schedule remains valid when only realized terminal tails are reported.

The dynamic-potential and CG-observable consequences of Theorem 1 are

$$\begin{aligned} R_T &\le 2\sqrt{(\sigma^2+G^2/\lambda)\Gamma_T^{\mathrm{dyn}} S_T} + 2E_T, \\ R_T &\le 2\sqrt{(\sigma^2+G^2/\lambda)\widehat{\Lambda^{\mathrm{C}}}_T S_T} + 2E_T. \end{aligned} \tag{35}$$

Corollary 10 (Corrected-center variant, no centering certificate)**.** *Replace the predictive center in Algorithm 1 by*

$$\widehat{\mu}_t^{\mathrm{corr}}(a) := \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a) + \mathbf{g}_t(a)^\top(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}_t) \tag{36}$$

and use $\omega_t = \bar{\beta}_t$. Under the hypotheses of Theorem 1 except Assumption 4, the bounds (15)–(35) *hold.*

Proof. Local linearization gives $\mu^*(\mathbf{x}_t, a) - \widehat{\mu}_t^{\mathrm{corr}}(a) = \mathbf{g}_t(a)^\top(\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) + \rho_t$. Lemma 29 bounds the first term by $\beta_t \bar{s}_t(a)$ and $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$, which is the first step of the regret proof with $\omega_t = \bar{\beta}_t$; all later steps are unchanged. □

Remark (Cost of the corrected center). *Exact evaluation requires a solve against $\bar{\mathbf{C}}_t$. A constant number of $O(d)$ CG vectors is insufficient by itself: one additionally needs stored historical frozen gradients, raw replay data with the historical parameter checkpoints needed to reconstruct them, a frozen-feature matrix–vector oracle, an explicit low-dimensional representation, or a certified sketch. Maintaining $\bar{\mathbf{C}}_t$ explicitly costs $O(d^2)$ memory. Approximate frozen-feature access introduces an error that must be returned to the centering certificate unless separately controlled.*

C.1 Assumptions

Assumption 1 (Sub-Gaussian noise). *The noise η_t is conditionally σ-sub-Gaussian given the context and action: $\mathbb{E}[\exp(s\eta_t) \mid \mathcal{G}_t] \le \exp(s^2\sigma^2/2)$ for all $s \in \mathbb{R}$, where $\mathcal{G}_t := \mathcal{H}_t \vee \sigma(a_t)$ is the pre-reward sigma-algebra of (105), available immediately after the action is selected and before the reward is revealed.*

Assumption 2 (Bounded gradient features). *For a constant $G > 0$,*

$$\sup_{t\le T}\sup_{s\le t}\sup_{a}\|\nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a)\| \;\le\; G.$$

This bounds both the current-round features $\mathbf{g}_t(\mathbf{x}_t, a)$ and the replayed features $\mathbf{g}_t(\mathbf{x}_s, a_s)$ appearing in the curvature matrix (5), as well as the frozen features $\mathbf{g}_s(\mathbf{x}_s, a_s)$ in (6).

Assumption 3 (Local linearization). *There is a known convex parameter region Θ containing $\boldsymbol{\theta}^*$ and every realized $\boldsymbol{\theta}_t$, on which $\nabla_\theta\mu_\theta(\mathbf{x}, a)$ is L_μ-Lipschitz uniformly over candidate $(\mathbf{x}, a)$. Thus*

$$\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) \;=\; \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) + \mathbf{g}_t(\mathbf{x}, a)^\top(\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \rho_t(\mathbf{x}, a, \boldsymbol{\theta}), \tag{37}$$

where the remainder satisfies $|\rho_t(\mathbf{x}, a, \boldsymbol{\theta})| \le \frac{L_\mu}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_t\|^2$ for $\boldsymbol{\theta} \in \Theta$. Evaluating at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ gives a time-varying linearization error

$$\varepsilon_{\mathrm{lin}}(t) \;:=\; \sup_{a\in\mathcal{A}_t}|\rho_t(\mathbf{x}_t, a, \boldsymbol{\theta}^*)| \;\le\; \frac{L_\mu}{2}\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\|^2. \tag{38}$$

For Theorem 1 to give a sublinear guarantee, we require the cumulative linearization error $E_T := \sum_{t=1}^{T}\varepsilon_{\mathrm{lin}}(t) = o(T)$; see Theorem 1 for the precise dependence. Concretely, $E_T = O(1)$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ (since $\varepsilon_{\mathrm{lin}}(t) = O(1/t^2)$ and $\sum t^{-2} < \infty$; achievable in convex or strongly convex regimes), and $E_T = O(\sqrt{T})$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$. In practice this is enforced by restricting fine-tuning depth, using trust-region updates, or freezing the backbone (which renders the last-layer model convex or near-linear with small L_μ). These rate examples are illustrative only. The present paper does not prove convergence of the nonconvex optimization path $\{\boldsymbol{\theta}_t\}$ to $\boldsymbol{\theta}^*$; a sublinear guarantee from this bound therefore requires this term to be controlled by an external stability or convergence argument.

Assumption 4 (Action-scaled centering certificate). *Lemma 11 below proves the geometric bound $\big|\mathbf{g}_t(\mathbf{x}_t, a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})\big| \le \bar{s}_t(a)\,\psi_t$ for every $a \in \mathcal{A}_t$, where*

$$\psi_t \;:=\; \frac{\zeta_t}{\sqrt{\lambda}} \;+\; \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}, \tag{39}$$

$\zeta_t \ge 0$ is an $\mathcal{H}_t$-measurable optimizer gradient-residual bound ($\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \le \zeta_t$) and M_t is the gradient mismatch vector defined there. We assume there is an $\mathcal{H}_t$-measurable upper bound $\bar{\psi}_t \ge \psi_t$ (the centering certificate*) such that, for every action $a \in \mathcal{A}_t$,*

$$\big|\mathbf{g}_t(\mathbf{x}_t, a)^\top\big(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}\big)\big| \;\le\; \bar{\psi}_t\,\bar{s}_t(a). \tag{40}$$

The discrepancy is thus action-scaled*: it multiplies the confidence width $\bar{s}_t(a)$ rather than entering additively. The certificate absorbs the same three sources as before—optimizer residual ζ_t, replayed-Jacobian drift inside M_t, and higher-order nonlinear terms—but now inside a single per-round scalar $\bar{\psi}_t$. In the convex-in-parameters regime (frozen backbone with a linear reward head), under exact optimization $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} = \boldsymbol{\theta}_t$ and $\bar{\psi}_t = 0$. The corrected-center variant of Corollary 10 removes $\bar{\psi}_t$ entirely, at the cost of an explicit frozen-feature solve.*

Remark (Status of Assumption 4). *Lemma 13 supplies a predictable valid $\bar{\psi}_t$ from the optimizer residual, Welford path moments, and observed residual energy. We do not prove that this conservative quantity is small outside a restricted path. It enters the exploration term through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ (Equation (96)), not as an additive $O(T)$ term.*

Assumption	Status	Use	Checkability	Experimental instantiation
Conditional sub-Gaussian noise	Standard	Self-normalized confidence	Distributional	Gaussian generator
Realizability, $\|\theta^*\| \leq S$	Restrictive	Bias and Taylor schedules	Not generally testable	Known synthetic teacher
Bounded gradients	Standard on compact sets	Width and condition bounds	Analytic or monitored	Analytic tanh bound
L_μ, L_g on convex Θ	Restrictive	Linearization, transfer, centering	Analytic for tanh; otherwise hard	Global tanh Hessian bound and projection
Optimizer residual ζ_t	Algorithmic	Centering certificate	Pre-action gradient norm	Exact full-objective gradient norm
Operator and CG certificates	Algorithmic	Optimism and solve error	Residual/Loewner proof required	Analytic condition bound; float64 audit
Residual-energy envelope	Rate-only	Near-linear scaling	Observable after collection	Reported, not assumed by policy
Fixed active rank r	Restrictive	Closes γ_T	Supplied subspace, not terminal audit	Constructed active embedding
Window excitation κ_w	Restrictive	Closes $\Lambda_T^{\mathcal{C}}$	Analytic construction or assumed	Cyclic construction after burn-in

Table 5: Assumption ledger. Analytic and pre-action quantities are policy available; teacher comparisons and exact whitened quantities are post-hoc only.

The next lemma gives the primitive geometric bound underlying Assumption 4, expressed through the certifiable quantity ψ_t.

Lemma 11 (Primitive upper bound for prediction discrepancy)**.** *Let $z_s = (\mathbf{x}_s, a_s)$, $\mathbf{g}_s = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(z_s)$, $\mathbf{g}_{s,t} = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(z_s)$, and $\delta_{s,t} = \mathbf{g}_{s,t} - \mathbf{g}_s$. Define*

$$\bar{\mu}_{s,t} := \mu_{\boldsymbol{\theta}_s}(z_s) + \mathbf{g}_s^\top(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s), \qquad e_{s,t} := \mu_{\boldsymbol{\theta}_t}(z_s) - \bar{\mu}_{s,t}, \qquad \bar{r}_{s,t} := \bar{\mu}_{s,t} - r_s.$$

Let $\bar{\mathcal{L}}_t$ be the frozen-feature linearized ridge objective whose minimizer is $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$, and define

$$M_t := \frac{1}{\sigma^2} \sum_{s<t} \left[\bar{r}_{s,t}\delta_{s,t} + e_{s,t}\mathbf{g}_s + e_{s,t}\delta_{s,t}\right].$$

If the nonlinear optimizer satisfies $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \leq \zeta_t$, then for every candidate action $a \in \mathcal{A}_t$,

$$\left|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})\right| \;\leq\; \bar{s}_t(a)\left(\frac{\zeta_t}{\sqrt{\lambda}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\right), \qquad \bar{s}_t(a) := \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}.$$

The left-hand side is exactly the geometric bound $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \leq \bar{s}_t(a)\,\psi_t$ of Assumption 4 with $\psi_t = \zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}$ (Equation (39)). Consequently, since $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \leq \|M_t\|_2/\sqrt{\lambda}$, a valid centering certificate is

$$\bar{\psi}_t \;:=\; \frac{\zeta_t + \|M_t\|_2}{\sqrt{\lambda}} \;\geq\; \psi_t.$$

If, in addition, on a fixed event, $\boldsymbol{\theta} \mapsto \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}}(z)$ is L_g-Lipschitz, $|\bar{r}_{s,t}| \leq B_{t,\delta}$ for all $s < t$, and $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$, then

$$\|M_t\|_2 \;\leq\; \frac{1}{\sigma^2} \sum_{s<t} \left[B_{t,\delta} L_g \Delta_{s,t} + \frac{G L_g}{2}\Delta_{s,t}^2 + \frac{L_g^2}{2}\Delta_{s,t}^3\right].$$

Proof. The linearized ridge objective satisfies $\nabla \bar{\mathcal{L}}_t(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) = 0$ and, by its quadratic form,

$$\nabla \bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \bar{\mathbf{C}}_t(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}).$$

Thus $\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}} = \bar{\mathbf{C}}_t^{-1} \nabla \bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)$. Using $\mu_{\boldsymbol{\theta}_t}(z_s) - r_s = \bar{r}_{s,t} + e_{s,t}$ and $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, the gradients of the nonlinear loss and the frozen-feature linearized loss satisfy

$$\nabla \mathcal{L}_t(\boldsymbol{\theta}_t) = \nabla \bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) + M_t, \qquad \text{so} \qquad \nabla \bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \nabla \mathcal{L}_t(\boldsymbol{\theta}_t) - M_t.$$

Therefore, by Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm,

$$|\mathbf{g}_t(a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{s}_t(a) \|\nabla \bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \le \bar{s}_t(a) \left(\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \right).$$

Since $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$, $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \le \zeta_t/\sqrt{\lambda}$, which gives the geometric inequality $|\mathbf{g}_t(a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{s}_t(a)\big(\zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\big) = \bar{s}_t(a)\, \psi_t$, i.e. the bound of (39). For the certificate, $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$ also gives $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \le \|M_t\|_2/\sqrt{\lambda}$, so $\psi_t \le (\zeta_t + \|M_t\|_2)/\sqrt{\lambda} =: \bar{\psi}_t \ge \psi_t$, the proposed certificate. For the final display, Lipschitzness gives $\|\delta_{s,t}\| \le L_g \Delta_{s,t}$ and the Taylor remainder bound $|e_{s,t}| \le (L_g/2)\Delta_{s,t}^2$; applying these together with $\|\mathbf{g}_s\| \le G$ (Assumption 2) and $|\bar{r}_{s,t}| \le B_{t,\delta}$ term by term in the definition of M_t gives the claimed bound on $\|M_t\|_2$. □

Lemma 12 (Bounded-output residual energy)**.** *Fix a horizon $T \ge 1$ and $\delta_R \in (0,1)$. Suppose $|\mu_{\boldsymbol{\theta}}(x,a)| \le B_\mu$ throughout the certified parameter region and $|\mu^*(x,a)| \le B_\mu$. Under Assumption 1, with probability at least $1-\delta_R$, simultaneously for every $t \le T+1$,*

$$R_t^{\text{col}} \le \left[8B_\mu^2 + 4\sigma^2 \log\left(\frac{2T}{\delta_R}\right) \right] (t-1). \tag{41}$$

This finite-horizon envelope is not claimed to be tight.

Proof. Conditional sub-Gaussianity gives

$$\Pr\Big(|\eta_s| > \sigma \sqrt{2\log(2T/\delta_R)} \mid \mathcal{G}_s\Big) \le \delta_R/T.$$

Taking expectations and a union bound over $s = 1, \dots, T$ yields an event of probability at least $1-\delta_R$ on which this bound holds for every noise term. On that event, $c_s = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) - \mu^*(x_s, a_s) - \eta_s$ and $(a+b)^2 \le 2a^2 + 2b^2$ give

$$c_s^2 \le 2(2B_\mu)^2 + 2\sigma^2\, 2\log(2T/\delta_R) = 8B_\mu^2 + 4\sigma^2 \log(2T/\delta_R).$$

Summing the first $t-1$ terms proves (41); at $t=1$ both sides are zero. □

Lemma 13 ($O(d)$-state path certificates)**.** *Suppose $\nabla_\theta \mu_\theta(z)$ is L_g-Lipschitz and bounded by G on a convex region containing the realized iterates. Before round t, let $n_t = t-1$, let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ summarize past iterates, and let $R_t^{\text{col}} = \sum_{s<t} c_s^2$ with $c_s = \mu_{\boldsymbol{\theta}_s}(z_s) - r_s$. At $t=1$ use the initialization $\bar{\boldsymbol{\theta}}_{<1} = \mathbf{0}$, $J_{<1} = Q_1 = R_1^{\text{col}} = 0$. Then, for all t,*

$$Q_t := J_{<t} + n_t \|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2 = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$$

and the following $\mathcal{H}_t$-measurable quantities are valid:

$$\bar{\chi}_t = \frac{L_g}{\sigma\sqrt{\lambda}} \sqrt{Q_t} \ge \chi_t, \tag{42}$$

$$\bar{M}_t = \frac{1}{\sigma^2} \left[L_g \sqrt{R_t^{\text{col}} Q_t} + \frac{3GL_g}{2} Q_t + \frac{L_g^2}{2} Q_t^{3/2} \right] \ge \|M_t\|_2. \tag{43}$$

If $\|\boldsymbol{\theta}_s\|, \|\boldsymbol{\theta}_t\| \le R$, the final term in (43) may be replaced by $L_g^2 R Q_t$. If a nonnegative $\mathcal{H}_t$-measurable optimizer certificate satisfies $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\| \le \zeta_t$, then

$$\bar{\psi}_t = (\zeta_t + \bar{M}_t)/\sqrt{\lambda} \ge \psi_t.$$

If prediction gradients are L_μ-Lipschitz on a convex region containing $\boldsymbol{\theta}_t$ and $\boldsymbol{\theta}^*$ and $\|\boldsymbol{\theta}^*\| \le S$, then

$$\bar{\epsilon}_t^{\text{lin}} := \frac{L_\mu}{2} (S + \|\boldsymbol{\theta}_t\|)^2 \ge \varepsilon_{\text{lin}}(t).$$

Consequently $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\text{lin}})^2 \ge F_t$ and $\bar{E}_T = \sum_{t \le T} \bar{\epsilon}_t^{\text{lin}} \ge E_T$.

Proof. The parallel-axis identity follows after writing $\boldsymbol{\theta}_t - \boldsymbol{\theta}_s = (\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}) + (\bar{\boldsymbol{\theta}}_{<t} - \boldsymbol{\theta}_s)$; the cross term sums to zero. Lemma 19 and $\|\nabla\mu_{\boldsymbol{\theta}_t}(z_s) - \nabla\mu_{\boldsymbol{\theta}_s}(z_s)\| \le L_g\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$ give (42). Let $d_{s,t} = \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. In Lemma 11,

$$|\bar{r}_{s,t}| \le |c_s| + Gd_{s,t}, \qquad \|\delta_{s,t}\| \le L_g d_{s,t}, \qquad |e_{s,t}| \le (L_g/2)d_{s,t}^2.$$

Thus each summand defining M_t has norm at most

$$L_g|c_s|d_{s,t} + \frac{3GL_g}{2}d_{s,t}^2 + \frac{L_g^2}{2}d_{s,t}^3.$$

Cauchy–Schwarz gives $\sum |c_s|d_{s,t} \le \sqrt{R_t^{\mathrm{col}}Q_t}$, while $\sum d_{s,t}^2 = Q_t$ and $\sum d_{s,t}^3 \le Q_t^{3/2}$, proving (43). Under the trust region, $d_{s,t} \le 2R$, so $\sum d_{s,t}^3 \le 2RQ_t$. The centering claim follows from Lemma 11 and $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$. Finally Taylor's theorem and $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| \le S + \|\boldsymbol{\theta}_t\|$ prove the remainder, F_t, and E_T claims. □

Filtration-order implementation. At round t, the summaries through $t-1$, $\boldsymbol{\theta}_t$, and the certificate ζ_t are available before selection. The learner computes $Q_t, \bar{\chi}_t, u_t, \bar{M}_t, \bar{\psi}_t, \bar{\epsilon}_t^{\mathrm{lin}}$ and $\bar{\beta}_t$, then solves every candidate system to the uniform tolerance and selects a_t. After observing r_t, it adds c_t^2 to R_{t+1}^{col}, adds $\boldsymbol{\theta}_t$ with the Welford update

$$d = \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}, \quad \bar{\boldsymbol{\theta}}_{\le t} = \bar{\boldsymbol{\theta}}_{<t} + d/t, \quad J_{\le t} = J_{<t} + d^\top(\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{\le t}),$$

and updates $\bar{F}_{t+1}$ and $\hat{\gamma}_t$. It then forms $\boldsymbol{\theta}_{t+1}$ and certifies the nonnegative bound ζ_{t+1} before the next action. No current reward enters the current score.

In frozen linear-head models Q_t may be nonzero while $L_g = L_\mu = 0$; then $\bar{\chi}_t = \bar{M}_t = \bar{\epsilon}_t^{\mathrm{lin}} = 0$, and exact optimization also gives $\bar{\psi}_t = 0$.

Corollary 14 (Bounded-path near-linear rate)**.** *Fix T and suppose for every $t \le T$*

$$\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R, \quad L_\mu \le c_\mu/\sqrt{W}, \quad L_g \le c_g/\sqrt{W}, \quad \zeta_t \le \zeta_0/\sqrt{W}, \quad R_t^{\mathrm{col}} \le C_R t P_T,$$

where $P_T \ge 1$ is an explicitly supplied residual-energy factor. For exact current-parameter curvature and a uniform CG tolerance $\bar{\varepsilon}_t \le \bar{\varepsilon} < 1$, set

$$x_T = \frac{2c_g R\sqrt{T-1}}{\sigma\sqrt{\lambda W}}, \quad \rho_- = (1-x_T)^2, \qquad \rho_+ = (1+x_T)^2.$$

If $x_T < 1$, then $\rho_-\bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq \rho_+\bar{\mathbf{C}}_t$ for all $t \le T$. Writing

$$\alpha_I = \sqrt{\frac{1+\bar{\varepsilon}}{1-\bar{\varepsilon}}}, \qquad A_{\inf} = \alpha_I^2\frac{\rho_+}{\rho_-},$$

the observable information schedule satisfies $\gamma_T \le \hat{\gamma}_T \le A_{\inf}\gamma_T$, and

$$R_T \le 2\alpha_I\sqrt{\rho_+/\rho_-}\,\omega_{\max,T}^{\mathrm{op}}\sqrt{(\sigma^2 + G^2/\lambda)T\gamma_T} + 2\bar{E}_T \tag{44}$$

holds with

$$\omega_{\max,T}^{\mathrm{op}} := \max_{t\le T}(\bar{\beta}_t + \bar{\psi}_t)$$

and

$$\begin{aligned}
\omega_{\max,T}^{\mathrm{op}} \le{}& \sqrt{A_{\inf}\gamma_T + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} \\
&+ \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda W}}, \\
A_T ={}& \zeta_0 + \sigma^{-2}\Big[2c_g R\sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1)\Big], \\
B_T ={}& 4c_g^2R^3(T-1)/\sigma^2.
\end{aligned}$$

In addition, (26) *holds and*

$$\bar{\psi}_t \le \frac{1}{\sqrt{\lambda}}\left\{\frac{\zeta_0}{\sqrt{W}} + \frac{1}{\sigma^2}\left[\frac{2c_g R\sqrt{C_R t(t-1)P_T} + 6Gc_g R^2(t-1)}{\sqrt{W}} + \frac{4c_g^2R^3(t-1)}{W}\right]\right\}.$$

Proof. The trust region gives $Q_t \le 4R^2(t-1)$; substitution in Lemma 13 proves all displayed path bounds. Corollary 21 gives the two-sided operator relation. The CG sandwich and $u_t \le (1+x_T)^2 = \rho_+$ imply that every logarithm in $\hat{\gamma}_T$ is at most $\log(1 + A_{\inf}\sigma^{-2}\bar{s}_t^2(a_t))$. For $A \ge 1$, $\log(1+Ax) \le A\log(1+x)$, proving the information comparison. Substitution into Corollary 39 gives the regret display. □

Corollary 15 (Corrected-center near-linear rate)**.** *Fix T and suppose, for every $t \le T$,*

$$\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R, \qquad L_\mu \le c_\mu/\sqrt{W}, \qquad L_g \le c_g/\sqrt{W},$$

where W is an abstract near-linearity scale. Use the corrected center, exact current curvature, and a uniform CG tolerance $\bar{\varepsilon}_t \le \bar{\varepsilon} < 1$. Fix $r \in \{0, \ldots, d\}$ and suppose a predictable prefix information envelope $\mathcal{G}_{n,r} \ge \gamma_n$ is supplied for $n = 0, \ldots, T$. Valid choices are the rank envelope $\Gamma_{n,r}$ under a known rank bound, or the spectral-tail envelope in Corollary 3 with supplied prefix tails $\bar{\Delta}_{n,r}$. Define its predictable monotone closure $\mathcal{G}^{\uparrow}_{n,r} := \max_{0 \le j \le n} \mathcal{G}_{j,r}$ and set

$$x_T := \frac{2c_g R\sqrt{T-1}}{\sigma\sqrt{\lambda W}}, \qquad \alpha_I := \sqrt{\frac{1+\bar{\varepsilon}}{1-\bar{\varepsilon}}},$$
$$\mathcal{B}^{\mathrm{corr}}_{T,r} := \sqrt{\mathcal{G}^{\uparrow}_{T,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}}.$$

If $x_T < 1$, the predictable radius

$$\bar{\beta}^{\mathrm{corr}}_t := \sqrt{\mathcal{G}^{\uparrow}_{t-1,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{t-1}}{\sigma\sqrt{W}} \tag{45}$$

is valid and the corrected-center policy satisfies

$$\begin{aligned} R_T \le 2\alpha_I \frac{1+x_T}{1-x_T}\mathcal{B}^{\mathrm{corr}}_{T,r}\sqrt{(\sigma^2 + G^2/\lambda)T\mathcal{G}^{\uparrow}_{T,r}} \\ + \frac{4c_\mu R^2 T}{\sqrt{W}}. \end{aligned} \tag{46}$$

In particular, bounded transfer is ensured by $W > 4c_g^2R^2(T-1)/(\lambda\sigma^2) = \Omega(T)$ up to problem constants, and $E_T \le 2c_\mu R^2T/\sqrt{W}$. If $\mathcal{G}^{\uparrow}_{T,r} = O(r\log T)$—for example, if $\max_{n\le T}\bar{\Delta}_{n,r} = O(r\log T)$ at fixed λ—and $W = \Omega(T)$ with a sufficiently large constant to keep x_T bounded away from one, then

$$R_T = O(r\sqrt{T}\log T) + O(\sqrt{T})$$

up to confidence and CG constants.

This corollary assumes neither an optimizer-residual rate nor an envelope on R^{col}_t. Its cost is frozen-feature access: exact corrected centers require an explicit $d \times d$ frozen Gram, stored frozen gradients, replay with historical checkpoints, a supplied low-dimensional representation, or a separately certified sketch. The result therefore does not have $O(d)$ total memory merely because current-curvature products are matrix-free.

Proof. The running maximum $\mathcal{G}^{\uparrow}_{n,r}$ is predictable and dominates both $\mathcal{G}_{n,r}$ and every earlier prefix envelope. The trust region gives $Q_t \le 4R^2(t-1)$, so Lemma 13 yields $\bar{\chi}_t \le 2c_gR\sqrt{t-1}/(\sigma\sqrt{\lambda W}) \le x_T$, $\bar{\epsilon}^{\mathrm{lin}}_t \le 2c_\mu R^2/\sqrt{W}$, $\bar{F}_t \le 4c_\mu^2R^4(t-1)/W$, and $\bar{E}_T \le 2c_\mu R^2T/\sqrt{W}$. Hence (45) dominates (99). Apply the corrected-center form of Corollary 2, bound $\alpha_t \le \alpha_I$, $(1+\bar{\chi}_t)/(1-\bar{\chi}_t) \le (1+x_T)/(1-x_T)$, and $\bar{\beta}^{\mathrm{corr}}_t \le \mathcal{B}^{\mathrm{corr}}_{T,r}$, then use $\gamma_T \le \mathcal{G}_{T,r} \le \mathcal{G}^{\uparrow}_{T,r}$. Finally substitute $2\bar{E}_T \le 4c_\mu R^2T/\sqrt{W}$. The stated orders follow directly. □

Remark (Why W is an abstract near-linearity scale)**.** *The all-weights MLP used in the nonlinear diagnostics has action-a mean $v_a^\top \tanh(Ux+b) + c_a$ and no $W^{-1/2}$ forward normalization. With $\bar{x} = (x,1)$, hidden vector $q_i = (U_i, b_i)$, and $z_i = q_i^\top\bar{x}$, the Hessian block on $(q_i, v_{a,i})$ is*

$$\begin{bmatrix} v_{a,i}\tanh''(z_i)\bar{x}\bar{x}^\top & \tanh'(z_i)\bar{x} \\ \tanh'(z_i)\bar{x}^\top & 0 \end{bmatrix}.$$

At $z_i = 0$ and $v_{a,i} = 0$ its operator norm is $\|\bar{x}\|$, independent of hidden width. Thus the executed all-weights parameterization does not imply $L_\mu, L_g = O(W^{-1/2})$. Such a conclusion would require a different explicitly normalized parameterization and coordinate constraints, which would also change the ridge penalty and GGN geometry. We therefore do not identify W with generic neural-network width or derive optimizer-residual scaling from it.

Corollary 16 (Conditional projected-Gram rate for a supplied subspace)**.** *Under Corollary 14, additionally suppose a subspace $U \subseteq \mathbb{R}^d$ with $\dim U \le r$ is fixed and supplied before interaction, and every candidate or replayed tangent feature through round T lies in U. Hence $\operatorname{rank}(\bar{\mathbf{C}}_t - \lambda \mathbf{I}) \le r$ for every $t \le T+1$. Let $r_U = \dim U$, let $Q_U \in \mathbb{R}^{d\times r_U}$ have orthonormal columns spanning U, and define*

$$\mathbf{z}_{t,s} = Q_U^\top \mathbf{g}_t(\mathbf{x}_s, a_s), \qquad \mathbf{z}_t(a) = Q_U^\top \mathbf{g}_t(\mathbf{x}_t, a).$$

Then the current-GGN width reduces exactly to

$$\mathbf{g}_t(a)^\top \mathbf{C}_t^{-1}\mathbf{g}_t(a) = \mathbf{z}_t(a)^\top \left(\lambda I_{r_U} + \sigma^{-2}\sum_{s<t} \mathbf{z}_{t,s}\mathbf{z}_{t,s}^\top\right)^{-1} \mathbf{z}_t(a), \tag{47}$$

with the analogous identity for $\bar{\mathbf{C}}_t$ using $Q_U^\top \mathbf{g}_s(\mathbf{x}_s, a_s)$. Thus a supplied basis permits an exact dense $r_U \times r_U$ Gram with $O(r_U^2)$ curvature state. Given tangent coordinates, forming the current Gram costs $O((t-1)r_U^2)$, factorization $O(r_U^3)$, and each width query $O(r_U^2)$; current-parameter replay, model training, and gradient generation can still be d-dimensional. Full-dimensional CG is not needed in this corollary's covered regime.

Define $\Gamma_{T,r}$ by (27). Then (28) holds with

$$\begin{aligned}
\Omega_{T,r} &:= \sqrt{A_{\inf}\Gamma_{T,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda}W},\\
A_T &= \zeta_0 + \sigma^{-2}\left[2c_g R\sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1)\right],\\
B_T &= 4c_g^2 R^3(T-1)/\sigma^2,
\end{aligned}$$

and $A_{\inf}, \rho_\pm$ are exactly those in Corollary 14. In particular, for fixed r and fixed problem constants, if $W \ge CT^2P_T$ for a constant C satisfying the explicit thresholds in (48), then

$$R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W}).$$

This rate is conditional on the ex ante subspace U and every bounded-path, near-linearity, residual, optimizer, transfer, and solver premise of Corollary 14. It neither learns U nor covers unrestricted representation learning or end-to-end neural-network training; it is an effective projected-feature closure, not a full-dimensional computational advantage for CC-UCB.

Proof. Extend Q_U to an orthogonal basis of $\mathbb{R}^d$. In that basis $\mathbf{C}_t$ is block diagonal with the $r_U \times r_U$ matrix in (47) and λI_{d-r_U}; the query has zero coordinates in the second block. This proves (47), and the same argument proves the frozen-Gram identity. Lemma 7 gives $\gamma_T \le \Gamma_{T,r}$. Every occurrence of γ_T in Corollary 14 is monotone, so direct substitution gives the display. Under the stated near-linearity scaling, x_T, the centering terms, and the linearization terms are bounded, while $\Gamma_{T,r} = O(r\log T)$. Hence $\Omega_{T,r} = O(\sqrt{r\log T} + 1)$ and the leading product is $O(r\sqrt{T}\log T)$. □

Remark (Linear-subclass lower bound)**.** *The exact-rank class contains a standard r-dimensional stochastic linear bandit. Take a constant context, a known orthonormal basis Q_U, the finite action set $z(a) \in \{\pm 1/\sqrt{r}\}^r$, linear mean $\mu_\vartheta(a) = z(a)^\top\vartheta$, and Gaussian noise. Choosing the usual hypercube parameter family with coordinates of order $\sqrt{r/T}$ keeps the parameter norm bounded when $T \gtrsim r^2$. Here $L_\mu = L_g = E_T = 0$, $\mathbf{C}_t = \bar{\mathbf{C}}_t$, and all features lie in U. The stochastic linear-bandit minimax expected-regret lower bound is $\Omega(r\sqrt{T})$ in this regime (Dani et al., 2008). Hence the leading $r\sqrt{T}$ dependence of the exact-rank upper bound is sharp up to logarithms on this linear subclass. This does not prove a lower bound for the nonlinear or spectral-tail classes, nor a computational lower bound.*

For explicit targets $\bar{x}_{\mathrm{drift}} \in (0, 1)$ and $\Psi, b_F, \eta_E > 0$, the sufficient conditions

$$\begin{aligned} &W \geq \frac{4c_g^2 R^2 (T-1)}{\lambda \sigma^2 \bar{x}_{\mathrm{drift}}^2}, \qquad W \geq \frac{4c_\mu^2 R^4 T}{\sigma^2 b_F^2}, \quad W \geq \frac{4c_\mu^2 R^4}{\eta_E^2}, \\ &W \geq \max\left\{\frac{4A_T^2}{\lambda \Psi^2}, \frac{2B_T}{\sqrt{\lambda}\Psi}\right\} \end{aligned} \tag{48}$$

respectively ensure $x_T \leq \bar{x}_{\mathrm{drift}} < 1$ (and therefore $\rho_- \geq (1-\bar{x}_{\mathrm{drift}})^2$ and $\rho_+ \leq (1+\bar{x}_{\mathrm{drift}})^2$), $\sqrt{\bar{F}_{T+1}}/\sigma \leq b_F$, $\bar{E}_T/T \leq \eta_E$, and $\bar{\psi}_T \leq \Psi$. Thus the centering requirement has leading sufficient scaling $W = \Omega(T^2 P_T)$; these conditions do not bound γ_T.

Corollary 17 (Nonlinear tanh-link Gaussian bandit)**.** *For $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $\|\phi(x, a)\| \leq B_\phi$, one may take*

$$G = B_\phi, \qquad L_\mu = L_g = \frac{4}{3\sqrt{3}} B_\phi^2.$$

On any certified trust region, Corollary 5 therefore gives an executable nonlinear Gaussian-reward CC-UCB policy using only the online path statistics, a pre-action optimizer-residual certificate, and the stated operator/CG certificates. Under realizability, both the model and teacher outputs have $B_\mu = 1$. Hence Lemma 12 removes the supplied residual-energy factor in Corollary 14: on its event one may take

$$C_R = 1, \qquad P_T = 8 + 4\sigma^2 \log(2T/\delta_R),$$

and allocate δ_R to the certificate failure budget.

Proof. $\nabla_\theta \mu = \mathrm{sech}^2(\phi^\top \theta)\phi$, so its norm is at most B_ϕ. The Hessian is $-2\tanh(z)\,\mathrm{sech}^2(z)\phi\phi^\top$. For $y = |\tanh z|$, $2y(1-y^2)$ is maximized at $y = 1/\sqrt{3}$ with value $4/(3\sqrt{3})$, proving both Lipschitz constants. Apply Lemma 13 and Corollary 5. □

Corollary 18 (Scaled-tanh relative-link rate)**.** *Consider the explicit zero-offset scalar-link family*

$$\mu_{\boldsymbol{\theta}, W}(z) = \sqrt{W}\,\tanh\left(\frac{\phi(z)^\top \boldsymbol{\theta}}{\sqrt{W}}\right), \qquad \|\phi(z)\| \leq B_\phi, \qquad \|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \leq R. \tag{49}$$

This is a scaled generalized-linear family in the displacement coordinates, not a claim that generic neural-network width provides the scale W. Let

$$c_h := \frac{4B_\phi^2}{3\sqrt{3}}, \qquad \rho_W := \exp\left(\frac{4B_\phi R}{\sqrt{W}}\right) - 1.$$

Then valid global constants on the certified region are

$$G = B_\phi, \qquad L_\mu = L_g = \frac{c_h}{\sqrt{W}}, qquad |\mu_{\boldsymbol{\theta}, W}(z)| \leq B_\phi R, qquad \rho_t \leq \rho_W. \tag{50}$$

The stated exponential relative-drift certificate obeys $\rho_W < 1$ exactly when

$$W > \left(\frac{4B_\phi R}{\log 2}\right)^2. \tag{51}$$

Fix a uniform CG certificate $\bar{\varepsilon}_t \leq \bar{\varepsilon} < 1$, let $\alpha_I = \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$, and suppose $\zeta_t \leq \zeta_0/\sqrt{W}$ and $R_t^{\mathrm{col}} \leq C_R t P_T$ for $P_T \geq 1$. Fix a rank index r and supply a predictable prefix information envelope $\mathcal{G}_{n,r} \geq \gamma_n$; write $\mathcal{G}_{n,r}^\uparrow = \max_{j \leq n} \mathcal{G}_{j,r}$. The envelope may be the exact-rank bound or either valid predictable spectral-tail schedule in Corollary 3. Define

$$\begin{aligned} \Psi_{T,W}^{\mathrm{rel}} := \frac{\zeta_0}{\sqrt{\lambda W}} + \frac{1}{\sigma}\Bigg\{ &\rho_W \left[\sqrt{C_R T P_T} + 2B_\phi R\sqrt{T-1}\right] \\ &+ \frac{2(1+\rho_W) c_h R^2 \sqrt{T-1}}{\sqrt{W}} \Bigg\}, \end{aligned} \tag{52}$$

$$\mathcal{B}_{T,r}^{\mathrm{rel}} := \sqrt{\mathcal{G}_{T,r}^\uparrow + 2\log(1/\delta)} + \sqrt{\lambda} R + \frac{2c_h R^2 \sqrt{T}}{\sigma\sqrt{W}} + \Psi_{T,W}^{\mathrm{rel}}. \tag{53}$$

If $\rho_W < 1$, exact current curvature with the original nonlinear center and the relative certificate of Lemma 20 satisfies

$$R_T \le 2\alpha_I \frac{1+\rho_W}{1-\rho_W}\mathcal{B}^{\mathrm{rel}}_{T,r}\sqrt{\left(\sigma^2 + \frac{B_\phi^2}{\lambda}\right)T\mathcal{G}^{\uparrow}_{T,r}} + \frac{4c_h R^2 T}{\sqrt{W}}. \tag{54}$$

More explicitly, choose a fixed target $\rho_0 \in (0,1)$ and suppose

$$W \ge CTP_T, \qquad C \ge C_{\rho_0} := \left(\frac{4B_\phi R}{\log(1+\rho_0)}\right)^2. \tag{55}$$

Then $\rho_W \le \rho_0$, and the mean-value theorem gives the displayed uniform certificate

$$\begin{aligned}\Psi^{\mathrm{rel}}_{T,W} \le \frac{\zeta_0}{\sqrt{\lambda W}} + \frac{1}{\sigma}\Bigg[&4(1+\rho_0)B_\phi R\sqrt{\frac{C_R}{C}} \\ &+ \frac{8(1+\rho_0)B_\phi^2 R^2 + 2(1+\rho_0)c_h R^2}{\sqrt{CP_T}}\Bigg].\end{aligned} \tag{56}$$

Furthermore $\bar{F}_{T+1} \le 4c_h^2 R^4 T/W$ and $\bar{E}_T \le 2c_h R^2 T/\sqrt{W}$. Hence, if $\mathcal{G}^{\uparrow}_{T,r} = O(r\log T)$ with fixed problem, confidence, and CG constants,

$$R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W}).$$

At $W = \Theta(TP_T)$ this is $O(r\sqrt{T}\log T + \sqrt{T/P_T})$, and therefore $O(r\sqrt{T}\log T + \sqrt{T})$ up to the displayed residual factor.

Under realizability, Lemma 12 supplies the premise with probability at least $1-\delta_R$ by taking

$$C_R = 1, \qquad P_T = \max\{1, 8B_\phi^2 R^2 + 4\sigma^2\log(2T/\delta_R)\}. \tag{57}$$

Thus this fully analytic residual choice makes the sufficient width $W = \Omega(T\log(T/\delta_R))$ at fixed problem constants. Its event is added to the certificate failure allocation. No claim is made that the residual or width constants are tight.

Proof. For $h_W(q) = \sqrt{W}\tanh(q/\sqrt{W})$, $h'_W(q) = \operatorname{sech}^2(q/\sqrt{W})$ and

$$h''_W(q) = -\frac{2}{\sqrt{W}}\tanh(q/\sqrt{W})\operatorname{sech}^2(q/\sqrt{W}).$$

The maximum of $2y(1-y^2)$ on $y \in [0,1]$ is $4/(3\sqrt{3})$, proving the gradient and Hessian bounds in (50); also $\sqrt{W}|\tanh(q/\sqrt{W})| \le |q| \le B_\phi R$. Moreover

$$\left|\frac{d}{dq}\log h'_W(q)\right| = \frac{2}{\sqrt{W}}|\tanh(q/\sqrt{W})| \le \frac{2}{\sqrt{W}}.$$

Since $|q_{s,t} - q_s| \le B_\phi\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\| \le 2B_\phi R$, $|\log c_{s,t}| \le 4B_\phi R/\sqrt{W}$. Hence $|c_{s,t} - 1| \le e^{4B_\phi R/\sqrt{W}} - 1 = \rho_W$, and solving $e^{4B_\phi R/\sqrt{W}} - 1 < 1$ gives (51) for this exponential certificate.

The trust region gives $Q_t \le 4R^2(t-1)$. Substitution in (75) proves $\bar{\psi}^{\mathrm{rel}}_t \le \Psi^{\mathrm{rel}}_{T,W}$. Taylor's theorem gives $\bar{\epsilon}^{\mathrm{lin}}_t \le 2c_h R^2/\sqrt{W}$, and hence the stated bounds on $\bar{F}_{T+1}, \bar{E}_T$ and the confidence radius (53). Apply Corollary 2, bound the time-varying factors by α_I, ρ_W, and $\mathcal{B}^{\mathrm{rel}}_{T,r}$, and use $\gamma_T \le \mathcal{G}^{\uparrow}_{T,r}$ to obtain (54).

Under (55), the exponential mean-value bound $e^x - 1 \le (1+\rho_0)x$ for $0 \le x \le \log(1+\rho_0)$ gives $\rho_W \le 4(1+\rho_0)B_\phi R/\sqrt{W}$. Substitute this and $W \ge CTP_T$ in (52) to prove (56); the rate follows. Finally Lemma 12 with $B_\mu = B_\phi R$ proves (57). □

Predictable algorithmic curvature under certified transfer. The analysis is conducted for a *generic predictable SPD algorithmic curvature sequence* $\{\mathcal{C}_t\}_{t\geq 1}$ satisfying

$$\mathcal{C}_t \text{ is symmetric}, \qquad \mathcal{C}_t \succeq \lambda \mathbf{I} \succ 0, \qquad \mathcal{C}_t \text{ is } \mathcal{H}_t\text{-measurable}. \tag{58}$$

This is the operator whose inverse the algorithm actually applies (via CG) to form widths. The exact full-history result is the special case $\mathcal{C}_t = \mathbf{C}_t$ (relinearized curvature (5)); approximate, subsampled, windowed, refreshed, or sketched operators $\hat{\mathbf{C}}_t$ are also represented by $\mathcal{C}_t$, but are covered only when the required transfer certificate below is available. Write the algorithmic width

$$s^2_{\mathcal{C},t}(a) \;:=\; \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a). \tag{59}$$

Assumption 5 (One-sided optimism transfer)**.** *For every round $t \leq T$ there is an $\mathcal{H}_t$-measurable factor $u_t(a) \geq 1$ such that, for every candidate action $a \in \mathcal{A}_t$,*

$$\bar{s}_t^2(a) \;\leq\; u_t(a)\, s^2_{\mathcal{C},t}(a). \tag{60}$$

This is the *only* spectral relation the main theorem needs, and it is one-sided: it dominates the predictable frozen-feature width $\bar{s}_t(a)$ (the object of the self-normalized confidence bound) by an inflated algorithmic width. No *lower* Loewner factor on $\mathcal{C}_t$ is required for optimism. The factor may be action-dependent; when a uniform per-round u_t satisfies (60) for every action we write u_t and state the action-dependent theorem with $u_t(a)$. For the exact-curvature instantiation $\mathcal{C}_t = \mathbf{C}_t$, the whitened relative-drift lemma below certifies (60) with the uniform factor $u_t = (1+\bar{\chi}_t)^2$; for an approximate operator, the composition of Lemma 19 with a one-sided spectral bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t} \mathbf{C}_t$ gives $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ (Theorem 34).

The following lemma supplies the exact-curvature transfer factor through a *whitened, relative* feature-drift bound: the drift is measured in the $\bar{\mathbf{C}}_t^{-1}$ metric and accumulated in ℓ_2, and the resulting one-sided bound needs *no* smallness condition.

Lemma 19 (Whitened relative feature-drift bound)**.** *Let $\mathbf{g}_s := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$, $\mathbf{g}_{s,t} := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$, $\delta_{s,t} := \mathbf{g}_{s,t} - \mathbf{g}_s$, and $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. Using the symmetric (principal) square root $\bar{\mathbf{C}}_t^{1/2}$, define the $d \times (t-1)$ whitened column matrices*

$$\mathsf{G}_t := \Big[\, \bar{\mathbf{C}}_t^{-1/2} \mathbf{g}_s / \sigma \,\Big]_{s<t}, \qquad \mathsf{D}_t := \Big[\, \bar{\mathbf{C}}_t^{-1/2} \delta_{s,t} / \sigma \,\Big]_{s<t}, \qquad \chi_t := \|\mathsf{D}_t\|_{\mathrm{op}}. \tag{61}$$

Writing $n_t = t-1$, also define the $(d+n_t) \times d$ stacked matrices

$$\mathbf{A}_t := \begin{bmatrix} \sqrt{\lambda}\, \bar{\mathbf{C}}_t^{-1/2} \\ \mathsf{G}_t^\top \end{bmatrix}, \qquad \mathbf{B}_t := \begin{bmatrix} \sqrt{\lambda}\, \bar{\mathbf{C}}_t^{-1/2} \\ (\mathsf{G}_t + \mathsf{D}_t)^\top \end{bmatrix}. \tag{62}$$

At $t=1$ the lower blocks have zero rows. All square roots and inverse square roots in this statement are principal. Then:

(a) $\mathsf{G}_t \mathsf{G}_t^\top = \mathbf{I} - \lambda\, \bar{\mathbf{C}}_t^{-1} \preceq \mathbf{I}$, *hence* $\|\mathsf{G}_t\|_{\mathrm{op}} \leq 1$.

(b) $\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2} = \mathsf{G}_t \mathsf{D}_t^\top + \mathsf{D}_t \mathsf{G}_t^\top + \mathsf{D}_t \mathsf{D}_t^\top$, *so* $\big\|\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}\big\|_{\mathrm{op}} \leq 2\chi_t + \chi_t^2$.

(c) The stacked designs obey

$$\mathbf{A}_t^\top \mathbf{A}_t = \mathbf{I}_d, \qquad \mathbf{B}_t^\top \mathbf{B}_t = \bar{\mathbf{C}}_t^{-1/2} \mathbf{C}_t \bar{\mathbf{C}}_t^{-1/2}, \qquad \|\mathbf{B}_t - \mathbf{A}_t\|_{\mathrm{op}} = \chi_t. \tag{63}$$

(d) ***(One-sided, no smallness condition.)*** *For every $\chi_t \geq 0$,*

$$\mathbf{C}_t \;\preceq\; (1+\chi_t)^2\, \bar{\mathbf{C}}_t, \qquad \text{hence} \qquad \bar{s}_t^2(a) \;\leq\; (1+\chi_t)^2\, s_t^2(a) \quad \forall a. \tag{64}$$

A predictable certificate $\bar{\chi}_t \geq \chi_t$ therefore satisfies Assumption 5 in the exact case $\mathcal{C}_t = \mathbf{C}_t$ with the uniform factor $u_t = (1+\bar{\chi}_t)^2$.

(e) ***(Whitened ℓ_2 primitive bound.)***

$$\chi_t \;\leq\; \|\mathsf{D}_t\|_F = \left[\sigma^{-2} \sum_{s<t} \|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}}\right]^{1/2} \;\leq\; \frac{L_g}{\sigma\sqrt{\lambda}} \left[\sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2\right]^{1/2}, \tag{65}$$

the last step under the L_g-Lipschitz-Jacobian assumption $\|\delta_{s,t}\| \leq L_g \Delta_{s,t}$.

Proof. *(a)* $\mathsf{G}_t\mathsf{G}_t^\top = \sigma^{-2}\sum_{s<t}\bar{\mathbf{C}}_t^{-1/2}\mathbf{g}_s\mathbf{g}_s^\top\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}(\sigma^{-2}\sum_{s<t}\mathbf{g}_s\mathbf{g}_s^\top)\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}(\bar{\mathbf{C}}_t-\lambda\mathbf{I})\bar{\mathbf{C}}_t^{-1/2} = \mathbf{I}-\lambda\bar{\mathbf{C}}_t^{-1}$. Since $\lambda\bar{\mathbf{C}}_t^{-1}\succeq 0$, $\mathsf{G}_t\mathsf{G}_t^\top\preceq\mathbf{I}$, and $\|\mathsf{G}_t\|_{\mathrm{op}}^2 = \|\mathsf{G}_t\mathsf{G}_t^\top\|_{\mathrm{op}}\le 1$.

(b) From $\mathbf{g}_{s,t}=\mathbf{g}_s+\delta_{s,t}$, $\mathbf{g}_{s,t}\mathbf{g}_{s,t}^\top-\mathbf{g}_s\mathbf{g}_s^\top = \mathbf{g}_s\delta_{s,t}^\top+\delta_{s,t}\mathbf{g}_s^\top+\delta_{s,t}\delta_{s,t}^\top$. Conjugating each summand by $\bar{\mathbf{C}}_t^{-1/2}$ and weighting by σ^{-2}, the three groups sum exactly to $\mathsf{G}_t\mathsf{D}_t^\top$, $\mathsf{D}_t\mathsf{G}_t^\top$, and $\mathsf{D}_t\mathsf{D}_t^\top$. By the triangle inequality and $\|\mathsf{G}_t\|_{\mathrm{op}}\le 1$, $\|\mathsf{G}_t\mathsf{D}_t^\top+\mathsf{D}_t\mathsf{G}_t^\top+\mathsf{D}_t\mathsf{D}_t^\top\|_{\mathrm{op}}\le 2\|\mathsf{G}_t\|_{\mathrm{op}}\|\mathsf{D}_t\|_{\mathrm{op}}+\|\mathsf{D}_t\|_{\mathrm{op}}^2\le 2\chi_t+\chi_t^2$.

(c) By part (a), $\mathbf{A}_t^\top\mathbf{A}_t=\lambda\bar{\mathbf{C}}_t^{-1}+\mathsf{G}_t\mathsf{G}_t^\top=\mathbf{I}_d$. Expanding the current replay Gram gives

$$\mathbf{B}_t^\top\mathbf{B}_t=\lambda\bar{\mathbf{C}}_t^{-1}+(\mathsf{G}_t+\mathsf{D}_t)(\mathsf{G}_t+\mathsf{D}_t)^\top=\bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2}.$$

The two stacks differ only in their bottom block, hence $\|\mathbf{B}_t-\mathbf{A}_t\|_{\mathrm{op}}=\|\mathsf{D}_t^\top\|_{\mathrm{op}}=\chi_t$. These identities also hold at $t=1$, with empty bottom blocks and $\chi_1=0$.

(d) The matrix $W_t:=\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t-\bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}$ is symmetric with $\|W_t\|_{\mathrm{op}}\le 2\chi_t+\chi_t^2$, hence $W_t\preceq(2\chi_t+\chi_t^2)\mathbf{I}$. Congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t-\bar{\mathbf{C}}_t\preceq(2\chi_t+\chi_t^2)\bar{\mathbf{C}}_t$, i.e. $\mathbf{C}_t\preceq(1+\chi_t)^2\bar{\mathbf{C}}_t$; no bound on χ_t is used. Both $\mathbf{C}_t,\bar{\mathbf{C}}_t\succ 0$, so inversion reverses the order: $\mathbf{C}_t^{-1}\succeq(1+\chi_t)^{-2}\bar{\mathbf{C}}_t^{-1}$. The quadratic form at $\mathbf{g}_t(a)$ gives $s_t^2(a)\ge(1+\chi_t)^{-2}\bar{s}_t^2(a)$, i.e. $\bar{s}_t^2(a)\le(1+\chi_t)^2 s_t^2(a)$. Since $x\mapsto(1+x)^2$ is increasing on $[0,\infty)$, replacing χ_t by any $\bar{\chi}_t\ge\chi_t$ preserves the inequality, giving $u_t=(1+\bar{\chi}_t)^2$.

(e) $\chi_t=\|\mathsf{D}_t\|_{\mathrm{op}}\le\|\mathsf{D}_t\|_F$, and $\|\mathsf{D}_t\|_F^2=\sum_{s<t}\|\bar{\mathbf{C}}_t^{-1/2}\delta_{s,t}/\sigma\|^2=\sigma^{-2}\sum_{s<t}\delta_{s,t}^\top\bar{\mathbf{C}}_t^{-1}\delta_{s,t}=\sigma^{-2}\sum_{s<t}\|\delta_{s,t}\|_{\bar{\mathbf{C}}_t^{-1}}^2$. Since $\bar{\mathbf{C}}_t^{-1}\preceq\lambda^{-1}\mathbf{I}$, $\|\delta_{s,t}\|_{\bar{\mathbf{C}}_t^{-1}}^2\le\|\delta_{s,t}\|^2/\lambda\le L_g^2\Delta_{s,t}^2/\lambda$; summing and taking square roots gives (65). ☐

Lemma 20 (Relative scalar-link drift and centering)**.** *Suppose the prediction model has the fixed-feature scalar-link form $\mu_{\boldsymbol{\theta}}(z)=h(\phi(z)^\top\boldsymbol{\theta})$, where $\phi(z)\in\mathbb{R}^d$, h is differentiable, and $h'(q)$ is nonzero throughout the certified scalar region. For $s<t$, write $\phi_s=\phi(z_s)$ and define*

$$q_s=\phi_s^\top\boldsymbol{\theta}_s,\qquad q_{s,t}=\phi_s^\top\boldsymbol{\theta}_t,\qquad c_{s,t}=\frac{h'(q_{s,t})}{h'(q_s)},\qquad \rho_t=\max_{s<t}|c_{s,t}-1|,$$

with $\rho_1=0$. Here ϕ_s,q_s, and $h'(q_s)$ are fixed before reward r_s, while $\boldsymbol{\theta}_t$ and every ratio above are $\mathcal{H}_t$-measurable before action a_t is selected. Then:

(a) The scalar-link gradients obey

$$\mathbf{g}_s=h'(q_s)\phi_s,\qquad \mathbf{g}_{s,t}=h'(q_{s,t})\phi_s=c_{s,t}\mathbf{g}_s,\qquad \delta_{s,t}=(c_{s,t}-1)\mathbf{g}_s. \tag{66}$$

Writing $\mathbf{c}_t=(c_{1,t},\dots,c_{t-1,t})^\top\in\mathbb{R}^{t-1}$, the $d\times(t-1)$ matrices of Lemma 19 satisfy

$$\mathsf{D}_t=\mathsf{G}_t\,\mathrm{Diag}(\mathbf{c}_t-\mathbf{1}),\qquad \chi_t\le\|\mathsf{G}_t\|_{\mathrm{op}}\rho_t\le\rho_t. \tag{67}$$

At $t=1$ the matrices have zero columns and the same identities hold under the empty-matrix convention.

(b) Exact current curvature therefore satisfies, without a smallness condition,

$$\mathbf{C}_t\preceq(1+\rho_t)^2\bar{\mathbf{C}}_t. \tag{68}$$

If $\rho_t<1$, it also satisfies

$$(1-\rho_t)^2\bar{\mathbf{C}}_t\preceq\mathbf{C}_t\preceq(1+\rho_t)^2\bar{\mathbf{C}}_t. \tag{69}$$

Any predictable upper certificate $\bar{\rho}_t\ge\rho_t$ may replace ρ_t in the one-sided result, and may replace it in both sides when $\bar{\rho}_t<1$.

(c) Reuse $\bar{r}_{s,t},e_{s,t}$, and M_t from Lemma 11, and define

$$v_{s,t}:=\bar{r}_{s,t}(c_{s,t}-1)+c_{s,t}e_{s,t},\qquad \mathbf{v}_t=(v_{1,t},\dots,v_{t-1,t})^\top\in\mathbb{R}^{t-1}. \tag{70}$$

Then the centering mismatch has the exact structured form

$$M_t=\frac{1}{\sigma^2}\sum_{s<t}v_{s,t}\mathbf{g}_s,\qquad \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\le\frac{\|\mathbf{v}_t\|_2}{\sigma}. \tag{71}$$

(d) If $\|\boldsymbol{\theta}_s\|, \|\boldsymbol{\theta}_t\| \le R$, the gradients are bounded by G and are L_g-Lipschitz on the radius-R region, then

$$\|\bar{\mathbf{r}}_t\|_2 \le \sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}, \qquad \bar{\mathbf{r}}_t = (\bar{r}_{1,t}, \dots, \bar{r}_{t-1,t})^\top, \tag{72}$$

$$\|\mathbf{e}_t\|_2 \le L_g R\sqrt{Q_t}, \qquad \mathbf{e}_t = (e_{1,t}, \dots, e_{t-1,t})^\top, \tag{73}$$

$$\|\mathbf{v}_t\|_2 \le \rho_t\left(\sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}\right) + (1+\rho_t)L_g R\sqrt{Q_t}. \tag{74}$$

Consequently a valid original-center certificate is

$$\bar{\psi}_t^{\mathrm{rel}} := \frac{\zeta_t}{\sqrt{\lambda}} + \frac{\rho_t(\sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}) + (1+\rho_t)L_g R\sqrt{Q_t}}{\sigma}. \tag{75}$$

The right side remains valid after replacing ρ_t by a predictable upper bound $\bar{\rho}_t$.

The exact ρ_t requires replaying ϕ_s and retaining at least the collection scalar q_s (or $h'(q_s)$) for every sample. Computing the exact $v_{s,t}$ additionally uses the stored reward and the same scalar-link replay; q_s suffices because both $\bar{r}_{s,t}$ and $e_{s,t}$ can be reconstructed from $q_s, q_{s,t}, h(q_s)$, and $h'(q_s)$. Thus the structured replay uses $O(t)$ scalar metadata in addition to raw-example replay. The Welford quantities Q_t, R_t^{col} remain $O(d)$ path-summary state, but neither implementation is $O(d)$ total memory.

Proof. Equation (66) is the chain rule. Applying it to each column of (61) gives (67). Since the operator norm of the diagonal matrix is ρ_t and $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$ by Lemma 19, $\chi_t \le \rho_t$. Equations (68) and (69) then follow respectively from the one-sided part of Lemma 19 and Corollary 21.

Substitute $\delta_{s,t} = (c_{s,t} - 1)\mathbf{g}_s$ in the definition of M_t:

$$\bar{r}_{s,t}\delta_{s,t} + e_{s,t}\mathbf{g}_s + e_{s,t}\delta_{s,t} = \{\bar{r}_{s,t}(c_{s,t} - 1) + c_{s,t}e_{s,t}\}\mathbf{g}_s.$$

This proves the first identity in (71). Since $M_t = \sigma^{-1}\mathsf{G}_t\mathbf{v}_t$ after left multiplication by $\bar{\mathbf{C}}_t^{-1/2}$,

$$\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} = \sigma^{-1}\|\mathsf{G}_t\mathbf{v}_t\|_2 \le \sigma^{-1}\|\mathbf{v}_t\|_2,$$

which proves the second identity and uses equivalently $\mathsf{G}_t^\top\mathsf{G}_t \preceq I_{t-1}$.

Finally let $d_{s,t} = \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. The collection residual $c_s = \mu_{\boldsymbol{\theta}_s}(z_s) - r_s$ gives $\bar{r}_{s,t} = c_s + \mathbf{g}_s^\top(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s)$. The triangle and Cauchy–Schwarz inequalities therefore yield $\|\bar{\mathbf{r}}_t\|_2 \le \sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}$. Taylor's theorem gives $|e_{s,t}| \le (L_g/2)d_{s,t}^2$; because both parameters lie in the radius-R region, $d_{s,t} \le 2R$, hence $|e_{s,t}| \le L_g R d_{s,t}$ and $\|\mathbf{e}_t\|_2 \le L_g R\sqrt{Q_t}$. Applying the triangle inequality to $\mathbf{v}_t = \mathrm{Diag}(\mathbf{c}_t - \mathbf{1})\bar{\mathbf{r}}_t + \mathrm{Diag}(\mathbf{c}_t)\mathbf{e}_t$, using $\|\mathbf{c}_t - \mathbf{1}\|_\infty = \rho_t$ and $\|\mathbf{c}_t\|_\infty \le 1 + \rho_t$, proves (74). Combine this with Lemma 11 and (71) to obtain (75). □

Corollary 21 (Sharp two-sided feature-drift sandwich)**.** *If $\chi_t < 1$, then*

$$(1-\chi_t)^2\,\bar{\mathbf{C}}_t \;\preceq\; \mathbf{C}_t \;\preceq\; (1+\chi_t)^2\,\bar{\mathbf{C}}_t. \tag{76}$$

Consequently, if a predictable certificate satisfies $\chi_t \le \bar{\chi}_t < 1$, then $(1-\bar{\chi}_t)^2\bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq (1+\bar{\chi}_t)^2\bar{\mathbf{C}}_t$.

Proof. Because $\mathbf{A}_t^\top\mathbf{A}_t = \mathbf{I}_d$, all d singular values of the $(d+n_t)\times d$ matrix $\mathbf{A}_t$ equal one. Singular-value perturbation and (63) give, for $j = 1, \dots, d$,

$$|\sigma_j(\mathbf{B}_t) - 1| \le \|\mathbf{B}_t - \mathbf{A}_t\|_{\mathrm{op}} = \chi_t.$$

For $\chi_t < 1$, every singular value of $\mathbf{B}_t$ therefore lies in $[1-\chi_t, 1+\chi_t]$. Squaring the singular values and using $\mathbf{B}_t^\top\mathbf{B}_t = \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2}$ yields $(1-\chi_t)^2\mathbf{I} \preceq \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2} \preceq (1+\chi_t)^2\mathbf{I}$. Congruence by the principal $\bar{\mathbf{C}}_t^{1/2}$ proves (76). If $0 \le \chi_t \le \bar{\chi}_t < 1$, then $(1-\bar{\chi}_t)^2 \le (1-\chi_t)^2$ and $(1+\chi_t)^2 \le (1+\bar{\chi}_t)^2$, proving the certificate version. □

The main theorem uses only the one-sided bound (64), which holds for *any* $\chi_t \geq 0$; the smallness condition $\chi_t < 1$ is needed only for the lower half of the two-sided sandwich (76), which in turn is used by the explicit near-linear corollary and by any optional analysis that explicitly assumes a two-sided sandwich. An unrescaled current-parameter window is handled by Proposition 33 through a one-sided relation and does not require this condition. The binding quantity is the *whitened ℓ_2 accumulation* $\chi_t \leq \|\mathsf{D}_t\|_F$ of replayed-Jacobian drift, measured in the $\bar{\mathbf{C}}_t^{-1}$ metric—a relative, whitened ℓ_2 bound that can be sharper than an ℓ_1 sum normalized only by λ when the drift occurs in well-observed directions (it credits directions the design has already explored), though it does not dominate uniformly. Together, Lemmas 11 and 19 identify exact whitened diagnostic targets. The operational upper bounds in Lemma 13 replace those targets by the predictable path statistics Q_t and R_t, the certified optimizer residual ζ_t, and analytic smoothness constants. Thus $\bar{\psi}_t$ and $u_t = (1 + \bar{\chi}_t)^2$ require only $O(d)$ additional state; the exact whitened quantities remain post-hoc diagnostics used to measure the conservatism of these certificates.

Positive definiteness of both $\mathbf{C}_t$ and $\bar{\mathbf{C}}_t$ follows from $\lambda > 0$ and the outer-product form; it is not an independent assumption. All theoretical results hold under these explicit assumptions; we do *not* claim a regret guarantee for the full nonlinear model outside the local-linearization regime.

C.2 Novelty Relative to Prior Neural Bandits

NeuralUCB/NeuralTS use historical-time full-gradient Grams in their stated algorithms, while NeuralLinear and last-layer Laplace restrict uncertainty to a learned representation or final layer (Zhou et al., 2020; Zhang et al., 2021; Riquelme et al., 2018; Kristiadi et al., 2020). Matrix-free full-network predictive-variance solves also predate this work (Maddox et al., 2021). CC-UCB's narrower adaptation is to replay every historical Jacobian at the current parameter, separate the predictable frozen confidence Gram from that algorithmic operator, and expose centering, drift, transfer, and residual-checked CG terms. We claim neither the CG/CVP primitive nor a prior assertion of uniform full-metric superiority as new.

The one-sided whitened result in Lemma 19 is the main technical step: optimism transfer needs no lower spectral comparison. The determinant formula is bookkeeping rather than a standalone rate, and the supplied-U closure is exactly the projected Gram in (47), not a concrete neural-training theorem. Appendix D compares NTK, regression, EKF/LO-FI, structured Laplace, and sketching alternatives.

C.3 Computable Curvature Operator

Applying $\mathbf{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s<t}\mathbf{g}_t(\mathbf{x}_s, a_s)\big[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}\big]$ *naively* requires one curvature–vector product (CVP) per historical sample per CG step—cost $O(It)$ CVPs per round. Each CVP evaluates one Jacobian–vector product (forward mode) followed by one vector–Jacobian product (reverse mode) through the network, at roughly the cost of one gradient evaluation (one JVP plus one VJP)[1]; we do not store per-sample gradient features $\mathbf{g}_s$, only a constant number of $O(d)$ CG vectors for CG, with CVPs computed on the fly. We describe three strategies that limit the per-step cost to $|\mathcal{S}_t|$ CVPs:

(i) Subsampled GGN (worst-case spectral baseline). At each round draw a uniform subsample $\mathcal{S}_t \subset \{1, \ldots, t-1\}$ of size n (without replacement) and set $w_s = (t-1)/n$ to obtain the rescaled operator. The following lemma is a worst-case uniform spectral guarantee.

Lemma 22 (Worst-case uniform spectral guarantee for subsampled GGN)**.** *Fix round t and set $m = t-1$. Let $Y_s = \sigma^{-2}\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ for $s = 1, \ldots, m$, so that $\mathbf{C}_t = \lambda\mathbf{I} + V$ with $V = \sum_{s=1}^m Y_s$. Each Y_s is PSD with $\|Y_s\|_{\mathrm{op}} \leq G^2/\sigma^2 =: B$. For $\delta \in (0, 1)$ and $1 \leq n \leq m$, draw $\mathcal{S}_t$ of size n uniformly without replacement from $\{1, \ldots, m\}$ and define $\hat{V} = \frac{m}{n}\sum_{s\in\mathcal{S}_t} Y_s$ and $\hat{\mathbf{C}}_t = \lambda\mathbf{I} + \hat{V}$. Write $L := \log(2d/\delta)$.*

(i) Deviation bound. *With probability $\geq 1 - \delta$,*

$$\|\hat{V} - V\|_{\mathrm{op}} \;\leq\; mB\sqrt{\frac{2L}{n}} \;+\; \frac{4\,mB\,L}{3\,n}. \tag{77}$$

[1] Implementations based on double-backward Hessian–vector products cost closer to two gradient-equivalents per CVP; the constant does not affect the $O(\cdot)$ accounting.

(ii) Sufficient sample size. *For any* $\varepsilon \in (0, 1)$*, if*

$$n \;\geq\; 8L\left[\frac{m^2\,B^2}{\varepsilon^2\,\lambda^2} \;+\; \frac{m\,B}{3\,\varepsilon\,\lambda}\right], \tag{78}$$

then $\Pr[\|\hat{V} - V\|_{\mathrm{op}} > \varepsilon\lambda] \leq \delta$ *and consequently* $(1{-}\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1{+}\varepsilon)\mathbf{C}_t$.

Proof. The $\{Y_s\}_{s=1}^m$ are fixed PSD self-adjoint matrices (deterministic conditional on the history $\mathcal{F}_{t-1}$ and the current parameter $\boldsymbol{\theta}_t$). The displayed proof is for *with-replacement* (i.i.d.) sampling. The without-replacement version follows from the domination argument of Gross and Nesme (2010), building on Hoeffding's reduction (Hoeffding, 1963): the matrix moment-generating function of a sum sampled without replacement from a fixed finite collection of self-adjoint matrices is dominated by that of the corresponding i.i.d. sum, so every tail bound derived from the matrix Laplace-transform method, including the matrix Bernstein inequality of Tropp (2015, Thm. 6.1.1), carries over with the same constants. In our setting the $\{Y_s\}$ are *fixed* PSD matrices (deterministic conditional on $\mathcal{F}_{t-1}$ and $\boldsymbol{\theta}_t$) and only the index set $\mathcal{S}_t$ is random, so the domination argument applies directly. We therefore state and prove the bound for i.i.d. draws $s_1, \ldots, s_n$ uniform on $\{1, \ldots, m\}$; the without-replacement guarantee is at least as strong. Define the centered random matrices $Z_k = (m/n)Y_{s_k} - V/n$ (mean zero, independent). The range satisfies $\|Z_k\|_{\mathrm{op}} \leq (m/n)B + (1/n)\|V\| \leq 2mB/n =: R$ (using $\|V\| \leq mB$). For the variance parameter, since $Y_s \preceq B\mathbf{I}$,

$$\mathbb{E}[Z_k Z_k^\top] \;\preceq\; \tfrac{m^2}{n^2}\mathbb{E}[Y_{s_1}^2] \;=\; \tfrac{m}{n^2}\sum_{s=1}^{m} Y_s^2 \;\preceq\; \tfrac{m}{n^2} B\, V \;\preceq\; \tfrac{m^2 B^2}{n^2}\mathbf{I},$$

so $v := \big\|\sum_{k=1}^n \mathbb{E}[Z_k Z_k^\top]\big\| \leq m^2B^2/n$. By the matrix Bernstein inequality (Tropp, 2015, Thm. 6.1.1), $\Pr[\|\hat{V}{-}V\| \geq t] \leq 2d\exp\!\big({-t^2}/(2v + 2Rt/3)\big)$. Choosing $t_0 = \sqrt{2vL} + \frac{2}{3}RL$ makes the matrix-Bernstein tail probability at most δ. Substituting $v \leq m^2B^2/n$ and $R = 2mB/n$ gives $t_0 \leq mB\sqrt{2L/n} + \frac{4mBL}{3n}$, yielding (77). For (78), require each summand in (77) to be $\leq \varepsilon\lambda/2$: $mB\sqrt{2L/n} \leq \varepsilon\lambda/2$ gives $n \geq 8m^2B^2L/(\varepsilon^2\lambda^2)$, and $4mBL/(3n) \leq \varepsilon\lambda/2$ gives $n \geq 8mBL/(3\varepsilon\lambda)$. Summing the two conditions yields (78); this is a conservative sufficient condition (using $a + b$ rather than $\max\{a, b\}$), not a sharp threshold. The multiplicative sandwich follows: $\|\hat{V}{-}V\| \leq \varepsilon\lambda$ gives $\mathbf{C}_t{-}\varepsilon\lambda\mathbf{I} \preceq \hat{\mathbf{C}}_t \preceq \mathbf{C}_t{+}\varepsilon\lambda\mathbf{I}$. Since $\mathbf{C}_t \succeq \lambda\mathbf{I}$, we have $\varepsilon\lambda\mathbf{I} \preceq \varepsilon\mathbf{C}_t$, so $\mathbf{C}_t{-}\varepsilon\lambda\mathbf{I} \succeq (1{-}\varepsilon)\mathbf{C}_t$ and $\mathbf{C}_t{+}\varepsilon\lambda\mathbf{I} \preceq (1{+}\varepsilon)\mathbf{C}_t$. □

With $B = G^2/\sigma^2$, the right-hand side of the sufficient condition (78) is $O\big(L \cdot [m^2G^4/(\varepsilon^2\lambda^2\sigma^4) + mG^2/(\varepsilon\lambda\sigma^2)]\big)$. The first (variance) term dominates unless $\varepsilon\lambda\sigma^2 \gtrsim mG^2$, so this worst-case sufficient threshold has a *quadratic leading dependence* on $m = t{-}1$ (it is a sufficient upper bound on the required sample size, not an exact or necessary sample complexity). The condition (78) is conservative and sufficient, not necessary, with worst-case leading dependence quadratic in m. If its threshold exceeds the population size $m = t{-}1$, it provides no nontrivial subsampling guarantee for $n < m$. Taking $n = m$ recovers the full population exactly for the separate reason that every summand is included.

This is a worst-case baseline guarantee, too conservative for practical use at large t. In particular, the bound above does not by itself justify computational savings at large t; it should be read as a worst-case spectral guarantee rather than as a practical scaling result. Subsampling reduces computation ($n \ll m$) only when m is moderate or $\varepsilon\lambda$ is large relative to mB; for large t the full history may be needed under the uniform Bernstein bound. Under additional spectral structure, leverage-score sampling or Frequent Directions sketches may yield useful approximations with smaller state. For linear bandits, Kuzborskij et al. (2019) give regret bounds in terms of the Frequent Directions spectral-tail error. Translating such structure into a uniform GGN sandwich and a concrete buffer or sketch size is left open. The Bernstein guarantee is not the mechanism by which we expect practical savings at scale. Practical savings require small buffers treated heuristically, spectral decay/effective dimension, ridge-leverage sampling, Frequent Directions, or empirical verification of the spectral sandwich. We leave structure-aware sampling and empirical spectral certification as future work. To obtain a spectral sandwich that holds simultaneously for all $t \leq T$, union-bound over t (replacing δ by δ/T). Theorem 34 then exposes the one-sided transfer factor and realized width complexity as operator-dependent terms along the realized path; it does not identify a standalone causal regret inflation.

(ii) Windowed curvature (unrescaled, current parameter). Maintain a sliding window $\mathcal{S}_t \subseteq \{1, \ldots, t{-}1\}$ of the most recent n interactions with unit weights ($w_s{=}1$), *re-evaluating the Jacobians at the current parameter* $\boldsymbol{\theta}_t$. This is a nonnegative-weighted *subset* of the same current-parameter outer products, so $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (Proposition 33);

Theorem 34 then applies with the *deterministic* approximate-to-full factor $\kappa_{+,t}$=1 and the full-to-frozen factor $u_t = (1 + \bar{\chi}_t)^2$—no two-sided sandwich is required, though the overall guarantee remains conditional on that transfer certificate. The cost of the smaller operator is not hidden: it surfaces as larger realized widths $s^2_{\mathcal{C},t}(a_t)$ and hence a larger dynamic width complexity $\Lambda^{\mathcal{C}}_T$. The same holds for any current-parameter downweighted subset with $w_s \in [0, 1]$ (including an unrescaled subsample). A *stale* window that reuses Jacobians from an older parameter is *not* covered; see (iii).

(iii) Periodic refresh (heuristic; stale operator). Recompute the curvature operator every M rounds on a cached buffer and reuse it for M consecutive actions. Because the reused operator evaluates Jacobians at an *older* parameter, it is no longer a subset of the current-parameter outer products, so the $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ argument of Proposition 33 does not apply and the stale operator can both over- and under-estimate $\mathbf{C}_t$. This remains a heuristic: Theorem 34 covers it only when a one-sided factor $\kappa_{+,t}$ (an explicit bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$) is separately verified, which we evaluate empirically. The rescaled subsample of (i), with weights $w_s = (t-1)/n > 1$, is likewise not covered by Proposition 33; it instead uses the two-sided matrix Bernstein bound of Lemma 22.

C.4 Computational Cost Model

Primitive assumptions. CC-UCB requires a single computational primitive: the *curvature–vector product* (CVP) $\mathbf{v} \mapsto \mathbf{C}_t\mathbf{v}$, which decomposes into per-sample operations $\mathbf{v} \mapsto \mathbf{g}_t(\mathbf{x}_s, a_s)\big[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}\big]$. Each such operation consists of:

1. a Jacobian–vector product (JVP) $\mathbf{v} \mapsto \mathbf{g}_t^\top\mathbf{v}$ (forward-mode AD, cost $\approx 1\times$ forward pass);
2. a vector–Jacobian product (VJP) $\alpha \mapsto \alpha\,\mathbf{g}_t$ (reverse-mode AD, cost $\approx 1\times$ backward pass).

Sharded autodiff can distribute JVP/VJP evaluation, but its communication pattern and sample parallelism are implementation-dependent and are not benchmarked here. For $m_t := |\mathcal{S}_t|$, one matrix-free oracle application consists of m_t sample-CVPs plus $O(d)$ damping and accumulation work. Thus $I_{t,a}$ CG recurrences for action a use $I_{t,a}m_t$ sample-CVPs; an explicit final check of the original-system residual $\mathbf{g}_t(a) - \mathcal{C}_t\tilde{\mathbf{u}}$ uses another m_t. The sequential Krylov state is $O(d)$ ($O(Kd)$ if right-hand sides are batched), excluding model and optimizer state, activations, and raw replay. Caching an explicit $m_t \times d$ Jacobian matrix instead uses $O(m_t d)$ additional state and is not the on-the-fly JVP/VJP implementation analyzed by this operation count.

Krylov methods for Hessian spectral analysis have been run on networks with $> 10^8$ parameters under distributed training (Granziol and Juarev, 2026), which indicates that curvature–vector primitives of this general form are implementable at scale. We do *not* claim foundation-model feasibility from this: that work times Hessian–vector products, not the exact per-example GGN JVP/VJP used here. Our real-`torch.func` study reaches roughly 10^7 parameters and $m = 512$ on one accelerator; it does not measure distributed execution or model training. Warm-starting CG from the previous round's solution can reduce the iteration count in practice.

Cost accounting. Let $C_F(W_{\rm net}), C_B(W_{\rm net}), C_J(W_{\rm net})$, and $C_V(W_{\rm net})$ denote the one-example forward, reverse, JVP, and VJP costs for model-compute scale $W_{\rm net}$, and define

$$C_{{\rm op},t} := m_t[C_J(W_{\rm net}) + C_V(W_{\rm net})] + O(d).$$

Parameter count d alone does not determine these costs when architecture, input length, or weight sharing changes. Let $J^{\rm opt}_{t+1}$ be the number of full-batch first-order steps after round t; $q^{\rm opt}_{t+1} \in \{0, 1\}$ records a separate full-gradient optimizer certificate, and $q^{\rm CG}_{t,a} \in \{0, 1\}$ records an explicit final CG residual application. Table 6 separates the training history N_t from the curvature buffer m_t.

Table 6: Per-round arithmetic cost. $N_t = t - 1$ is the training-history size; $m_t = |\mathcal{S}_t|$ is the curvature-buffer size.

Component	Cost
Candidate means/features	$O(K[C_F + C_B])$
CG recurrences/checks	$C_{{\rm op},t}\sum_{a=1}^K(I_{t,a} + q^{\rm CG}_{t,a})$
Parameter update/certificate	$(J^{\rm opt}_{t+1} + q^{\rm opt}_{t+1})\{(N_t + 1)[C_F + C_B] + O(d)\}$
Path statistics/scoring	$O(d + K)$

This is an operation count, not a wall-clock benchmark. For illustration, $K = 10$, $I = 25$, and $m_t = 256$ give 64,000 sample-CVPs for the recurrences and 66,560 when every solve receives one explicit final residual application. Both numbers exclude candidate gradients, full-history model training, and optimizer certification. Backtracking adds full-history objective passes, and second-order optimization has additional Hessian costs. A linear sufficient-statistic update may substitute a lower special-case cost, but must still supply the theorem's optimizer certificate.

Cost of the analyzed configuration. The full-history instantiation $\mathcal{C}_t = \mathbf{C}_t$, $\mathcal{S}_t = \{1, \ldots, t-1\}$ with adaptive CG stopping has the certificate $\bar{\kappa}_t = 1 + (t-1)G^2/(\lambda\sigma^2)$, so maintaining $\max_a \varepsilon_{t,a} \le \bar{\varepsilon}_t$ is certified by $I_t = O\big(\sqrt{t}\, G/(\sigma\sqrt{\lambda})\, \log(1/\bar{\varepsilon}_t)\big)$ iterations per solve (Eq. (112)), so the per-round CG cost is $K\, I_t\, (t-1) = O(K\, t^{3/2})$ sample-CVPs, i.e. $O(K\, T^{5/2})$ cumulatively—the worst-case width-solve sample-CVP count under the stated CG certificate. End-to-end work additionally contains $\sum_{t \le T}(J_t^{\mathrm{opt}} + q_t^{\mathrm{opt}})N_t$ full-data sample-gradient evaluations. This is $O((J+q)T^2)$ only when those counts are uniformly bounded, and curvature windowing does not reduce it. Table 6 and the numerical example describe fixed-buffer operators ($m_t \ll N_t$), which reduce width-solve work but are covered by Theorem 1 only under the corrected window/subsample conditions of §C.3: an unrescaled current-parameter buffer has the deterministic approximate-to-full factor $\kappa_{+,t}{=}1$ (Proposition 33)—the overall guarantee remaining conditional on the full-to-frozen transfer and the other certificates of Theorem 1—while a rescaled or stale buffer additionally requires the verified transfer factor of Theorem 34.

For the growing window of Theorem 6, one solve costs $O(m_t^{3/2} \log(1/\bar{\varepsilon}))$ sample-CVPs under the analytic condition-number bound. The following table uses the sufficient fixed-burn-in setting $\tau_w = O(1)$, $H_T = \widetilde{O}(1)$, and $E_T = \widetilde{O}(T^{1-q/2})$; $E_T = o(T)$ alone is insufficient. The sample-CVP column counts only width solves. With uniformly bounded full-batch optimizer/certificate passes, the final column adds the T^2 full-history training term. These are achieved upper bounds for this implementation, not a Pareto frontier, lower bound, or wall-clock measurement (logarithmic factors suppressed):

q	conditional regret	width-solve CVPs	full-policy sample work
$\frac{1}{2}$	$T^{\frac{3}{4}}$	$KT^{\frac{7}{4}}$	T^2
$\frac{2}{3}$	$T^{\frac{2}{3}}$	KT^2	T^2
1	$T^{\frac{1}{2}}$	$KT^{\frac{5}{2}}$	$T^{\frac{5}{2}}$

C.5 An Exact Log-Determinant Bookkeeping Identity

The regret bound is driven not by γ_T but by the realized *width complexity* of the widths the algorithm actually uses,

$$\Lambda_T^{\mathcal{C}} \;:=\; \sum_{t=1}^{T} \log\!\big(1 + \sigma^{-2}\, s_{\mathcal{C},t}^2(a_t)\big). \tag{79}$$

Unlike $\bar{\mathbf{C}}_t$, the algorithmic curvature $\mathcal{C}_t$ need *not* be monotone in t: relinearization, subsampling, windowing, and periodic refresh can all *shrink* it, so the elementary determinant telescoping of (31) no longer applies. The following lemma is an exact pathwise bookkeeping identity for a general predictable SPD sequence. It is not a standalone rate result: its upper bound is informative only when the terminal log determinant and determinant-decreasing refresh charge are independently controlled, as in Lemma 24.

Lemma 23 (Dynamic log-determinant potential)**.** *Let $\{\mathcal{C}_t\}_{t\ge 1}$ satisfy* (58) *with $\mathcal{C}_1 = \lambda\mathbf{I}$. For the played feature $\mathbf{h}_t := \mathbf{g}_t(\mathbf{x}_t, a_t)$ define the rank-one update $\mathcal{C}_t^+ := \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$ and, using the symmetric (principal) root $(\mathcal{C}_t^+)^{-1/2}$, the normalized perturbation*

$$\Xi_t \;:=\; (\mathcal{C}_t^+)^{-1/2}\,(\mathcal{C}_{t+1} - \mathcal{C}_t^+)\,(\mathcal{C}_t^+)^{-1/2}, \tag{80}$$

so that $\mathbf{I} + \Xi_t = (\mathcal{C}_t^+)^{-1/2}\mathcal{C}_{t+1}(\mathcal{C}_t^+)^{-1/2}$ is SPD; equivalently, every eigenvalue of Ξ_t exceeds -1 and every eigenvalue of $\mathbf{I} + \Xi_t$ is positive (with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$). Fix any end-of-round-$T$ *terminal operator $\mathcal{C}_{T+1}$ constructed only from data and randomness available through round T ($\mathcal{F}_T$-measurable, hence not a function of $\mathbf{x}_{T+1}$, $\mathcal{A}_{T+1}$, or future rewards; e.g. the canonical rank-one update $\mathcal{C}_{T+1} = \mathcal{C}_T^+$, or the operator the algorithm commits for round $T{+}1$). The identity below is then an a posteriori statement about the completed sequence $\mathcal{C}_1, \ldots, \mathcal{C}_{T+1}$; the quantities Ξ_T, $V_T^{\mathcal{C}}$, and Γ_T^{dyn} are defined relative to this choice, and the* *bound* $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ *holds for* every *such valid choice. Then*

$$\Lambda_T^{\mathcal{C}} \;=\; \log\frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} \;-\; \sum_{t=1}^{T}\log\det(\mathbf{I}+\Xi_t). \tag{81}$$

With the determinant-destroying variation charge and the dynamic potential

$$V_T^{\mathcal{C}} := \sum_{t=1}^{T}\big[-\log\det(\mathbf{I}+\Xi_t)\big]_+, \qquad \Gamma_T^{\mathrm{dyn}} := \log\frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} + V_T^{\mathcal{C}}, \tag{82}$$

we have $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$*, and, using* $s^2_{\mathcal{C},t}(a_t) \le G^2/\lambda$ *and* $\log(1+x) \ge x/(1+x)$*,*

$$\sum_{t=1}^{T} s^2_{\mathcal{C},t}(a_t) \;\le\; (\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}} \;\le\; (\sigma^2 + G^2/\lambda)\,\Gamma_T^{\mathrm{dyn}}. \tag{83}$$

Proof. By the matrix determinant lemma applied to $\mathcal{C}_t^+ = \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$,

$$\frac{\det(\mathcal{C}_t^+)}{\det(\mathcal{C}_t)} = 1 + \sigma^{-2}\mathbf{h}_t^\top\mathcal{C}_t^{-1}\mathbf{h}_t = 1 + \sigma^{-2}s^2_{\mathcal{C},t}(a_t). \tag{84}$$

Writing $B_t := (\mathcal{C}_t^+)^{-1/2}$, we have $\mathbf{I}+\Xi_t = B_t\mathcal{C}_t^+B_t + B_t(\mathcal{C}_{t+1}-\mathcal{C}_t^+)B_t = B_t\mathcal{C}_{t+1}B_t$, a congruence of the SPD matrix $\mathcal{C}_{t+1}$ by the invertible symmetric B_t, hence SPD (in particular its eigenvalues exceed -1 with no smallness condition on the jump $\mathcal{C}_{t+1}-\mathcal{C}_t^+$), and

$$\frac{\det(\mathcal{C}_{t+1})}{\det(\mathcal{C}_t^+)} = \det\!\big(B_t^{-1}(\mathbf{I}+\Xi_t)B_t^{-1}\big)\big/\det(\mathcal{C}_t^+) = \det(\mathbf{I}+\Xi_t), \tag{85}$$

using $\det(\mathcal{C}_{t+1}) = \det(\mathcal{C}_t^+)\det(\mathbf{I}+\Xi_t)$. Taking logs of (84)–(85) and adding,

$$\log\!\big(1+\sigma^{-2}s^2_{\mathcal{C},t}(a_t)\big) = \log\det(\mathcal{C}_{t+1}) - \log\det(\mathcal{C}_t) - \log\det(\mathbf{I}+\Xi_t).$$

Summing over $t = 1,\dots,T$ telescopes the first difference to $\log\det(\mathcal{C}_{T+1}) - \log\det(\mathcal{C}_1)$; with $\mathcal{C}_1 = \lambda\mathbf{I}$ this is $\log[\det(\mathcal{C}_{T+1})/\det(\lambda\mathbf{I})]$, proving (81). Since $-\log\det(\mathbf{I}+\Xi_t) \le [-\log\det(\mathbf{I}+\Xi_t)]_+$ term by term, $-\sum_t\log\det(\mathbf{I}+\Xi_t) \le V_T^{\mathcal{C}}$, so $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. For (83): $\mathcal{C}_t \succeq \lambda\mathbf{I}$ gives $s^2_{\mathcal{C},t}(a_t) = \mathbf{h}_t^\top\mathcal{C}_t^{-1}\mathbf{h}_t \le \|\mathbf{h}_t\|^2/\lambda \le G^2/\lambda =: c_{\max}\sigma^2$. With $x_t := \sigma^{-2}s^2_{\mathcal{C},t}(a_t) \in [0, c_{\max}]$ and $\log(1+x) \ge x/(1+x) \ge x/(1+c_{\max})$, $\Lambda_T^{\mathcal{C}} \ge (1+c_{\max})^{-1}\sum_t x_t$, i.e. $\sum_t s^2_{\mathcal{C},t}(a_t) \le (\sigma^2+G^2/\lambda)\Lambda_T^{\mathcal{C}}$; the second inequality is $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. □

Remark (Interpretation and scope of the dynamic potential)**.** *Identity* (81) *is pathwise.* $V_T^{\mathcal{C}}$ *charges only changes that reduce determinant relative to the rank-one update; determinant-increasing changes contribute zero. If* $\mathcal{C}_1 \ne \lambda\mathbf{I}$*, the denominator becomes* $\det(\mathcal{C}_1)$*. Neither* $V_T^{\mathcal{C}}$ *nor the endpoint term is generally small. Bounding both is sufficient, but not necessary because the exact identity may cancel them. Thus the identity does not imply sublinear regret; outside Corollary 16 and Theorem 6, controlling* $\Lambda_T^{\mathcal{C}}$ *for an adapting backbone remains open.*

Lemma 24 (Rank-sensitive refresh bound)**.** *Let* $\mathcal{R} = \{t : \mathcal{C}_{t+1} \ne \mathcal{C}_t^+\}$*. Suppose that for every* $t \in \mathcal{R}$*, the negative part* $(\Xi_t)_- = (-\Xi_t)_+$ *has rank at most* r_t *and* $\lambda_{\min}(\mathbf{I}+\Xi_t) \ge 1-\nu_t$ *for* $0 \le \nu_t < 1$*. Then*

$$V_T^{\mathcal{C}} \le \sum_{t\in\mathcal{R}} r_t\log\frac{1}{1-\nu_t}. \tag{86}$$

If additionally $\mathcal{C}_{T+1} = \lambda\mathbf{I} + A$*, where* $A \succeq 0$*,* $\mathrm{rank}(A) \le r_{\mathrm{end}}$*, and* $\mathrm{tr}(A) \le W_{\mathrm{end}}G^2/\sigma^2$ *for a specified total endpoint weight* $W_{\mathrm{end}} \ge 0$*, then, for* $r_{\mathrm{end}} > 0$*,*

$$\Lambda_T^{\mathcal{C}} \le r_{\mathrm{end}}\log\left(1 + \frac{W_{\mathrm{end}}G^2}{r_{\mathrm{end}}\lambda\sigma^2}\right) + \sum_{t\in\mathcal{R}} r_t\log\frac{1}{1-\nu_t}. \tag{87}$$

For $r_{\mathrm{end}} = 0$*, the first term is defined as zero.*

Proof. Let $\xi_{t,i}$ be the eigenvalues of Ξ_t. At most r_t are negative, each is at least $-\nu_t$, and every nonnegative eigenvalue makes a nonpositive contribution to $-\sum_i \log(1+\xi_{t,i})$. Hence $[-\log\det(\mathbf{I}+\Xi_t)]_+ \le r_t \log(1/(1-\nu_t))$, proving (86). List the nonzero eigenvalues of A and pad with zeros to obtain $a_1,\dots,a_{r_{\text{end}}}$. Concavity gives

$$\sum_i \log(1+a_i/\lambda) \le r_{\text{end}} \log\left(1+\frac{\sum_i a_i}{r_{\text{end}}\lambda}\right),$$

and the trace premise plus Lemma 23 proves (87). ☐

Consequences. Sequential Gram updates have $\Xi_t = 0$. If canonical rank-one updates are used outside J_T refresh rounds, and each refresh has $r_t \le r$, $\nu_t \le \nu < 1$, then $V_T^{\mathcal{C}} \le J_T r \log(1/(1-\nu))$; a geometric schedule is therefore polylogarithmic only when these rank and spectral-floor bounds hold. For a replacement $\mathcal{C}_{t+1} - \mathcal{C}_t^+ = A_t^{\text{new}} - A_t^{\text{old}}$ with both terms PSD and $\operatorname{rank}(A_t^{\text{old}}) \le r$, Sylvester inertia gives $r_t \le r$, but a spectral floor is still required.

For a fixed-feature window of length m, after warm-up each deletion has rank one. Writing $N_{\text{drop}} = \max\{0, T-m\}$, Sherman–Morrison gives the following exact charge for each $t > m$. Let $v_t^{\text{drop}} = \sigma^{-1}\mathbf{g}_{t-m}(\mathbf{x}_{t-m}, a_{t-m})$ and let $D_t = \mathcal{C}_t^+ - v_t^{\text{drop}}(v_t^{\text{drop}})^\top = \mathcal{C}_{t+1}$ be the remaining-window matrix. Then $-\log\det(\mathbf{I}+\Xi_t) = \log[1+(v_t^{\text{drop}})^\top D_t^{-1} v_t^{\text{drop}}]$, and therefore

$$V_T^{\mathcal{C}} \le N_{\text{drop}} \log\left(1+\frac{G^2}{\lambda\sigma^2}\right).$$

This can be linear in T. A relinearized window need not be a rank-one replacement. Likewise, merely holding a stale operator fixed does not make non-refresh transitions canonical rank-one updates, so geometric refresh timing alone does not imply a logarithmic charge.

Corollary 25 (Observable CG upper bound on the width complexity)**.** *If the played-action CG solve satisfies $\varepsilon_{t,a_t} \le \bar{\varepsilon}_t < 1$ so that $\tilde{s}_t^2(a_t) \ge (1-\bar{\varepsilon}_t)\, s_{\mathcal{C},t}^2(a_t)$ (Lemma 26), then*

$$\Lambda_T^{\mathcal{C}} \;\le\; \widehat{\Lambda^{\mathcal{C}}}_T \;:=\; \sum_{t=1}^{T} \log\!\Big(1+\frac{\tilde{s}_t^2(a_t)}{\sigma^2\,(1-\bar{\varepsilon}_t)}\Big). \tag{88}$$

Proof. $s_{\mathcal{C},t}^2(a_t) \le \tilde{s}_t^2(a_t)/(1-\bar{\varepsilon}_t)$ and $x \mapsto \log(1+\sigma^{-2}x)$ is increasing, so each summand of $\Lambda_T^{\mathcal{C}}$ is dominated term by term. ☐

This gives an observable upper bound on the dynamic width-complexity term, conditional on the certified CG tolerances; $V_T^{\mathcal{C}}$ itself is *not* observable in general (it requires determinant estimation of the curvature sequence).

C.6 CG Approximation Lemma

Lemma 26 (CG multiplicative accuracy)**.** *Fix a round t and an action a with right-hand side $\mathbf{g} := \mathbf{g}_t(a) \ne \mathbf{0}$. Let $\mathbf{u} = \mathcal{C}_t^{-1}\mathbf{g}$ and let $\tilde{\mathbf{u}}$ be the iterate after $I_{t,a}$ steps of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$, with $\mathcal{C}_t$ held* fixed *across all iterations. The relative energy-norm error is the per-action quantity $\varepsilon_{t,a}$ of* (9)*, and*

$$\varepsilon_{t,a} \;=\; \frac{\|\mathbf{u}-\tilde{\mathbf{u}}\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} \;\le\; 2\left(\frac{\sqrt{\bar{\kappa}_t}-1}{\sqrt{\bar{\kappa}_t}+1}\right)^{I_{t,a}}, \tag{89}$$

where $\kappa_t := \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number and $\bar{\kappa}_t \ge \kappa_t$ is a certified $\mathcal{H}_t$-measurable upper bound. The convergence rate in (89) *is right-hand-side independent, so it holds for every action with the same $\bar{\kappa}_t$. For a* nonnegative weighted outer-product *operator $\mathcal{C}_t = \lambda\mathbf{I} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\,\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ ($w_s \ge 0$, so $\mathcal{C}_t \succeq \lambda\mathbf{I} \succ 0$ and CG is well-posed),*

$$\kappa_t \le \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \le 1 + \frac{W_t G^2}{\lambda\sigma^2} =: \bar{\kappa}_t,$$

with $W_t = \sum_{s\in\mathcal{S}_t} w_s$ and $\|\mathbf{g}_t(\mathbf{x}_s, a_s)\| \le G$ from Assumption 2; here $W_t = n$ for unrescaled windowed/refresh buffers, giving $\bar{\kappa}_t$ and α_t independent of T, and $W_t = t-1$ for full history and for rescaled uniform subsampling with $w_s = (t-1)/n$. For a general SPD operator that is not such an outer product, $\bar{\kappa}_t$ must be certified separately. Write $\tilde{s}_t^2(a) = \mathbf{g}^\top\tilde{\mathbf{u}}$ and $s_{\mathcal{C},t}^2(a) = \mathbf{g}^\top\mathcal{C}_t^{-1}\mathbf{g}$.

Assume $\varepsilon_{t,a} < 1$ *(see Appendix G for the minimum $I_{t,a}$ ensuring this as a function of $\bar{\kappa}_t$; Algorithm 1 enforces a known* per-round *tolerance $\bar{\varepsilon}_t$ uniformly over the scored actions,* $\max_a \varepsilon_{t,a} \le \bar{\varepsilon}_t < 1$*, via adaptive residual stopping or a fixed budget under a certified condition-number bound; see §3). Then*

$$(1 - \varepsilon_{t,a})\, s^2_{\mathcal{C},t}(a) \;\le\; \tilde{s}^2_t(a) \;\le\; (1 + \varepsilon_{t,a})\, s^2_{\mathcal{C},t}(a). \tag{90}$$

For $\mathbf{g}_t(a) = \mathbf{0}$ the algorithm sets $\tilde{s}^2_t(a) = 0 = s^2_{\mathcal{C},t}(a)$ and (90) holds trivially.

Proof. The CG minimax property gives $\|e_I\|_{\mathcal{C}_t}/\|e_0\|_{\mathcal{C}_t} \le \min_{p\in\mathcal{P}_I,\, p(0)=1} \max_{x\in[\lambda_{\min},\lambda_{\max}]} |p(x)|$. Choosing the scaled Chebyshev polynomial bounds this maximum by $2[(\sqrt{\kappa_t}-1)/(\sqrt{\kappa_t}+1)]^I$; this ratio is increasing in κ_t, so $\bar{\kappa}_t \ge \kappa_t$ proves (89). $\tilde{s}^2_t(a) - s^2_{\mathcal{C},t}(a) = \mathbf{g}^\top(\tilde{\mathbf{u}} - \mathbf{u}) = (\mathcal{C}_t\mathbf{u})^\top(\tilde{\mathbf{u}} - \mathbf{u}) = \mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})$. By Cauchy–Schwarz in the $\mathcal{C}_t$-norm, $|\mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})| \le \|\mathbf{u}\|_{\mathcal{C}_t}\|\tilde{\mathbf{u}} - \mathbf{u}\|_{\mathcal{C}_t} \le \varepsilon_{t,a}\|\mathbf{u}\|^2_{\mathcal{C}_t} = \varepsilon_{t,a}\, s^2_{\mathcal{C},t}(a)$. For the outer-product special case the condition-number bound follows from $\lambda_{\max}(\mathcal{C}_t) \le \lambda + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\|\mathbf{g}_t(\mathbf{x}_s, a_s)\|^2 \le \lambda + W_t G^2/\sigma^2$. □

Lemma 27 (Fixed-preconditioner CG accuracy)**.** *Fix round t and let $P_t \in \mathbb{R}^{d\times d}$ be an $\mathcal{H}_t$-measurable SPD preconditioner, held fixed and applied as the same exact linear map throughout every iteration of a solve. For $\mathbf{g} = \mathbf{g}_t(a) \ne 0$, define the symmetric transformed system*

$$\widehat{C}_t\mathbf{y} = \widehat{\mathbf{g}}, \qquad \widehat{C}_t = P_t^{-1/2}\mathcal{C}_t P_t^{-1/2}, \qquad \widehat{\mathbf{g}} = P_t^{-1/2}\mathbf{g}, \tag{91}$$

using principal square roots. If CG starts from zero on this system and $\tilde{\mathbf{y}}_I$ is its Ith iterate for $I \ge 1$, set $\mathbf{u} = \mathcal{C}_t^{-1}\mathbf{g}$, $\mathbf{y} = P_t^{1/2}\mathbf{u}$, and $\tilde{\mathbf{u}}_I = P_t^{-1/2}\tilde{\mathbf{y}}_I$. Then

$$\frac{\|\mathbf{u} - \tilde{\mathbf{u}}_I\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} = \frac{\|\mathbf{y} - \tilde{\mathbf{y}}_I\|_{\widehat{C}_t}}{\|\mathbf{y}\|_{\widehat{C}_t}} \le 2\left(\frac{\sqrt{\bar{\kappa}^P_t} - 1}{\sqrt{\bar{\kappa}^P_t} + 1}\right)^I, \tag{92}$$

where $\bar{\kappa}^P_t \ge \kappa(\widehat{C}_t)$ is a certified predictable bound. In particular, positive $\mathcal{H}_t$-measurable certified quantities ℓ_t, L_t satisfying the spectral equivalence $\ell_t P_t \preceq \mathcal{C}_t \preceq L_t P_t$ with $0 < \ell_t \le L_t$ give $\bar{\kappa}^P_t = L_t/\ell_t$.

Writing the left side of (92) as $\varepsilon^P_{t,a}$, the approximate width $\tilde{s}^2_{P,t}(a) = \mathbf{g}^\top\tilde{\mathbf{u}}_I$ satisfies, whenever $\varepsilon^P_{t,a} < 1$,

$$(1 - \varepsilon^P_{t,a})s^2_{\mathcal{C},t}(a) \le \tilde{s}^2_{P,t}(a) \le (1 + \varepsilon^P_{t,a})s^2_{\mathcal{C},t}(a). \tag{93}$$

There is also a directly recomputable residual certificate. For $\mathbf{r}_I = \mathbf{g} - \mathcal{C}_t\tilde{\mathbf{u}}_I$,

$$\varepsilon^P_{t,a} \le \sqrt{\bar{\kappa}^P_t}\left(\frac{\mathbf{r}_I^\top P_t^{-1}\mathbf{r}_I}{\mathbf{g}^\top P_t^{-1}\mathbf{g}}\right)^{1/2}. \tag{94}$$

Thus stopping only after the preconditioned residual ratio is at most $\bar{\varepsilon}_t/\sqrt{\bar{\kappa}^P_t}$ certifies $\varepsilon^P_{t,a} \le \bar{\varepsilon}_t$. For $\mathbf{g} = 0$, both widths are defined as zero. Taking $P_t = I$ recovers ordinary CG. The lemma makes no claim that a particular preconditioner improves the condition number; that requires a proved or measured spectral-equivalence bound. Iteration-varying or inexact nonlinear preconditioners are outside this fixed transformed-CG argument.

Proof. The substitutions $\mathbf{u} = P_t^{-1/2}\mathbf{y}$ and $\tilde{\mathbf{u}}_I = P_t^{-1/2}\tilde{\mathbf{y}}_I$ give

$$\|\mathbf{u} - \tilde{\mathbf{u}}_I\|^2_{\mathcal{C}_t} = \|\mathbf{y} - \tilde{\mathbf{y}}_I\|^2_{\widehat{C}_t}, \qquad \|\mathbf{u}\|^2_{\mathcal{C}_t} = \|\mathbf{y}\|^2_{\widehat{C}_t}.$$

The standard zero-start CG Chebyshev bound on the SPD matrix $\widehat{C}_t$ proves (92). Congruence of $\ell_t P_t \preceq \mathcal{C}_t \preceq L_t P_t$ by $P_t^{-1/2}$ puts the spectrum of $\widehat{C}_t$ in $[\ell_t, L_t]$.

Also $\mathbf{g}^\top(\tilde{\mathbf{u}}_I - \mathbf{u}) = \mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}}_I - \mathbf{u})$, so Cauchy–Schwarz in the $\mathcal{C}_t$ energy norm proves (93), exactly as in Lemma 26. Finally the transformed residual is $\widehat{\mathbf{r}}_I = P_t^{-1/2}\mathbf{r}_I$. If $m_t = \lambda_{\min}(\widehat{C}_t)$ and $M^P_t = \lambda_{\max}(\widehat{C}_t)$, then

$$\|\mathbf{y} - \tilde{\mathbf{y}}_I\|_{\widehat{C}_t} \le \frac{\|\widehat{\mathbf{r}}_I\|_2}{\sqrt{m_t}}, \qquad \|\mathbf{y}\|_{\widehat{C}_t} \ge \frac{\|\widehat{\mathbf{g}}\|_2}{\sqrt{M^P_t}}.$$

Their ratio is at most $\sqrt{\kappa(\widehat{C}_t)}\|\widehat{\mathbf{r}}_I\|_2/\|\widehat{\mathbf{g}}\|_2$, which yields (94) after using the certified upper bound. □

The algorithmic bonus. With the *per-round* certified tolerance $\bar{\varepsilon}_t \geq \max_a \varepsilon_{t,a}$ and the one-sided transfer factor $u_t(a) \geq 1$ of Assumption 5, Algorithm 1 plays the bonus

$$w_t(a) \;:=\; \omega_t \sqrt{\frac{u_t(a)}{1-\bar{\varepsilon}_t}}\,\sqrt{\tilde{s}_t^2(a)}, \tag{95}$$

where $\bar{\beta}_t \geq \beta_t$ is the predictable confidence upper bound, $\bar{\psi}_t$ the centering certificate (Assumption 4), and

$$\omega_t := \bar{\beta}_t + \bar{\psi}_t \tag{96}$$

is the combined confidence-plus-centering width factor. Define the *per-round* CG inflation factor $\alpha_t := \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$.

Lemma 28 (Bonus sandwiching)**.** *Under Assumption 5 and* $\max_{a\in\mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1$, *the bonus* (95) *satisfies, for every* $a \in \mathcal{A}_t$,

$$\omega_t\,\bar{s}_t(a) \;\leq\; w_t(a) \;\leq\; \alpha_t\,\omega_t\,\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a). \tag{97}$$

Proof. For $\mathbf{g}_t(a) = \mathbf{0}$ all three terms vanish and the bound is trivial. Otherwise Lemma 26 with $\varepsilon_{t,a} \leq \bar{\varepsilon}_t$ gives $(1-\bar{\varepsilon}_t)s^2_{\mathcal{C},t}(a) \leq \tilde{s}_t^2(a) \leq (1+\bar{\varepsilon}_t)s^2_{\mathcal{C},t}(a)$. The lower bound and $\omega_t \geq 0$ give $\sqrt{\tilde{s}_t^2(a)} \geq \sqrt{1-\bar{\varepsilon}_t}\,s_{\mathcal{C},t}(a)$, so $w_t(a) \geq \omega_t\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a)$; then (60) gives $\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a) \geq \bar{s}_t(a)$, i.e. $w_t(a) \geq \omega_t\bar{s}_t(a)$. The upper bound gives $\sqrt{\tilde{s}_t^2(a)} \leq \sqrt{1+\bar{\varepsilon}_t}\,s_{\mathcal{C},t}(a)$, whence $w_t(a) \leq \omega_t\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}\sqrt{1+\bar{\varepsilon}_t}\,s_{\mathcal{C},t}(a) = \alpha_t\omega_t\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a)$. Both bounds are per-round; no horizon maximum is taken before summing. □

C.7 Confidence and Discrepancy Lemmas

Lemma 29 (Confidence for the linearized ridge estimator)**.** *Suppose Assumptions 1 to 3 hold and* $\mu^* = \mu_{\boldsymbol{\theta}^*}$ *with* $\|\boldsymbol{\theta}^*\| \leq S$. *Define the linearized ridge estimator* $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ *by* (107), *the frozen-feature variance* $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a)}$, *and the cumulative squared linearization error* $F_t := \sum_{s=1}^{t-1} \varepsilon_{\text{lin}}(s)^2$. *Then with probability at least* $1-\delta$, *for all* $t \geq 1$ *and all* $a \in \mathcal{A}_t$ *simultaneously,*

$$\left|\mathbf{g}_t(a)^\top\big(\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^*\big)\right| \;\leq\; \bar{s}_t(a)\,\beta_t, \tag{98}$$

where

$$\beta_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\,S \;+\; \frac{1}{\sigma}\sqrt{F_t}. \tag{99}$$

(Here $\gamma_{t-1} = \log\det(\bar{\mathbf{C}}_t)/\det(\lambda\mathbf{I})$ *is the information gain through round* $t-1$.*)*

The quantity β_t *in* (99) *is a theoretical radius. The algorithm may use any predictable upper bound* $\bar{\beta}_t \geq \beta_t$*; all results below remain valid with* $\bar{\beta}_t$ *in place of* β_t. *This is necessary because* β_t *depends on the unknown cumulative linearization term* F_t.

Proof. From (106), $y_s = \mathbf{g}_s^\top\boldsymbol{\theta}^* + \rho_s + \eta_s$ with $|\rho_s| \leq \varepsilon_{\text{lin}}(s)$. The closed form of $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ gives

$$\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^* = -\lambda\bar{\mathbf{C}}_t^{-1}\boldsymbol{\theta}^* + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\rho_s + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\eta_s.$$

Taking the inner product with $\mathbf{g}_t(a)$ and bounding in absolute value:

(i) *Prior bias.* $|\lambda\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\boldsymbol{\theta}^*| \leq \lambda\,\bar{s}_t(a)\,\|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \leq \bar{s}_t(a)\sqrt{\lambda}\,S$. (Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm; $\lambda\|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \leq \lambda\|\boldsymbol{\theta}^*\|/\sqrt{\lambda_{\min}(\bar{\mathbf{C}}_t)} \leq \sqrt{\lambda}\,S$ since $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$.)

(ii) *Linearization bias.* Define $\Phi = [\mathbf{g}_1/\sigma, \ldots, \mathbf{g}_{t-1}/\sigma]$ (the $d \times (t-1)$ design matrix) and $\hat{\rho} = (\rho_1, \ldots, \rho_{t-1})^\top$. Then $\sigma^{-2}\sum\mathbf{g}_s\rho_s = \sigma^{-1}\Phi\hat{\rho}$. Since $\bar{\mathbf{C}}_t \succeq \Phi\Phi^\top$, we have $\Phi^\top\bar{\mathbf{C}}_t^{-1}\Phi \preceq \mathbf{I}$, so $\|\Phi\hat{\rho}\|_{\bar{\mathbf{C}}_t^{-1}} = \|\hat{\rho}\|_{\Phi^\top\bar{\mathbf{C}}_t^{-1}\Phi} \leq \|\hat{\rho}\| = \sqrt{F_t}$. Therefore $|\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\sigma^{-1}\Phi\hat{\rho}| \leq \bar{s}_t(a)\sigma^{-1}\sqrt{F_t}$.

(iii) *Noise.*

$$\left\| \sum_{s<t} \frac{\mathbf{g}_s}{\sigma} \frac{\eta_s}{\sigma} \right\|_{\bar{\mathbf{C}}_t^{-1}} \le \sqrt{\gamma_{t-1} + 2\log(1/\delta)}$$

holds for all $t \ge 1$ with probability at least $1-\delta$ by the time-uniform vector self-normalized martingale bound (Abbasi-Yadkori et al., 2011). Here $x_s = \mathbf{g}_s/\sigma$ is predictable with respect to the pre-reward sigma-algebra $\mathcal{G}_s$, and $\xi_s = \eta_s/\sigma$ is conditionally 1-sub-Gaussian given $\mathcal{G}_s$. For any candidate action a, Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm gives

$$\left| \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \sum_{s<t} \frac{\mathbf{g}_s \eta_s}{\sigma^2} \right| \le \bar{s}_t(a) \sqrt{\gamma_{t-1} + 2\log(1/\delta)}.$$

Summing gives (98)–(99). □

Corollary 30 (Predictable confidence schedule)**.** *Suppose a nonnegative $\mathcal{H}_t$-measurable sequence $\{\bar{F}_t\}_{t\ge1}$ satisfies $F_t \le \bar{F}_t$ for all t. Then Algorithm 1 may set*

$$\bar{\beta}_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\, S \;+\; \sigma^{-1}\sqrt{\bar{F}_t},$$

which is predictable and satisfies $\bar{\beta}_t \ge \beta_t$.

Proof. $\bar{F}_t \ge F_t$ term-by-term implies each summand in (99) is dominated, so $\bar{\beta}_t \ge \beta_t$. Predictability holds because γ_{t-1}, S, $\bar{F}_t$, λ, σ, and δ are all determined before round t. □

Lemma 13 supplies the computable choice $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\mathrm{lin}})^2$; a tighter externally certified predictable sequence may replace it.

Corollary 31 (Observable information-gain surrogate via the one-sided transfer factor)**.** *Run Algorithm 1 with a predictable algorithmic curvature sequence $\mathcal{C}_s$, per-round CG tolerance with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s < 1$ for the played action, and an $\mathcal{H}_s$-measurable one-sided optimism-transfer factor $u_s(a) \ge 1$ satisfying $\bar{s}_s^2(a) \le u_s(a)\, s_{\mathcal{C},s}^2(a)$ (Assumption 5). Suppose also that the nonnegative predictable sequence $\bar{F}_t$ satisfies $\bar{F}_t \ge F_t$. Define the observable information-gain surrogate*

$$\hat{\gamma}_{t-1} \;:=\; \sum_{s=1}^{t-1} \log\!\Big(1 + \sigma^{-2} \,\frac{u_s(a_s)}{1-\bar{\varepsilon}_s}\, \tilde{s}_s^2(a_s)\Big),$$

where $\tilde{s}_s^2(a_s) = \mathbf{g}_s(a_s)^\top \tilde{\mathbf{u}}_s$ is the CG variance proxy computed at round s for the played action. Then $\hat{\gamma}_{t-1}$ is $\mathcal{H}_t$-measurable and $\hat{\gamma}_{t-1} \ge \gamma_{t-1}$, so

$$\hat{\beta}_t \;:=\; \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\, S \;+\; \sigma^{-1}\sqrt{\bar{F}_t}$$

is a predictable schedule with $\hat{\beta}_t \ge \beta_t$. Played-action CG accuracy is sufficient for this information comparison. If the same tolerance also holds uniformly for every candidate action before selection, then Theorem 1 holds with $\bar{\beta}_t = \hat{\beta}_t$.

Proof. The transfer factor gives $\bar{s}_s^2(a_s) \le u_s(a_s)\, s_{\mathcal{C},s}^2(a_s)$, and Lemma 26 with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s$ gives $s_{\mathcal{C},s}^2(a_s) \le \tilde{s}_s^2(a_s)/(1-\bar{\varepsilon}_s)$; hence $\bar{s}_s^2(a_s) \le u_s(a_s)\, \tilde{s}_s^2(a_s)/(1-\bar{\varepsilon}_s)$. By monotonicity of $x \mapsto \log(1+\sigma^{-2}x)$, each summand of γ_{t-1} in (31) is dominated by the corresponding summand of $\hat{\gamma}_{t-1}$. Measurability holds because $\tilde{s}_s^2(a_s)$, $u_s(a_s)$, and $\bar{\varepsilon}_s$ are all $\mathcal{H}_s$-measurable. The schedule property then follows term by term as in Corollary 30. □

This makes the information-gain component observable when $u_s(a_s)$ is certified. Together with Lemma 13, it renders the full confidence and centering schedules operational; it does not make them tight.

Remark (Computability and dimension dependence of β_t)**.** *Two caveats about β_t deserve emphasis. First, γ_{t-1} is defined through the frozen-feature Gram matrix $\bar{\mathbf{C}}_t$, whose per-sample Jacobians the matrix-free implementation deliberately does not store; γ_{t-1} is therefore not directly computable in the stated memory model. Second, the worst-case substitute $\gamma_{t-1} \le d\log\big(1 + (t-1)G^2/(d\lambda\sigma^2)\big)$ does not exploit low realized information gain and can be order t when $d \gg t$, rendering the resulting bonus and theorem bound vacuous. Corollary 31 addresses both issues with an observable surrogate built from quantities the algorithm already computes; the resulting guarantee is data-dependent, and its tightness still depends on the realized surrogate together with the other theorem factors.*

Lemma 32 (Action-scaled centering discrepancy)**.** *Under Assumption 4,* $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \le \bar{\psi}_t\, \bar{s}_t(a)$ *for all* $a \in \mathcal{A}_t$.

Proof. Immediate from (40). □

> *Remark (status of the centering certificate $\bar{\psi}_t$).* Lemma 13 computes a predictable $\bar{\psi}_t \ge \psi_t$ from the certified optimizer residual, Welford path moments, observed collection residuals, and analytic smoothness constants. The schedule is operational but can be loose; the corrected-center variant (Corollary 10) removes it at the cost of frozen-feature access.

In the convex last-layer regime, under exact optimization, $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ and $\bar{\psi}_t = 0$. The full prediction error at $(\mathbf{x}_t, a)$ is bounded by the triangle inequality, with the centering discrepancy *action-scaled*:

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)| \;\le\; (\beta_t + \bar{\psi}_t)\, \bar{s}_t(a) \;+\; \varepsilon_{\mathrm{lin}}(t) \;\le\; \omega_t\, \bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t), \tag{100}$$

using (98), Lemma 32, the per-round linearization remainder $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$, and $\bar{\beta}_t \ge \beta_t$.

C.8 Cost of Approximate Curvature

Proposition 33 (Unrescaled current-parameter subset)**.** *Let* $\mathcal{S}_t \subseteq \{1, \ldots, t-1\}$ *and* $0 \le w_s \le 1$*, and define the current-parameter operator*

$$\hat{\mathbf{C}}_t \;=\; \lambda \mathbf{I} + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^\top,$$

using the same relinearized Jacobians $\mathbf{g}_t(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$ *that define* $\mathbf{C}_t$ *(5). Then* $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$*, so Theorem 34 applies with* $\kappa_{+,t} = 1$ *and, in the exact-transfer case,* $u_t = (1 + \bar{\chi}_t)^2$ *(Lemma 19).*

Proof. $\mathbf{C}_t - \hat{\mathbf{C}}_t = \sigma^{-2} \sum_{s \in \mathcal{S}_t} (1 - w_s) \mathbf{g}_t \mathbf{g}_t^\top + \sigma^{-2} \sum_{s \notin \mathcal{S}_t} \mathbf{g}_t \mathbf{g}_t^\top$, a sum of PSD outer products (each $1 - w_s \ge 0$), hence $\mathbf{C}_t - \hat{\mathbf{C}}_t \succeq 0$. With $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (so $\kappa_{+,t} = 1$ in (101)) and $\hat{\mathbf{C}}_t \succeq \lambda \mathbf{I}$, the hypotheses of Theorem 34 hold. □

This covers an unrescaled current-parameter sliding window, an unrescaled current-parameter subsample, and any current-parameter downweighted subset with weights in $[0, 1]$. It does *not* cover rescaled subsampling with weights $(t-1)/n > 1$ (Lemma 22), a stale periodic-refresh operator evaluated at an older parameter, or an arbitrary randomized sketch (e.g. an eigenspace Lanczos surrogate), for which a factor $\kappa_{+,t}$ must be certified separately.

Theorem 34 (Approximate curvature via one-sided transfer)**.** *Adopt the hypotheses of Theorem 1 with the algorithmic curvature* $\mathcal{C}_t = \hat{\mathbf{C}}_t$*, an approximate operator that is symmetric,* $\mathcal{H}_t$*-measurable, and satisfies* $\hat{\mathbf{C}}_t \succeq \lambda \mathbf{I}$*. Suppose the two* one-sided *bounds*

$$\hat{\mathbf{C}}_t \;\preceq\; \kappa_{+,t}\, \mathbf{C}_t, \qquad \mathbf{C}_t \;\preceq\; (1 + \bar{\chi}_t)^2\, \bar{\mathbf{C}}_t \tag{101}$$

hold for every $t \le T$*, where* $\kappa_{+,t} \ge 1$ *is* $\mathcal{H}_t$*-measurable and* $\bar{\chi}_t \ge \chi_t$ *is the whitened-drift certificate of Lemma 19. Then, for every action* a*,*

$$\bar{s}_t^2(a) \;\le\; \kappa_{+,t}\, (1 + \bar{\chi}_t)^2\, \hat{s}_t^2(a), \tag{102}$$

so Assumption 5 holds with the uniform factor $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ *and Theorem 1 applies verbatim with* $\mathcal{C}_t = \hat{\mathbf{C}}_t$ *and* $s_{\mathcal{C},t} = \hat{s}_t$*:*

$$R_T \;\le\; 2 \sqrt{(\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}} \sum_{t=1}^{T} \alpha_t^2\, u_t(a_t)\, \omega_t^2} \;+\; 2 E_T.$$

No lower *spectral factor on* $\hat{\mathbf{C}}_t$ *is required for optimism; the cost of an excessively small, stale, or nonmonotone approximate operator is charged through its realized width complexity* $\Lambda_T^{\mathcal{C}}$ *(and its variation* $V_T^{\mathcal{C}}$ *inside* Γ_T^{dyn}*), not through a lower Loewner bound.*

Proof. $\hat{\mathbf{C}}_t \preceq \kappa_{+,t} \mathbf{C}_t$ with both sides SPD inverts to $\hat{\mathbf{C}}_t^{-1} \succeq \kappa_{+,t}^{-1} \mathbf{C}_t^{-1}$, so $\hat{s}_t^2(a) \ge \kappa_{+,t}^{-1} s_t^2(a)$; and (64) gives $\bar{s}_t^2(a) \le (1 + \bar{\chi}_t)^2 s_t^2(a)$. Chaining, $\bar{s}_t^2(a) \le (1 + \bar{\chi}_t)^2 \kappa_{+,t} \hat{s}_t^2(a)$, which is (102) with $s_{\mathcal{C},t} = \hat{s}_t$. Theorem 1 applies with $\mathcal{C}_t = \hat{\mathbf{C}}_t$. The two-sided $\sqrt{\rho_+/\rho_-}$ inflation—which instead assumes a lower spectral factor—is retained for comparison in Appendix H; the two results trade different assumptions (a lower spectral bound versus dynamic width complexity), and neither numerically dominates the other in general. □

Proposition 35 (Roundwise scalar-width invariance). *If two width functions satisfy $\hat{s}_t(a) = c_t s_t(a)$ for every candidate action, where $c_t > 0$ is action independent, then replacing a scalar bonus coefficient b_t by b_t/c_t gives identical UCB scores and therefore the same selected action under the same tie rule.*

Proof. $\widehat{\mu}_t(a) + (b_t/c_t)\hat{s}_t(a) = \widehat{\mu}_t(a) + b_t s_t(a)$ for every a. $\square$

This is a roundwise statement. If two policies also have identical centers, histories, optimization, and coupled exogenous randomness, if c_t is predictable, and if proportionality holds at every subsequent common history, induction gives identical actions and rewards after the coefficient adjustment. Absent those conditions the proposition makes no regret claim. Accordingly, regret differences can reflect anisotropic width distortion or a difference in damping, center, adaptive path, optimization, calibration, or solve accuracy.

Proposition 36 (Off-diagonal linear-Gram witness). *Let $\sigma = 1$, $\mathbf{u} = (1,1)/\sqrt{2}$, $\mathbf{v} = (1,-1)/\sqrt{2}$, the action set be $\{\mathbf{u}, \mathbf{v}\}$, and $\boldsymbol{\theta}^* = \epsilon\mathbf{u} + \Delta\mathbf{v}$ with $\Delta > \epsilon > 0$. Rewards are noiseless, the ridge is $\lambda\mathbf{I}$, all policies use the same full-ridge mean center, and ties select $\mathbf{u}$. If*

$$\max\left\{\sqrt{\lambda(\epsilon^2 + \Delta^2)}, \epsilon\sqrt{\lambda}\left(1 + \sqrt{\frac{\lambda}{\lambda+1}}\right)\right\} < b \leq \left(\Delta - \frac{\epsilon}{\lambda+1}\right)\sqrt{\lambda+1}, \tag{103}$$

then full-Gram UCB plays $\mathbf{u}$ once and $\mathbf{v}$ thereafter, so $R_T^{\text{full}} = \Delta - \epsilon$. The raw diagonal policy and the policy inflated by the uniform analytic factor $\kappa_n = (\lambda + n/2)/\lambda$ play $\mathbf{u}$ forever and have $R_T^{\text{diag}} = T(\Delta - \epsilon)$.

Proof. After n pulls of $\mathbf{u}$ and no pull of $\mathbf{v}$,

$$C_n = \lambda\mathbf{I} + n\mathbf{u}\mathbf{u}^\top, \qquad \hat{\boldsymbol{\theta}}_n = \frac{n\epsilon}{\lambda+n}\mathbf{u}.$$

Thus the full scores are $n\epsilon/(\lambda+n) + b/\sqrt{\lambda+n}$ and $b/\sqrt{\lambda}$. The strict lower inequality in (103) makes the second score larger at $n = 1$. The confidence lower bound also implies $b > \Delta\sqrt{\lambda}$; after any number of $\mathbf{v}$ pulls its score is strictly larger than Δ, whereas the upper inequality makes the fixed $\mathbf{u}$ score at most Δ. Hence there is no return. In contrast, $D_n = \operatorname{diag}(C_n) = (\lambda + n/2)\mathbf{I}$, so both diagonal widths are equal and the center gives $\mathbf{u}$ a strict advantage after round one. Moreover $D_n \preceq \kappa_n C_n$ analytically, but multiplying both diagonal widths by $\sqrt{\kappa_n}$ leaves them equal; uniform certified inflation therefore does not change the diagonal action. For example, $(\lambda, \epsilon, \Delta, b) = (1, 0.1, 1, 1.1)$ satisfies the interval. $\square$

This is a two-dimensional linear-Gram existence witness. For a linear predictor with Gaussian squared loss, this Gram is exactly the GGN/Fisher operator, so the construction isolates a GGN-relevant geometric mechanism. It is neither a nonlinear-network result nor a uniform dominance result. A dense action-dependent factor recovers the full widths, but it uses the reference geometry that the diagonal approximation was meant to avoid.

Composing subsampling with feature drift. Lemma 22 guarantees a sandwich for the *relinearized* $\mathbf{C}_t$: $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$. Only the *upper* half is needed here: $\hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$ gives $\kappa_{+,t} = 1 + \varepsilon$, and composing with the one-sided whitened bound $\mathbf{C}_t \preceq (1 + \bar{\chi}_t)^2\bar{\mathbf{C}}_t$ (Lemma 19) yields $u_t = (1+\varepsilon)(1+\bar{\chi}_t)^2$ in Theorem 34. A union bound over $t \leq T$ (replacing δ' by δ'/T in Lemma 22) makes the subsampling event time-uniform and contributes to δ_{cert}.

For an *unrescaled current-parameter window* (or any current-parameter downweighted subset with weights in $[0,1]$), Proposition 33 gives the *deterministic* upper factor $\kappa_{+,t} = 1$ with *no* distributional assumption, so Theorem 34 applies directly with $u_t = (1 + \bar{\chi}_t)^2$. Only the *stale periodic-refresh* operator (§C.3(iii)), which reuses Jacobians from an older parameter, lacks a deterministic $\kappa_{+,t}$; there Theorem 34 provides a guarantee only when a one-sided bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ is separately verified, which we evaluate empirically.

What remains open. The path schedules close the computability gap, but their constants can be extremely loose. The approximate-rank result replaces exact rank by a realized or predictably bounded spectral tail, but does not show that this tail or feature drift remains small during unrestricted training. Useful control of these two quantities for a genuinely adapting backbone, tighter smoothness/residual certificates, and verified numerical enclosures for floating-point solver and spectral calculations remain open.

D Related Work

Neural and linear contextual bandits. LinUCB (Abbasi-Yadkori et al., 2011) and Thompson sampling (Thompson, 1933; Agrawal and Goyal, 2013) provide the analytical backbone for contextual bandit exploration. Neural extensions—NeuralUCB (Zhou et al., 2020), NeuralTS (Zhang et al., 2021), neural-linear models (Riquelme et al., 2018)—use different neural feature covariances. NeuralUCB and NeuralTS accumulate full neural-gradient Grams in their stated algorithms (and diagonalize them in experiments), while NeuralLinear maintains Bayesian linear uncertainty over learned last-layer features. CC-UCB instead re-evaluates every replayed Jacobian at the current parameter and accesses the resulting full operator implicitly. Neural-linear designs that train a deep representation but explore only in the last layer (Xu et al., 2022a) occupy the opposite design point. Recent alternatives sharpen different parts of this landscape. Salgia (2023) analyze overparameterized smooth-activation networks through non-asymptotic network–NTK approximation and give an efficient sublinear-regret algorithm. Deb et al. (2024) instead reduce contextual bandits to online neural regression, using perturbed predictions to obtain their regression guarantees. Bui et al. (2025) combine a learned deep representation with a variance-aware linear-UCB exploration layer. These are, respectively, an NTK analysis, a regression-oracle route, and a learned last-layer geometry; none analyzes the dynamically relinearized full GGN and its CG error. Our low-rank/excitation result is correspondingly narrower than generic neural regression, but controls that current-parameter operator.

Online Bayesian and EKF-style neural bandits. Durán-Martín et al. (2022) apply extended Kalman filter (EKF) updates in a learned or random low-dimensional affine parameter subspace. They maintain a dense or diagonal covariance in that subspace, inducing a low-rank covariance over parameters spanning all network layers, with memory independent of the number of rounds. This differs from applying exact curvature–vector products in the full parameter space. More directly, Chang et al. (2023) maintain a diagonal-plus-low-rank approximation to the full-parameter posterior precision through recursive EKF and online-SVD updates, and use it for Thompson sampling in an MNIST contextual bandit. Their LO-FI method is a one-pass compressed-posterior approach with fixed-rank cost linear in parameter count and no replay of past data. CC-UCB instead relinearizes the full replay history at the current parameter and accesses the uncompressed GGN through residual-checked CG. Their bandit study is empirical; our conditional UCB analysis explicitly accounts for CG truncation, but does not establish a practical advantage over LO-FI.

Last-layer and structured Laplace approximations. Last-layer Laplace fixes the backbone and restricts the approximate posterior to the final layer (Kristiadi et al., 2020), omitting backbone-weight uncertainty by construction. This is only one scalable option: all-layer block-diagonal Kronecker curvature has also been used (Ritter et al., 2018), and Daxberger et al. (2021) systematize all-weight, subnetwork, and last-layer variants with full, low-rank, KFAC, or diagonal curvature. CC-UCB uses the uncompressed current-parameter GGN as its reference operator without materializing the covariance matrix.

Sampling-based full-parameter exploration. Langevin Monte Carlo Thompson sampling (LMC-TS) (Xu et al., 2022b) explores with approximate samples of a loss-defined parameter posterior, obtained by noisy gradient steps on the regularized loss; like CC-UCB it never materializes a covariance matrix. Its regret guarantee is for linear contextual bandits; its neural-bandit evidence is empirical. The two routes pay for exploration differently: LMC-TS through the mixing of a per-round sampling chain, CC-UCB through CG truncation and spectral distortion of an explicit curvature metric, which yields deterministic widths, an optimism-style analysis, and a fixed curvature operator reused across separate per-action CG solves.

Efficient nonlinear UCB. Kassraie and Krause (2022) analyze NTK-UCB and propose NN-UCB and its convolutional variant CNN-UCB. NN-UCB combines a trained-network mean with confidence widths from the empirical NTK feature map at initialization. They prove sublinear regret for the SupNN-UCB variant under NTK-RKHS, sufficient-exploration, and network-width assumptions. Their fixed-feature uncertainty differs from CC-UCB's replayed GGN/Fisher curvature at the current parameter $\boldsymbol{\theta}_t$, accessed matrix-free via CG. CC-UCB thus targets the regime where the practitioner has a specific pretrained model and wants to use its curvature as-is, rather than constructing a surrogate NTK-based estimator.

Spectral sketching for linear bandits. Kuzborskij et al. (2019) analyze linear bandits where the Gram matrix is replaced by a Frequent Directions sketch, deriving regret bounds that depend on the spectral approximation quality. Wen et al. (2026) show that fixed-size sketch regret bounds can degenerate to the trivial linear rate under heavy spectral tails, and propose an adaptive dyadic block sketch. Our spectral-distortion theorem (Theorem 34) instantiates the same conceptual lever in the matrix-free GGN/CVP setting by exposing transfer and realized-width terms along the realized trajectory. We do not claim novelty for the sandwich-to-regret reduction itself; the contribution is adapting it to the CG/GGN context with explicit CG truncation and feature-drift factors.

Laplace approximation in deep learning. The Laplace approximation (MacKay, 1992; Ritter et al., 2018) provides a Gaussian posterior centered at a MAP estimate with covariance given by the inverse curvature. Scalable variants use diagonal, KFAC (Martens and Grosse, 2015), or subnetwork approximations (Daxberger et al., 2021). Matrix-free full-Jacobian/Fisher predictive variance using implicit products and CG predates this work (Maddox et al., 2021); CC-UCB adapts that computational route to an online, current-relinearized GGN operator and supplies bandit-specific transfer and solver-error accounting.

GGN–vector products, Hessian spectral analysis, and influence functions. Pearlmutter's algorithm (Pearlmutter, 1994) computes exact Hessian–vector products; the analogous GGN–vector product has the same Jacobian-transpose-times-Jacobian-times-vector structure (Schraudolph, 2002). Recent work on Krylov methods for Hessian spectral analysis (Granziol and Juarev, 2026) demonstrates related distributed HVP/Krylov infrastructure at $> 10^8$-parameter scale, but does not benchmark the exact per-example GGN operator or end-to-end CC-UCB latency. Influence functions (Koh and Liang, 2017) use inverse-curvature–vector products to attribute predictions to training points; the CG-based implicit solve we employ is the same computational primitive.

Optimal experimental design. Information-theoretic exploration objectives—such as maximizing $\log\det(\bar{\mathbf{C}}_t)$ or minimizing the trace of $\bar{\mathbf{C}}_t^{-1}$—connect bandit exploration to the classical D-optimal and A-optimal design criteria (Valko et al., 2013). CC-UCB's use of the full GGN/Fisher curvature enables stochastic estimates of these quantities via Hutchinson-type trace estimators (Ubaru et al., 2017), without forming the matrix.

Filtration. We separate *pre-action* randomness, revealed before the agent commits to a_t, from *post-reward* randomness, drawn only after r_t is observed. Let Ω_t collect all pre-action algorithmic randomness used at round t: curvature subsampling indices, randomized sketches, and stochastic solver seeds. The randomized sketch or subsample that defines the curvature operator is drawn into Ω_t and *fixed before action selection*; it is not re-randomized during any conjugate-gradient (CG) solve at round t. (Standard CG requires a fixed symmetric positive definite matrix–vector oracle across its iterations and does not apply to a stochastic oracle that changes between iterations.) Let U_{t+1} collect the *post-reward* randomness used to form the next parameter estimate $\boldsymbol{\theta}_{t+1}$ (minibatch order, dropout, perturbed warm starts). Define $\mathcal{F}_0 = \sigma(\boldsymbol{\theta}_1)$ and, for $t \geq 1$, the recursively built σ-algebra

$$\mathcal{F}_t \;:=\; \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t, \Omega_t, a_t, r_t, U_{t+1}\big), \tag{104}$$

so that $\boldsymbol{\theta}_{t+1}$—computed from the data through round t together with the post-reward randomness U_{t+1}—is $\mathcal{F}_t$-measurable. Define the *pre-action history* σ-algebra and the pre-reward σ-algebra

$$\mathcal{H}_t^- \;:=\; \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t\big), \qquad \mathcal{H}_t \;:=\; \mathcal{H}_t^- \vee \sigma(\Omega_t), \qquad \mathcal{G}_t \;:=\; \mathcal{H}_t \vee \sigma(a_t). \tag{105}$$

At round t the context $\mathbf{x}_t$, the finite action set $\mathcal{A}_t$, and the pre-action randomness Ω_t are revealed; the agent selects a_t; then the reward $r_t = \mu^*(\mathbf{x}_t, a_t) + \eta_t$ is observed; finally U_{t+1} is drawn and $\boldsymbol{\theta}_{t+1}$ is computed. $\mathcal{H}_t^-$ is the history immediately before the randomized certificate draw; after that draw, its realized operator and schedules are $\mathcal{H}_t$-measurable. By construction of Algorithm 1 the selection rule is a deterministic function of $\mathcal{H}_t$-measurable quantities, so the chosen action a_t is itself $\mathcal{H}_t$-measurable and hence $\mathcal{G}_t = \mathcal{H}_t$; we retain the symbol $\mathcal{G}_t$ where the played action is used. The following are all $\mathcal{H}_t$-measurable (i.e. determined before the reward r_t): the candidate action set $\mathcal{A}_t$ and every candidate feature $\mathbf{g}_t(\mathbf{x}_t, a)$, $a \in \mathcal{A}_t$; the chosen action a_t; the algorithmic curvature operator $\mathcal{C}_t$ (Section 3); the confidence schedule $\bar{\beta}_t$; the centering certificate $\bar{\psi}_t$ (Assumption 4); the per-round CG tolerance $\bar{\varepsilon}_t$ together with any condition-number certificate used by the CG stopping rule; and the one-sided optimism-transfer factor $u_t(a)$ (Assumption 5). Since $\boldsymbol{\theta}_t$ is $\mathcal{F}_{t-1} \subseteq \mathcal{H}_t$-measurable, each candidate design vector $\mathbf{g}_t(\mathbf{x}_t, a)$ is $\mathcal{H}_t$-measurable and the realized design vector $\mathbf{g}_t(\mathbf{x}_t, a_t)$ is $\mathcal{G}_t$-measurable. We assume $\{U_s\}_{s\geq 1}$ is independent of the reward noise $\{\eta_s\}_{s\geq 1}$, so that revealing U_{t+1} and forming $\boldsymbol{\theta}_{t+1}$ after r_t preserves the conditional sub-Gaussian property of Assumption 1 for the future noise $\{\eta_s\}_{s>t}$. Index every randomized certificate source (operator sketch, transfer bound, condition bound, or other randomized oracle) by $j \in \mathcal{J}$.

For its round-t validity event $\mathcal{E}_{j,t}$ choose in advance $\delta_{j,t} \geq 0$ such that its stated conditional guarantee gives $\Pr(\mathcal{E}_{j,t}^c \mid \mathcal{H}_t^-) \leq \delta_{j,t}$ and

$$\sum_{j\in\mathcal{J}} \sum_{t=1}^{T} \delta_{j,t} \leq \delta_{\text{cert}}, \qquad \mathcal{E}_{\text{cert}} := \bigcap_{j\in\mathcal{J}} \bigcap_{t=1}^{T} \mathcal{E}_{j,t}.$$

The tower property and union bound yield $\Pr(\mathcal{E}_{\text{cert}}) \geq 1 - \delta_{\text{cert}}$. Deterministic or almost-sure certificates use $\delta_{j,t} = 0$. The self-normalized confidence event is separate and receives probability δ. When Lemma 12 supplies the residual-energy envelope, include its time-uniform event $\mathcal{E}_R$ as one certificate source and reserve δ_R inside δ_{cert} (equivalently, require $\delta_R + \sum_{j,t} \delta_{j,t} \leq \delta_{\text{cert}}$ for the remaining sources). The residual event is never counted again in the self-normalized failure probability δ.

Auxiliary linearized ridge estimator. For the analysis we define a *linearized ridge estimator* based on frozen features. Set the predictable intercept $b_s := \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s) - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}_s$ and the pseudo-response $y_s := r_s - b_s$. Under Assumption 3 and realizability ($\mu^* = \mu_{\boldsymbol{\theta}^*}$),

$$y_s \;=\; \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}^* \;+\; \rho_s(\mathbf{x}_s, a_s, \boldsymbol{\theta}^*) \;+\; \eta_s, \tag{106}$$

where $|\rho_s| \leq \varepsilon_{\text{lin}}(s)$. The linearized estimator is

$$\hat{\boldsymbol{\theta}}_t^{\text{lin}} \;:=\; \arg\min_{\boldsymbol{\theta}} \left[\frac{1}{2\sigma^2} \sum_{s=1}^{t-1} \big(y_s - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}\big)^2 \;+\; \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 \right] \;=\; \bar{\mathbf{C}}_t^{-1} \, \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s)\, y_s. \tag{107}$$

This is a standard linear ridge regression with predictable design vectors $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$, so the self-normalized martingale bound (Abbasi-Yadkori et al., 2011) applies to it directly (Lemma 29).

GGN versus Hessian (context only). The regret theorem uses the GGN/Fisher curvature as the exploration metric. It does not require the GGN to be an accurate Hessian approximation. For squared loss (3) the full Hessian of the data-fit term is

$$\mathbf{H}_t \;=\; \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \Big[\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \;-\; \big(r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\big)\nabla^2 \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s) \Big] \;+\; \lambda \mathbf{I}.$$

The curvature matrix $\mathbf{C}_t$ drops the residual-times-second-derivative term, retaining only the outer-product part that is guaranteed PSD. The dropped residual-times-second-derivative term $D_t = \sigma^{-2} \sum_{s<t} (r_s - \mu_{\boldsymbol{\theta}_t}) \nabla^2 \mu_{\boldsymbol{\theta}_t}$ has operator norm at most $(t-1)\epsilon_r H/\sigma^2$, where ϵ_r bounds the residuals and H the Hessian norms. When this quantity is small relative to λ, $\mathbf{C}_t$ approximates the full Hessian in the Loewner sense (a formal statement with proof is given in Appendix I). This comparison is *not* invoked in the regret analysis (Theorem 1 uses $\mathbf{C}_t$ directly as the exploration metric); it serves only to contextualize $\mathbf{C}_t$ as a proxy for the full Hessian. Outside the small-residual regime, the GGN is not an exact Hessian approximation; it is the PSD Fisher/GGN metric used by the algorithm.

Remark (Fisher, empirical Fisher, and GGN). Three curvature objects appear in the literature and must be distinguished. The *empirical Fisher* is the realized score outer product $\sum_s \nabla_{\boldsymbol{\theta}} \log p(r_s \mid \mathbf{x}_s, a_s; \boldsymbol{\theta}) \, \nabla_{\boldsymbol{\theta}} \log p(r_s \mid \mathbf{x}_s, a_s; \boldsymbol{\theta})^\top$; it depends on observed rewards and is *not* the operator used by CC-UCB. The *true (model) Fisher* is the model expectation of the score outer product, $F(\boldsymbol{\theta}) = \mathbb{E}_{r\sim p(\cdot|\mathbf{x},a;\boldsymbol{\theta})}[\nabla_{\boldsymbol{\theta}} \log p \; \nabla_{\boldsymbol{\theta}} \log p^\top]$. The *generalized Gauss–Newton* (GGN) drops the residual-times-Hessian term from the full Hessian, retaining only the outer-product Jacobian part that is guaranteed PSD (Martens, 2020; Schraudolph, 2002). For the fixed-variance Gaussian working model underlying the squared-loss quasi-likelihood used here, the model Fisher and GGN are identical, and both reduce to the outer-product curvature $\sigma^{-2} \sum_s \mathbf{g}_s \mathbf{g}_s^\top$ used in (5). More generally, for exponential-family negative log-likelihoods parameterized by natural parameters $z_{\boldsymbol{\theta}}(\mathbf{x}, a)$, the true Fisher equals the GGN: $J_{\boldsymbol{\theta}}^\top \nabla^2 A(z) \, J_{\boldsymbol{\theta}}$, where $J_{\boldsymbol{\theta}} = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}}$ and $\nabla^2 A$ is the Hessian of the log-partition function (Martens, 2020; Pascanu and Bengio, 2013). Bernoulli rewards yield scalar weight $p(1-p)$; categorical rewards yield $\mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$. See Appendix J for the extension.

We access $\mathbf{C}_t$ only through curvature–vector products $\mathbf{v} \mapsto \mathbf{C}_t \, \mathbf{v}$, computed via forward-mode Jacobian–vector products composed with reverse-mode vector–Jacobian products (Pearlmutter, 1994; Schraudolph, 2002).

Extension to exponential-family likelihoods. The same matrix-free CG/CVP machinery extends algorithmically to exponential-family likelihoods (e.g. Bernoulli or categorical rewards) via the Fisher/GGN curvature $J_t^\top(\nabla^2 A)\,J_t$, but no regret guarantee beyond Gaussian squared loss is claimed in this draft. Details of the curvature definitions are in Appendix J.

Per-round CG tolerance $\bar{\varepsilon}_t$. The algorithm uses a *known* per-round tolerance $\bar{\varepsilon}_t \in (0,1)$ in the UCB bonus (line 8), not the true per-action energy-norm errors $\varepsilon_{t,a}$ (which are unobservable without the exact solves). It suffices that $\max_{a\in\mathcal{A}_t}\varepsilon_{t,a} \le \bar{\varepsilon}_t$ at every round ((10); zero-gradient actions have $\varepsilon_{t,a} = 0$ by convention and are trivially within tolerance); the bonus is then conservative. Two strategies ensure this:

1. *Adaptive residual stopping (recommended).* In exact arithmetic, for each action monitor the computable Euclidean relative residual $\|r_I\|/\|\mathbf{g}_t(a)\|$ at each CG step (skipping actions with $\mathbf{g}_t(a) = \mathbf{0}$). Since $\varepsilon_{t,a} \le \sqrt{\bar{\kappa}_t}\cdot\|r_I\|/\|\mathbf{g}_t(a)\|$ (standard CG bound, with $\bar{\kappa}_t$ a certified upper bound on the condition number of $\mathcal{C}_t$), stopping that solve when $\|r_I\|/\|\mathbf{g}_t(a)\| \le \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$ guarantees $\varepsilon_{t,a} \le \bar{\varepsilon}_t$, hence (10). This is conservative by the factor $\sqrt{\bar{\kappa}_t}$ but requires no access to $\mathcal{C}_t^{-1}$ and adapts automatically to the per-round condition number. A floating-point implementation needs a verified residual enclosure to inherit this certification literally.
2. *Fixed budget (requires a uniform condition-number bound).* If an a priori bound $\bar{\kappa}_t \le \kappa_{\max}$ holds for *all rounds* $t \le T$, choose I via Eq. (112) at $\kappa_{\max}$ and a fixed target $\bar{\varepsilon}_t \equiv \bar{\varepsilon}$. For a *nonnegative weighted outer-product* operator $\mathcal{C}_t = \lambda\mathbf{I} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\,\mathbf{g}_t(\mathbf{x}_s,a_s)\mathbf{g}_t(\mathbf{x}_s,a_s)^\top$ the condition number is bounded in closed form, $\bar{\kappa}_t \le 1 + W_tG^2/(\lambda\sigma^2)$ with $W_t = \sum_{s\in\mathcal{S}_t} w_s$; a fixed buffer of size n ($W_t = n$, unrescaled) makes this independent of T. For the full-history operator ($W_t = t-1$), and for rescaled uniform subsampling with $w_s = (t-1)/n$ (so $W_t = \sum_{s\in\mathcal{S}_t} w_s = t-1$), the bound grows with t. For a known finite horizon, a fixed budget at $\kappa_{\max} = 1 + (T-1)G^2/(\lambda\sigma^2)$ is valid but scales as $O(\sqrt{T}\log(1/\bar{\varepsilon}))$; a horizon-independent fixed budget is not uniform, so adaptive stopping or an increasing I_t is preferable (see Appendix G). For a *general* SPD operator that is not a nonnegative weighted outer product (e.g. a randomized sketch), the closed-form bound $1 + W_tG^2/(\lambda\sigma^2)$ does not apply, and a separately certified $\mathcal{H}_t$-measurable condition-number upper bound $\bar{\kappa}_t$ must be supplied.

In the regret bound (Theorem 1), the CG inflation enters through the per-round factor $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, inside the sum of squares $\sum_t \alpha_t^2 u_t(a_t)\omega_t^2$; no horizon maximum over $\bar{\varepsilon}_t$ is taken.

Action screening (practical heuristic). When the action set $\mathcal{A}_t$ is large, computing the CG solve for every action is prohibitive ($|\mathcal{A}_t| \times I$ CG steps per round). A practical heuristic is to screen the top-K actions by mean prediction $\widehat{\mu}_t(a)$ (forward passes only), then compute UCB widths for the K candidates. This reduces cost to $K \times I$ CG solves. *The regret guarantee in Theorem 1 analyzes the full-enumeration case ($K = |\mathcal{A}_t|$); screening can drop the optimal arm and is not covered by the theory unless one adds an assumption that $a_t^* \in \mathcal{A}_t^K$ for all t.*

Algorithm 2 Operational CC-UCB with path certificates

Require: Reward predictor $\mu_{\boldsymbol{\theta}}$, damping $\lambda > 0$, noise proxy σ, confidence level δ, radius S, analytic constants G, L_g, L_μ; algorithmic curvature mode $\mathcal{C}_t$ (exact full-history $\mathbf{C}_t$ or approximate $\hat{\mathbf{C}}_t$) with certified one-sided factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; per-round CG tolerance $\bar{\varepsilon}_t < 1$

Initialize $\boldsymbol{\theta}_1$, certified $\zeta_1 \geq \|\nabla \mathcal{L}_1(\boldsymbol{\theta}_1)\|$, $n \leftarrow 0$, $\bar{\boldsymbol{\theta}} \leftarrow \mathbf{0}$, $J \leftarrow 0$, $R^{\mathrm{col}} \leftarrow 0$, $\bar{F} \leftarrow 0$, $\hat{\gamma} \leftarrow 0$, and $\mathcal{D}_0 \leftarrow \emptyset$

for $t = 1, 2, \ldots, T$ **do**

  Observe context $\mathbf{x}_t$ and action set $\mathcal{A}_t$; draw pre-action randomness Ω_t {$\mathcal{H}_t$ of (105)}

  $Q_t \leftarrow J + n\|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}\|^2$; $\bar{\chi}_t \leftarrow L_g\sqrt{Q_t}/(\sigma\sqrt{\lambda})$; $u_t \leftarrow \kappa_{+,t}(1+\bar{\chi}_t)^2$

  $\bar{M}_t \leftarrow \sigma^{-2}[L_g\sqrt{R^{\mathrm{col}}Q_t} + (3GL_g/2)Q_t + (L_g^2/2)Q_t^{3/2}]$; $\bar{\psi}_t \leftarrow (\zeta_t + \bar{M}_t)/\sqrt{\lambda}$ {in a radius-R region use $L_g^2 R Q_t$ for the last term}

  $\bar{\epsilon}_t^{\mathrm{lin}} \leftarrow (L_\mu/2)(S + \|\boldsymbol{\theta}_t\|)^2$; $\bar{\beta}_t \leftarrow \sqrt{\hat{\gamma} + 2\log(1/\delta)} + \sqrt{\lambda}S + \sigma^{-1}\sqrt{\bar{F}}$; $\omega_t \leftarrow \bar{\beta}_t + \bar{\psi}_t$ {corrected center uses $\omega_t = \bar{\beta}_t$}

  Form or receive a fixed $\mathcal{H}_t$-measurable SPD matrix–vector oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$, $\mathcal{C}_t \succeq \lambda\mathbf{I}$, held fixed across all CG solves this round {exact/buffered special case $\mathcal{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \mathbf{v}]$, weights w_s per §C.3}

  for each $a \in \mathcal{A}_t$ **do**

    $\widehat{\mu}_t(a) \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {mean prediction (or corrected center (36))}

    $\mathbf{g}_t(a) \leftarrow \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {gradient feature}

    if $\mathbf{g}_t(a) = \mathbf{0}$ **then**

      $\tilde{s}_t^2(a) \leftarrow 0$ {skip CG; no residual ratio}

    else

      $\tilde{\mathbf{u}} \leftarrow \mathrm{CG}(\mathcal{C}_t,\ \mathbf{g}_t(a))$; stop only once $\|r_I\|/\|\mathbf{g}_t(a)\| \leq \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$, or use a certified budget I_t from (112) {recompute the original residual in floating point; a point check is not an enclosure}

      $\tilde{s}_t^2(a) \leftarrow \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}$ {approx. predictive variance}

    end if

  end for

  $a_t \leftarrow \arg\max_{a\in\mathcal{A}_t}\left[\widehat{\mu}_t(a) + \omega_t\sqrt{\dfrac{u_t(a)}{1-\bar{\varepsilon}_t}}\,\sqrt{\tilde{s}_t^2(a)}\right]$ {bonus (95); ties broken by a fixed rule (e.g. lowest index) or by Ω_t}

  Play a_t, observe r_t

  $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{x}_t, a_t, r_t)\}$

  $c_t \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a_t) - r_t$; $R^{\mathrm{col}} \leftarrow R^{\mathrm{col}} + c_t^2$; $\bar{F} \leftarrow \bar{F} + (\bar{\epsilon}_t^{\mathrm{lin}})^2$; $\hat{\gamma} \leftarrow \hat{\gamma} + log[1 + \sigma^{-2}u_t\tilde{s}_t^2(a_t)/(1-\bar{\varepsilon}_t)]$

  $n \leftarrow n+1$; $d \leftarrow \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}$; $\bar{\boldsymbol{\theta}} \leftarrow \bar{\boldsymbol{\theta}} + d/n$; $J \leftarrow J + d^\top(\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}})$ {vector Welford update}

  Run J_{t+1}^{opt} warm-started optimizer steps on the *full* data set $\mathcal{D}_t$ to obtain $\boldsymbol{\theta}_{t+1}$, projecting to the certified region if required

  Certify $\zeta_{t+1} \geq \|\nabla\mathcal{L}_{t+1}(\boldsymbol{\theta}_{t+1})\|$; unless the terminal optimizer evaluation supplies this quantity, this requires one additional full-data gradient evaluation

end for

For the exact-curvature theorem, Algorithm 1 is run with $\mathcal{S}_t = \{1, \ldots, t-1\}$, $w_s = 1$, full action enumeration, $\mathcal{C}_t = \mathbf{C}_t$, and the transfer factor $u_t = (1+\bar{\chi}_t)^2$ from Lemma 19. For an approximate operator $\mathcal{C}_t = \hat{\mathbf{C}}_t$, use $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ from Theorem 34. The selection rule needs only the *one-sided* upper transfer factor $u_t(a)$ for optimism; no lower spectral factor is required (the cost of the operator is charged through the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$). Algorithm 2 computes u_t, $\bar{\psi}_t$, and $\bar{\beta}_t$ from pre-action state. The operator-specific inputs are the proved factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; exact full and unrescaled-subset operators use $\kappa_{+,t} = 1$. Windowed/refresh curvature and action screening are practical heuristics unless the corresponding one-sided transfer factor or optimal-arm-survival condition is separately verified.

Corrected-center remark. Replacing the center $\widehat{\mu}_t(a)$ by the corrected center $\widehat{\mu}_t^{\mathrm{corr}}(a)$ of (36) sets $\bar{\psi}_t = 0$ (so $\omega_t = \bar{\beta}_t$) and removes the centering-stability assumption exactly (Corollary 10), at the cost of frozen-feature access for $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$; this departs from the $O(d)$ additional curvature-state-memory, no-stored-feature design unless a frozen-feature oracle, replay mechanism, or certified sketch is supplied (see the remark after Corollary 10).

CC-UCB is the analyzed method throughout.

How to read the theorem. The theorem is a transfer result for a general predictable SPD curvature sequence. Lemma 13 and Corollary 5 instantiate its nonlinear confidence, centering, and feature-transfer inputs from $O(d)$ path state, observed collection residuals, analytic smoothness constants, and a certified optimizer residual. These schedules are executable but can be very loose. A sublinear guarantee from this bound still requires control of the realized dynamic width complexity and a restricted parameter path; unrestricted end-to-end fine-tuning is not covered. No horizon-uniform lower transfer is required by Theorem 1; the lower sandwich appears only in the explicit near-linear corollary.

Discussion of terms.

- *Exploration term.* Write $S_T := \sum_{t=1}^{T} \alpha_t^2 \, u_t(a_t) \, \omega_t^2$ for the sum of squares of the per-round factors. The exploration term is $2\sqrt{(\sigma^2 + G^2/\lambda) \, \Lambda_T^{\mathcal{C}} \, S_T}$. A sufficient condition for sublinear regret is

$$\Lambda_T^{\mathcal{C}} \, S_T = o(T^2), \qquad E_T = o(T).$$

  Concretely this holds whenever the per-round factors are polylogarithmically bounded ($\sup_t \alpha_t^2 u_t(a_t) \omega_t^2 = O(\operatorname{polylog} T)$, so $S_T = O(T \operatorname{polylog} T)$) and the realized dynamic complexity is small ($\Lambda_T^{\mathcal{C}} = O(\operatorname{polylog} T)$, e.g. via $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ with both the variation charge $V_T^{\mathcal{C}}$ and the endpoint $\log\det(\mathcal{C}_{T+1}/\lambda \mathbf{I})$ polylogarithmically bounded). Note ω_t^2 contains the cumulative squared linearization error through $\bar{F}_t$ (via β_t) and the centering certificate $\bar{\psi}_t$, so a small $\Lambda_T^{\mathcal{C}}$ alone is insufficient. α_t is bounded whenever the CG tolerance $\bar{\varepsilon}_t$ is bounded away from 1; in the exact-curvature case $u_t = (1 + \bar{\chi}_t)^2$ is bounded whenever the whitened-drift certificate $\bar{\chi}_t$ is bounded (Lemma 19); the whitened ℓ_2 accumulation $\chi_t \le \|\mathsf{D}_t\|_F$ makes this a near-stationarity condition on replayed-Jacobian drift. The unstable-curvature cost is isolated in $\Lambda_T^{\mathcal{C}}$ (through $V_T^{\mathcal{C}}$), not hidden in a spectral constant. The factor $\sqrt{\sigma^2 + G^2/\lambda}$ arises from the width-sum bound of the dynamic width-complexity lemma (Lemma 23, Eq. (83)); when $G^2 \le \lambda\sigma^2$ it is at most $\sigma\sqrt{2}$.
- *Linearization remainder.* The bound involves $E_T = \sum_t \varepsilon_{\mathrm{lin}}(t)$, not $T\varepsilon_{\mathrm{lin}}$. Near-linear regimes also require the squared-error sum $F_T = \sum_t \varepsilon_{\mathrm{lin}}(t)^2$ (through $\bar{F}_t$, hence ω_t) to be controlled. $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ gives $E_T = O(1)$; $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$ gives $E_T = O(\sqrt{T})$. The additive role of E_T mirrors misspecified linear bandits, where a uniformly bounded per-round model error commonly contributes an additive regret term that scales linearly in the cumulative approximation error (Ghosh et al., 2017; Lattimore et al., 2020). The displayed bound requires $E_T/T \to 0$; a nonvanishing uniform per-round error bound alone yields an $O(T)$ additive term.
- *Centering.* The centering certificate enters *multiplicatively*, through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ inside the sum of squares S_T, rather than as a separate additive term. It vanishes ($\bar{\psi}_t = 0$) when $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ (convex last-layer models with exact optimization), and the corrected-center variant (Corollary 10) removes it in general at the cost of a frozen-feature solve.
- *Regret improvement over diagonal/last-layer surrogates.* The bound itself does not prove that full curvature yields strictly lower regret than diagonal or last-layer methods in any specific regime; it provides an upper bound with explicit per-round factors. *Demonstrating regret improvement is an empirical hypothesis* (Section 5). The theorem exposes the transfer factor u_t and realized width complexity $\Lambda_T^{\mathcal{C}}$ as operator-dependent terms along the realized trajectory; it does not identify a causal regret cost because the adaptive path and other factors may also change.
- *Scope.* The bound is proved for Gaussian/squared-loss rewards. Extending it to exponential-family likelihoods (Bernoulli, categorical) would require a weighted frozen-feature analysis accounting for the data-dependent Fisher weights $\nabla^2 A(z)$ and is left as future work (Appendix J).

Corollary 37 (Sanity-check: frozen features recover linear UCB)**.** *Suppose the backbone is frozen and the reward head is linear in its parameters, the ridge objective is optimized exactly, and take the algorithmic curvature to be the sequential frozen Gram matrix, $\mathcal{C}_t = \bar{\mathbf{C}}_t$. Then: $\varepsilon_{\mathrm{lin}}(t) = 0$ for all t ($L = 0$ in Assumption 3), so $E_T = F_T = 0$; $\bar{\psi}_t = 0$ (under exact optimization the MAP and linearized estimators coincide, so Assumption 4 holds with equality), giving $\omega_t = \bar{\beta}_t$; $u_t(a) = 1$ for all a (indeed $\bar{s}_t^2(a) = s_{\mathcal{C},t}^2(a)$, so Assumption 5 holds with equality); and, since $\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top$ by (7), the update is exactly the rank-one step, i.e. $\mathcal{C}_{t+1} = \mathcal{C}_t^+$, so $\Xi_t = 0$, $V_T^{\mathcal{C}} = 0$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$. Then Theorem 1 gives*

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\gamma_T \sum_{t=1}^{T} \alpha_t^2 \, \bar{\beta}_t^2}.$$

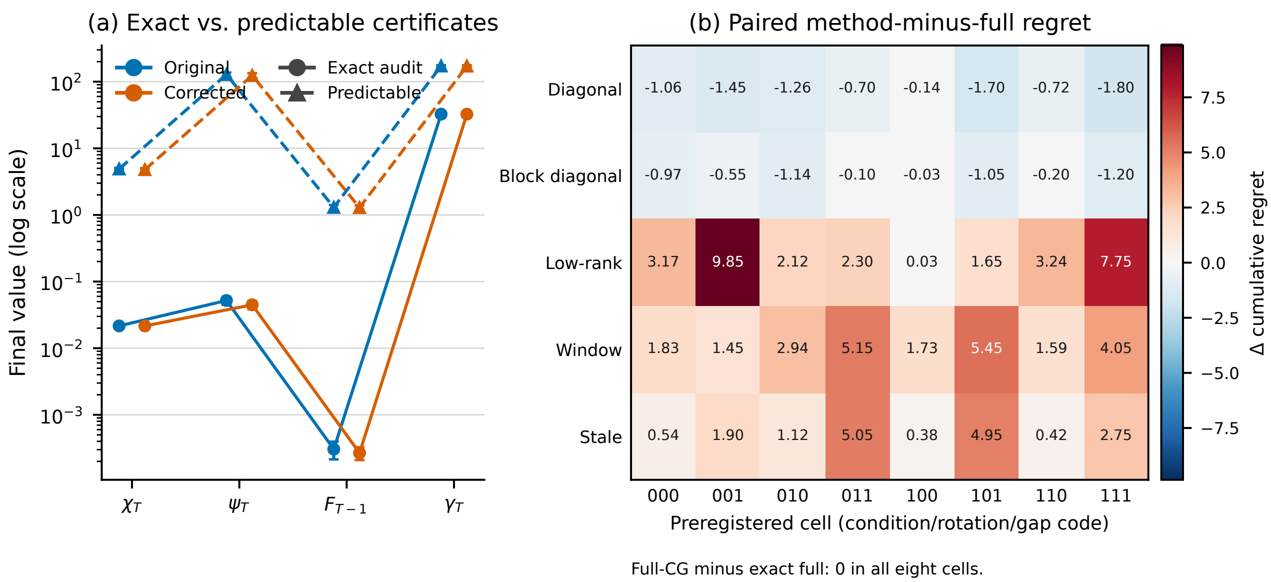

Figure 9: Supplementary certificate-tightness and curvature-mechanism overview. Panel (a) compares final-round float64 audit quantities with predictable schedules in the 50-seed tanh execution. Panel (b) gives paired regret differences in the predeclared curvature phase map. These are descriptive point audits.

If in addition $\bar{\varepsilon}_t \le \bar{\varepsilon}$ so that $\alpha_t \le \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$, the looser Corollary 38 recovers the familiar LinUCB form $R_T \le 2\,\alpha_I\,\bar{\beta}_{\max,T}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T}$, where $\bar{\beta}_{\max,T} := \max_{t\le T}\bar{\beta}_t$.

Proof. Substitute $\varepsilon_{\text{lin}}(t) = 0$, $\bar{\psi}_t = 0$, $u_t(a) = 1$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$, $V_T^{\mathcal{C}} = 0$ into (15); the frozen Gram is monotone, so $\Xi_t = 0$ and the dynamic potential collapses to the information gain γ_T. The LinUCB form is Corollary 38 with $\alpha_t \le \alpha_I$, $u_t = 1$, $\omega_t = \bar{\beta}_t \le \bar{\beta}_{\max,T}$. □

E Executed-Policy Audit Protocols

Interpretation and statistics. Primary regret results are generated by online policies executed from scratch. Synthetic evaluators know the teacher only to compute pseudo-regret, optimism failures, and post-hoc theorem-event checks; the action rule never receives it. Hyperparameter selection, where used, is performed on the declared tuning seeds before evaluation. Common random numbers pair methods within an evaluation seed. We report arithmetic means, marginal 95% Student-t intervals, and paired Student-t intervals for method differences. No multiplicity adjustment is applied, so the intervals are descriptive. Scaling-curve bands instead use pointwise percentile bootstrap intervals. Their slope intervals resample whole seed trajectories, reaverage the same five nested prefixes, and refit the specified unweighted OLS model; they measure seed sensitivity conditional on that finite grid, not whether a power law is appropriate. An offline common-trajectory diagnostic uses one logged action and checkpoint sequence for all operators and is never mixed with online regret.

The legacy grid uses a predeclared $\zeta_t = 10^{-6}$ and at most 40 optimizer iterations rather than enforcing the envelope. Its full CG row is also a one-step diagonal-basis equivalence test. Non-diagonal active-coordinate widths are ambient-invariant in exact arithmetic, while the coordinate-diagonal control depends on the embedding. These limitations motivated the rotated replacement; the legacy slopes in Table 7 remain descriptive only.

Predictable tanh execution. The analytically bounded tanh instance is $\mu_\theta(x,a) = \tanh(\phi(x,a)^\top\theta)$ with $d = 6$, four actions, $\|\phi\| \le 0.75$, Gaussian noise $\sigma = 0.20$, teacher radius $S = 0.35$, and an optimizer projected to radius 0.50. We use the global constants $G = 0.75$ and $L_g = L_\mu = 4(0.75)^2/(3\sqrt{3})$, $\lambda = 1$, $\delta = 0.05$, eight clipped gradient steps per round, and a CG energy target 0.05 with the analytic condition bound $1 + (t-1)G^2/(\lambda\sigma^2)$. At every round the policy records Q_t, R_t^{col}, $\bar{F}_t$, $\hat{\gamma}_t$, $\bar{\chi}_t$, $\bar{M}_t$, $\bar{\psi}_t$, $\bar{\beta}_t$, the optimizer gradient norm, all candidate

r	Method	Regret slope [95% interval]	R^2	Max. $\lvert \log R - \widehat{\log R}\rvert$	Λ slope [95% interval]
4	Exact current	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Full CG	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.105 [0.104, 0.106]
4	Window $q = 1/2$	0.851 [0.843, 0.859]	0.9937	8.445e-02	0.465 [0.464, 0.465]
4	Window $q = 2/3$	0.309 [0.292, 0.326]	0.8529	1.550e-01	0.334 [0.333, 0.334]
4	Window $q = 1$	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Frozen	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.106 [0.106, 0.106]
4	Diagonal	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.673 [0.673, 0.673]
4	Greedy	0.000 [-0.000, 0.000]	1.0000	5.551e-17	0.115 [0.114, 0.115]
8	Exact current	0.134 [0.115, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Full CG	0.141 [0.122, 0.161]	0.6969	1.247e-01	0.099 [0.098, 0.099]
8	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.383 [0.383, 0.383]
8	Window $q = 2/3$	0.994 [0.993, 0.995]	1.0000	7.659e-03	0.304 [0.304, 0.304]
8	Window $q = 1$	0.134 [0.116, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Frozen	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.115 [0.115, 0.115]
8	Diagonal	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.671 [0.671, 0.671]
8	Greedy	0.037 [0.018, 0.057]	0.5542	4.606e-02	0.125 [0.124, 0.126]
16	Exact current	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Full CG	0.829 [0.815, 0.842]	0.9841	1.418e-01	0.118 [0.117, 0.118]
16	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.406 [0.406, 0.406]
16	Window $q = 2/3$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.278 [0.278, 0.278]
16	Window $q = 1$	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Frozen	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.125 [0.125, 0.125]
16	Diagonal	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.669 [0.669, 0.669]
16	Greedy	0.273 [0.229, 0.315]	0.9024	1.086e-01	0.140 [0.138, 0.141]

Table 7: Five-point OLS diagnostics for every method at $d = 2048$. Slope brackets are obtained from 2,000 paired resamples of whole seed trajectories. R^2 and maximum absolute log residual describe the fit to the five selected prefix means; they are not tests of a power-law model or asymptotic exponents.

residuals, and whether any post-hoc exact check fails. No post-hoc field is read by action selection. Seeds 60–69 run a predeclared eight-cell one-factor sensitivity grid; the fixed reference configuration is rerun from scratch on evaluation seeds 160–209 for both centers. The 50-seed means and intervals in Appendix B.4 come only from those evaluation runs.

Linear theorem-event audit. The context is $x \in \{-1, +1\}^8/\sqrt{8}$, there are five actions, and $\phi(x, a) = [x, e_a, x \otimes e_a]$ has dimension 53 and norm $\sqrt{3}$. The fixed teacher contains shared, arm-intercept, and context–arm interaction blocks; exact enumeration of all 2^8 contexts confirms that multiple arms are optimal. Noise is $\mathcal{N}(0, 0.25^2)$. Evaluation uses seeds 100–119 after tuning on seeds 0–9. The tuning grid is $\lambda \in \{0.5, 1, 2\}$ and $c_{\text{bonus}} \in \{1, 1.25\}$; a candidate is eligible only if every tuning run passes the recorded confidence and small-scale numerical checks. Fresh evaluation policies use dense full Gram, CG full Gram, diagonal, window 64, rescaled subsample 64, rank-16 Lanczos–Ritz, and refresh period 20. The CG target is 0.05 with at most 106 iterations. Fixed-reference and independently tuned comparisons are both saved.

Linear policy checks and bound scale. The implemented base radius is

$$\beta_t^{\text{base}} = \sqrt{d \log\left(1 + \frac{(t-1)G^2}{d\lambda\sigma^2}\right) + 2\log(1/\delta)} + \sqrt{\lambda}\, S, \qquad \bar{\beta}_t = c_{\text{bonus}}\beta_t^{\text{base}}, \quad c_{\text{bonus}} \geq 1,$$

with $d = 53$, $G = \sqrt{3}$, $S = \|\theta^*\|_2$, and $\delta = 0.05$; hence tuning $c_{\text{bonus}} \in \{1, 1.25\}$ can only inflate the valid radius. Because $\bar{\psi}_t = 0$, the policy uses $\omega_t = \bar{\beta}_t$. Dense, CG, unrescaled window, and stale-refresh policies use the analytic factor $u_t = 1$. Diagonal, rescaled-subsample, and Lanczos policies materialize dense float64 small matrices before action selection and use $u_t = \max\{1, (1 + 32\epsilon_{\text{mach}})\lambda_{\max}(\bar{C}_t^{-1/2}\mathcal{C}_t\bar{C}_t^{-1/2})\}$. The multiplicative $1 + 32\epsilon_{\text{mach}}$ is a heuristic inflation, not a verified rounding-error enclosure. The CG policy likewise sets $\bar{\kappa}_t$ to a float64 point estimate of $\text{cond}(\mathcal{C}_t)$ before action selection and enforces $\bar{\varepsilon}_t = 0.05$ using relative residual target $\bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$; the other policies have $\bar{\varepsilon}_t = 0$. These pre-action computations are deterministic conditional on each seeded operator draw, but predictability does not turn an unenclosed point estimate into an upper certificate. The analytic $u_t = 1$ transfer statements for dense full Gram, unrescaled window, and stale-refresh are unaffected; diagonal, rescaled-subsample, Lanczos, and full-Gram CG remain numerical audits of theorem semantics rather than certified executions. Tuning-run event checks are a selection gate, not the mathematical basis of the evaluation checks. The machine-readable per-policy trace is `results/derived/certification_audit.json`.

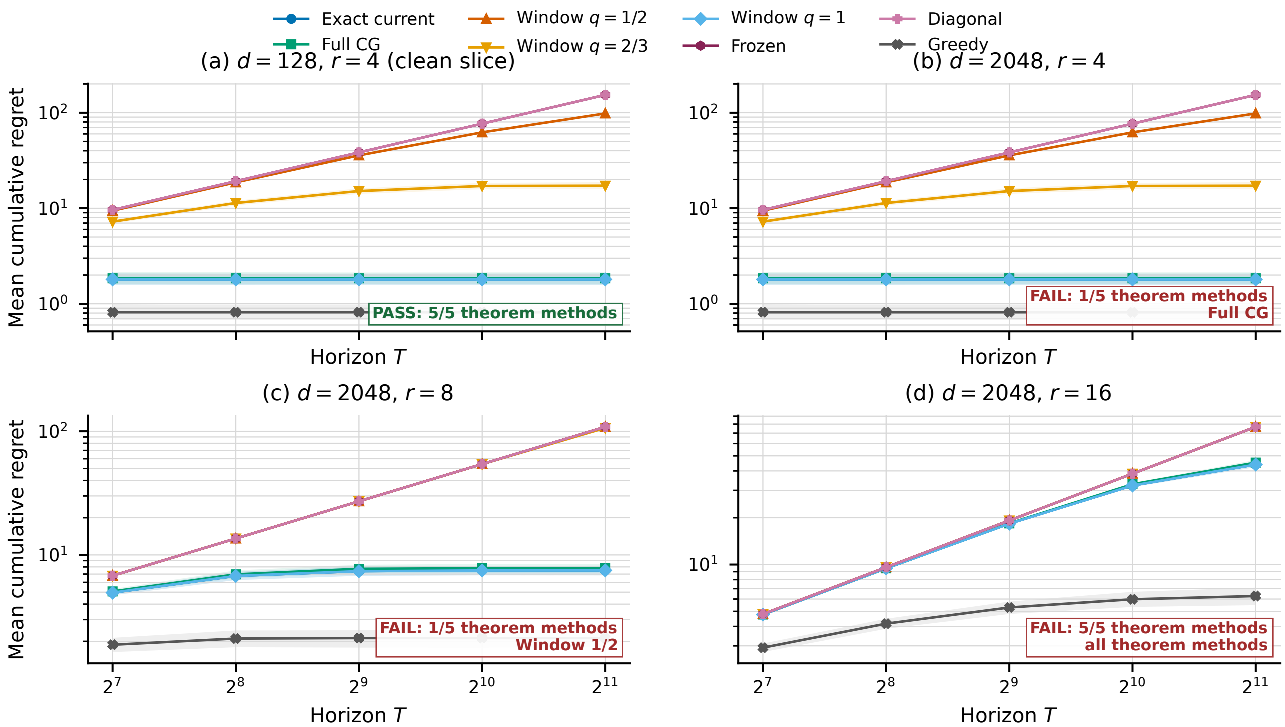

Figure 10: Legacy finite-horizon scaling diagnostic retained for disclosure. The first panel is the event-clean $d = 128, r = 4$ reference; remaining panels use $d = 2048$. Labels report the conjunction of float64 event fields. Failed cells are not interpreted as tests of Theorem 6; the main replacement is the premise-clean construction in Figure 5.

Nonlinear drift stress test. The teacher and student are one-hidden-layer width-four tanh networks with four inputs, five outputs, and 45 displacement parameters. They share the initialized backbone, making the exact frozen-head ridge solve a 25-parameter linear regime. Full-network regimes use respectively $(\text{learning rate}, \text{updates}, \text{step cap}) = (0.002, 2, 0.005), (0.004, 3, 0.020), (0.008, 4, 0.080)$. Each original- or corrected-center policy executes for 100 rounds on seeds 110–129 with a fixed, time-only schedule; no evaluation-seed tuning occurs. Dense float64 replay computes E_T, F_T, the dense-reference centering ratio and primitive ψ_t, χ_t, generalized transfer factors, condition numbers, dense-reference and residual-based CG errors, $\Lambda_T^{\mathcal{C}}$, endpoint log determinant, $V_T^{\mathcal{C}}$, Γ_T^{dyn}, and all-action optimism failures. Policy-available and post-hoc fields are separated in every JSONL record. The corrected-center identity is checked directly.

A secondary post-hoc analysis across the eight dependent regime–center cell means gives descriptive Spearman correlation 0.83 between the post-hoc decomposition score and regret, versus 0.57 for relative curvature change. With $n = 8$, reused seeds, and no seed-level sensitivity or inferential analysis, this is neither a primary result nor a causal or inferential claim. This stress test is secondary to the predictable tanh execution because its schedules fail.

Predeclared curvature-mechanism map. The fixed-gap bounded-linear family has $d = 10$, four actions, $T = 30$, and Gaussian noise 0.2. Its eight-cell fractional design crosses active-spectrum condition number $\{3, 50\}$, rotation $\{0, 60°\}$, action gap $\{0.15, 0.50\}$, nuisance strength $\{0.4, 1.2\}$, effective rank $\{3, 8\}$, and damping $\{0.5, 2\}$ according to the generators stored in the configuration; all cells are fixed before execution and no tuning is run. Exact full, residual-checked full CG, diagonal, block-diagonal, rank-three Lanczos, window-eight, and refresh-period-five policies execute independently on seeds 200–229. The exact-full trajectory is additionally replayed through every operator. These common-trajectory records contain per-action width ratios, cross-action coefficient of variation, Spearman and Kendall ranks, raw and scalar-rescaled top-score disagreement, normalized decision-margin distortion, candidate-gradient alignment with retained and discarded eigenspaces, global and

Cell (d/r)	Exact	CG	$q = 1/2$	$q = 2/3$	$q = 1$	Frozen	Diag.	Greedy
128/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/8	7.50	7.82	108.61	106.44	7.50	108.61	108.61	2.13
2048/16	43.95	45.24	76.80	76.80	43.95	76.80	76.80	6.27

	Theorem-event status and failed round–event categories			
Method	$d = 128, r = 4$	$d = 2048, r = 4$	$d = 2048, r = 8$	$d = 2048, r = 16$
Exact current	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]
Full CG	PASS	FAIL [O=1]	PASS	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 1/2$	PASS	PASS	FAIL [O=1]	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 2/3$	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 1$	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]

PASS means that all recorded float64 theorem-event fields passed for that method and cell.
Failure counts are round–event fields: O is optimizer residual, P_e is ψ-excitation, and P_λ is ψ-lambda.

Table 8: Legacy scaling summary. Panel (a) reports terminal mean regret; panel (b) decomposes failed float64 fields by method. O is the prescribed optimizer envelope, P_e the path bound using excitation, and P_λ the damping path bound. Any nonzero entry precludes a theorem-test interpretation.

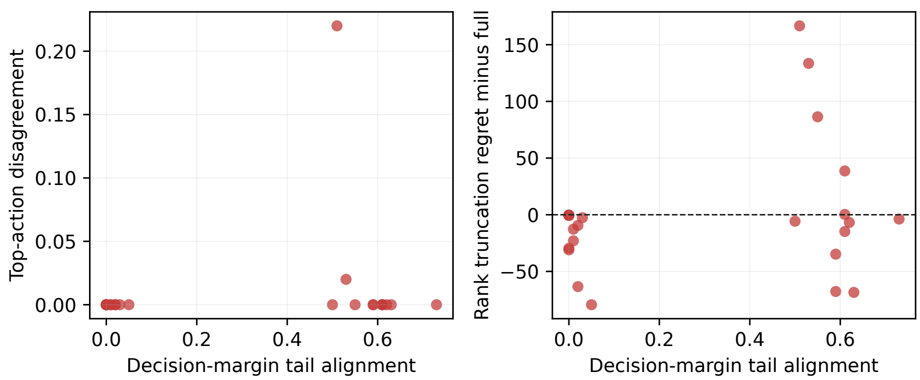

Figure 11: Decision effects of discarded spectral directions over all 24 predeclared rank/power/alignment cells. The horizontal coordinate measures alignment of the decision margin with the discarded tail. Regret differences are paired by evaluation seed and compare rank truncation with exact full; the figure is a mechanism diagnostic, not a dominance claim.

action-set transfer, effective rank, and condition number. Paired intervals are method-minus-exact-full; they are descriptive and unadjusted across cells.

Coverage-matched extension. The extension retains the same eight cells and operator settings, but adds tuning seeds 2000–2019 and evaluation seeds 2100–2149. Coefficients are pooled over every tuning cell, never selected per evaluation cell. The three protocols use an identical theoretical coefficient, the empirical 95th percentile of all one-step prediction-error/width ratios, or the ratio needed to match the full operator's mean bonus on common tuning trajectories. The latter two coefficients are frozen before any evaluation run. Holm adjustment is over all $3 \times 8 \times 7 = 168$ prespecified surrogate comparisons. Each marginal p-value comes from a two-sided paired Student-t test of the 50 seed-level terminal-regret differences; Holm controls familywise error at 0.05. For zero variance, an identically zero difference is assigned $p = 1$ and a constant nonzero difference the limiting value $p = 0$. Because this is a fixed-feature linear model, current and historical full-Gram rows are algebraically identical. An official recursive LO-FI implementation is not included because the repository lacks a compatible recursive-precision implementation; the seventh surrogate is the batch-refit low-rank-plus-diagonal LO-FI-style control described in Appendix B.5.

Balanced contextual benchmark. Normalized Rademacher contexts have dimension four, five available actions, and Gaussian noise 0.10; exact enumeration of the 16 contexts verifies that every arm is optimal somewhere. Each method selects $(\lambda, c_{\text{bonus}})$ from $\{0.5, 1\} \times \{0.5, 1, 2\}$ by mean pseudo-regret over tuning seeds 70–79 at $T = 80$, then runs from scratch for $T = 200$ on seeds 260–289 with shared contexts and reward noise.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.023	0.000
10	Full CG	PASS	0.023	2.002
10	Window $q = 1/2$	PASS	0.023	1.486
10	Window $q = 2/3$	PASS	0.023	1.658
10	Window $q = 1$	PASS	0.023	2.002
10	Frozen	PASS	0.023	0.000
10	Diagonal	control	0.023	0.000
10	Greedy	control	0.023	0.000
100	Exact current	PASS	0.049	0.000
100	Full CG	PASS	0.049	2.002
100	Window $q = 1/2$	PASS	0.049	1.486
100	Window $q = 2/3$	PASS	0.049	1.658
100	Window $q = 1$	PASS	0.049	2.002
100	Frozen	PASS	0.049	0.000
100	Diagonal	control	0.049	0.000
100	Greedy	control	0.049	0.000
1000	Exact current	PASS	0.063	0.000
1000	Full CG	PASS	0.063	2.002
1000	Window $q = 1/2$	PASS	0.063	1.486
1000	Window $q = 2/3$	PASS	0.063	1.658
1000	Window $q = 1$	PASS	0.063	2.002
1000	Frozen	PASS	0.063	0.000
1000	Diagonal	control	0.063	0.000
1000	Greedy	control	0.063	0.000

Table 9: Finite-grid log–log slope diagnostics for $d = 128, r = 4$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.079	0.000
10	Full CG	PASS	0.079	2.002
10	Window $q = 1/2$	PASS	0.079	1.496
10	Window $q = 2/3$	PASS	0.079	1.660
10	Window $q = 1$	PASS	0.079	2.002
10	Frozen	PASS	0.079	0.000
10	Diagonal	control	0.079	0.000
10	Greedy	control	0.079	0.000
100	Exact current	PASS	0.103	0.000
100	Full CG	PASS	0.103	2.002
100	Window $q = 1/2$	PASS	0.103	1.496
100	Window $q = 2/3$	PASS	0.103	1.660
100	Window $q = 1$	PASS	0.103	2.002
100	Frozen	PASS	0.103	0.000
100	Diagonal	control	0.103	0.000
100	Greedy	control	0.103	0.000
1000	Exact current	PASS	0.142	0.000
1000	Full CG	PASS	0.142	2.026
1000	Window $q = 1/2$	PASS	0.142	1.498
1000	Window $q = 2/3$	PASS	0.142	1.674
1000	Window $q = 1$	PASS	0.142	2.026
1000	Frozen	PASS	0.142	0.000
1000	Diagonal	control	0.142	0.000
1000	Greedy	control	0.142	0.000

Table 10: Finite-grid log–log slope diagnostics for $d = 128, r = 8$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.154	0.000
10	Full CG	PASS	0.154	2.025
10	Window $q = 1/2$	PASS	0.154	1.564
10	Window $q = 2/3$	PASS	0.154	1.678
10	Window $q = 1$	PASS	0.154	2.025
10	Frozen	PASS	0.154	0.000
10	Diagonal	control	0.154	0.000
10	Greedy	control	0.154	0.000
100	Exact current	PASS	0.221	0.000
100	Full CG	PASS	0.221	2.012
100	Window $q = 1/2$	PASS	0.221	1.621
100	Window $q = 2/3$	PASS	0.221	1.692
100	Window $q = 1$	PASS	0.221	2.012
100	Frozen	PASS	0.221	0.000
100	Diagonal	control	0.221	0.000
100	Greedy	control	0.221	0.000
1000	Exact current	PASS	0.237	0.000
1000	Full CG	PASS	0.237	2.022
1000	Window $q = 1/2$	PASS	0.237	1.639
1000	Window $q = 2/3$	PASS	0.237	1.745
1000	Window $q = 1$	PASS	0.237	2.022
1000	Frozen	PASS	0.237	0.000
1000	Diagonal	control	0.237	0.000
1000	Greedy	control	0.237	0.000

Table 11: Finite-grid log–log slope diagnostics for $d = 128, r = 16$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.024	0.000
10	Full CG	PASS	0.024	2.002
10	Window $q = 1/2$	PASS	0.024	1.486
10	Window $q = 2/3$	PASS	0.024	1.658
10	Window $q = 1$	PASS	0.024	2.002
10	Frozen	PASS	0.024	0.000
10	Diagonal	control	0.024	0.000
10	Greedy	control	0.024	0.000
100	Exact current	PASS	0.050	0.000
100	Full CG	PASS	0.050	2.002
100	Window $q = 1/2$	PASS	0.050	1.486
100	Window $q = 2/3$	PASS	0.050	1.658
100	Window $q = 1$	PASS	0.050	2.002
100	Frozen	PASS	0.050	0.000
100	Diagonal	control	0.050	0.000
100	Greedy	control	0.050	0.000
1000	Exact current	PASS	0.076	0.000
1000	Full CG	PASS	0.076	2.002
1000	Window $q = 1/2$	PASS	0.076	1.486
1000	Window $q = 2/3$	PASS	0.076	1.658
1000	Window $q = 1$	PASS	0.076	2.002
1000	Frozen	PASS	0.076	0.000
1000	Diagonal	control	0.076	0.000
1000	Greedy	control	0.076	0.000

Table 12: Finite-grid log–log slope diagnostics for $d = 512, r = 4$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.085	0.000
10	Full CG	PASS	0.085	2.002
10	Window $q = 1/2$	PASS	0.085	1.496
10	Window $q = 2/3$	PASS	0.085	1.660
10	Window $q = 1$	PASS	0.085	2.002
10	Frozen	PASS	0.085	0.000
10	Diagonal	control	0.085	0.000
10	Greedy	control	0.085	0.000
100	Exact current	PASS	0.087	0.000
100	Full CG	PASS	0.087	2.002
100	Window $q = 1/2$	PASS	0.087	1.496
100	Window $q = 2/3$	PASS	0.087	1.660
100	Window $q = 1$	PASS	0.087	2.002
100	Frozen	PASS	0.087	0.000
100	Diagonal	control	0.087	0.000
100	Greedy	control	0.087	0.000
1000	Exact current	PASS	0.111	0.000
1000	Full CG	PASS	0.111	2.026
1000	Window $q = 1/2$	PASS	0.111	1.497
1000	Window $q = 2/3$	PASS	0.111	1.675
1000	Window $q = 1$	PASS	0.111	2.026
1000	Frozen	PASS	0.111	0.000
1000	Diagonal	control	0.111	0.000
1000	Greedy	control	0.111	0.000

Table 13: Finite-grid log–log slope diagnostics for $d = 512, r = 8$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.175	0.000
10	Full CG	PASS	0.175	2.025
10	Window $q = 1/2$	PASS	0.175	1.564
10	Window $q = 2/3$	PASS	0.175	1.678
10	Window $q = 1$	PASS	0.175	2.025
10	Frozen	PASS	0.175	0.000
10	Diagonal	control	0.175	0.000
10	Greedy	control	0.175	0.000
100	Exact current	PASS	0.214	0.000
100	Full CG	PASS	0.214	2.012
100	Window $q = 1/2$	PASS	0.214	1.621
100	Window $q = 2/3$	PASS	0.214	1.692
100	Window $q = 1$	PASS	0.214	2.012
100	Frozen	PASS	0.214	0.000
100	Diagonal	control	0.214	0.000
100	Greedy	control	0.214	0.000
1000	Exact current	PASS	0.224	0.000
1000	Full CG	PASS	0.224	2.023
1000	Window $q = 1/2$	PASS	0.224	1.639
1000	Window $q = 2/3$	PASS	0.224	1.744
1000	Window $q = 1$	PASS	0.224	2.023
1000	Frozen	PASS	0.224	0.000
1000	Diagonal	control	0.224	0.000
1000	Greedy	control	0.224	0.000

Table 14: Finite-grid log–log slope diagnostics for $d = 512, r = 16$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.041	0.000
10	Full CG	PASS	0.041	2.002
10	Window $q = 1/2$	PASS	0.041	1.486
10	Window $q = 2/3$	PASS	0.041	1.658
10	Window $q = 1$	PASS	0.041	2.002
10	Frozen	PASS	0.041	0.000
10	Diagonal	control	0.041	0.000
10	Greedy	control	0.041	0.000
100	Exact current	PASS	0.021	0.000
100	Full CG	PASS	0.021	2.002
100	Window $q = 1/2$	PASS	0.021	1.486
100	Window $q = 2/3$	PASS	0.021	1.658
100	Window $q = 1$	PASS	0.021	2.002
100	Frozen	PASS	0.021	0.000
100	Diagonal	control	0.021	0.000
100	Greedy	control	0.021	0.000
1000	Exact current	PASS	0.028	0.000
1000	Full CG	PASS	0.028	2.002
1000	Window $q = 1/2$	PASS	0.028	1.486
1000	Window $q = 2/3$	PASS	0.028	1.658
1000	Window $q = 1$	PASS	0.028	2.002
1000	Frozen	PASS	0.028	0.000
1000	Diagonal	control	0.028	0.000
1000	Greedy	control	0.028	0.000

Table 15: Finite-grid log–log slope diagnostics for $d = 2048, r = 4$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.046	0.000
10	Full CG	PASS	0.046	2.002
10	Window $q = 1/2$	PASS	0.046	1.496
10	Window $q = 2/3$	PASS	0.046	1.660
10	Window $q = 1$	PASS	0.046	2.002
10	Frozen	PASS	0.046	0.000
10	Diagonal	control	0.046	0.000
10	Greedy	control	0.046	0.000
100	Exact current	PASS	0.093	0.000
100	Full CG	PASS	0.093	2.002
100	Window $q = 1/2$	PASS	0.093	1.496
100	Window $q = 2/3$	PASS	0.093	1.660
100	Window $q = 1$	PASS	0.093	2.002
100	Frozen	PASS	0.093	0.000
100	Diagonal	control	0.093	0.000
100	Greedy	control	0.093	0.000
1000	Exact current	PASS	0.093	0.000
1000	Full CG	PASS	0.093	2.026
1000	Window $q = 1/2$	PASS	0.093	1.498
1000	Window $q = 2/3$	PASS	0.093	1.674
1000	Window $q = 1$	PASS	0.093	2.026
1000	Frozen	PASS	0.093	0.000
1000	Diagonal	control	0.093	0.000
1000	Greedy	control	0.093	0.000

Table 16: Finite-grid log–log slope diagnostics for $d = 2048, r = 8$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

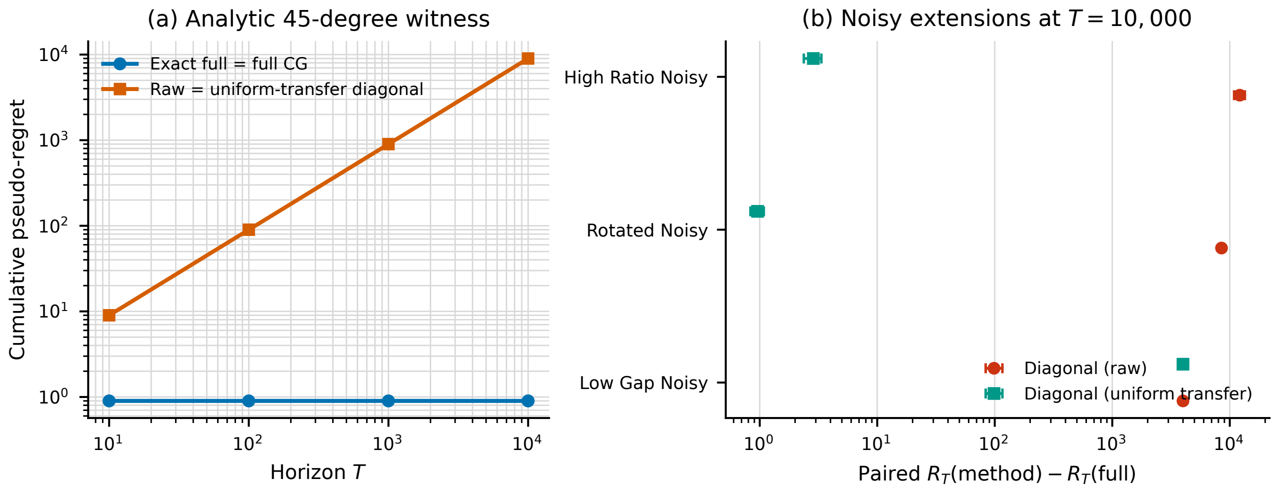

Figure 12: Off-diagonal linear-Gram witness. The analytic cell is constructive; the noisy cells are uncertified extensions. Paired differences are descriptive and establish no uniform ordering of curvature surrogates.

Full-network methods use the same width-four tanh backbone, one clipped float64 update per round, learning rate 0.01, and model ridge 0.01. The CC-UCB row relinearizes all selected historical samples at the current parameter and solves one fixed full-GGN operator per action with explicit residual checks. The local NeuralUCB row is the historical-time full-network gradient-Gram baseline: it stores each selected gradient at its collection parameter rather than relinearizing it. NeuralLinear is a Bayesian Gaussian head on the frozen initialized representation; the frozen-last-layer row uses the same posterior with a UCB rule. NeuralUCB and NeuralTS are identified as local matched implementations rather than exact reproductions of every published training detail. The one-sample clipped updates do not optimize (3) or certify its full-gradient residual, so these benchmark rows are practical comparisons rather than theorem instantiations. LinUCB, linear Thompson sampling, diagonal full-network UCB, greedy, Gaussian UCB1, and Gaussian context-free Thompson sampling complete the comparison. Arithmetic means, marginal intervals, paired differences, selected hyperparameters, run times, and all seed-level records are retained.

Secondary operator and CG audits. The operator factorial varies windows 16/64, uniform resamples 16/64, Lanczos ranks 4/16, and refresh periods 5/20 in addition to full, frozen, and diagonal operators. On both the bounded linear task and the medium-drift, corrected-center tanh task, twenty online evaluation seeds (120–139) and matching offline common trajectories record exact global and action-set transfer, widths, disagreement, determinant terms, regret, runtime, and memory. The nonlinear policies use the fixed schedule and remain post-hoc audits rather than certified executions. The CG audit uses $d = 64, n = 256, K = 5$, condition numbers 10/100/1000, five target errors, two initializations, and two preconditioners on seeds 130–139. Its solver audit contains 600,000 dense-reference action solves; its separate policy audit executes 200 full-enumeration linear-bandit policies for 200 rounds with the same five tolerances and start/preconditioner factorial. Unlike the one-step scaling construction, the rotated-SPD right-hand sides exercise nontrivial Krylov convergence. At target energy error 0.1 with zero start and no preconditioner, mean iteration counts are 6.026, 22.643, and 55.043 for condition numbers 10, 10^2, and 10^3; mean measured energy errors are 2.23×10^{-2}, 6.87×10^{-3}, and 2.45×10^{-3}, and mean relative width errors are 2.54×10^{-4}, 2.43×10^{-5}, and 3.28×10^{-6}. The residual certificate is at most the prescribed 0.1 in every retained solve. These solver audits do not establish a method-speed advantage or a policy benefit.

Systems audit. Five evaluation seeds (140–144) cross $d \in \{32, 64, 128\}$, $n \in \{32, 128, 512\}$, $K \in \{5, 10\}$, and $I \in \{5, 15, 30\}$. We measure dense exact, separate full CG, diagonal, final-coordinate block, Lanczos–Ritz, row-batched independent CG, and row-batched Jacobi-PCG. Every action retains its own original-system residual; batched methods share only the operator evaluation. An additional matrix-free CPU grid uses $d \in \{512, 2048, 8192\}$, $n = 32$, $K = 4$, and $I = 16$ for the two batched solvers, with no dense allocation. Records

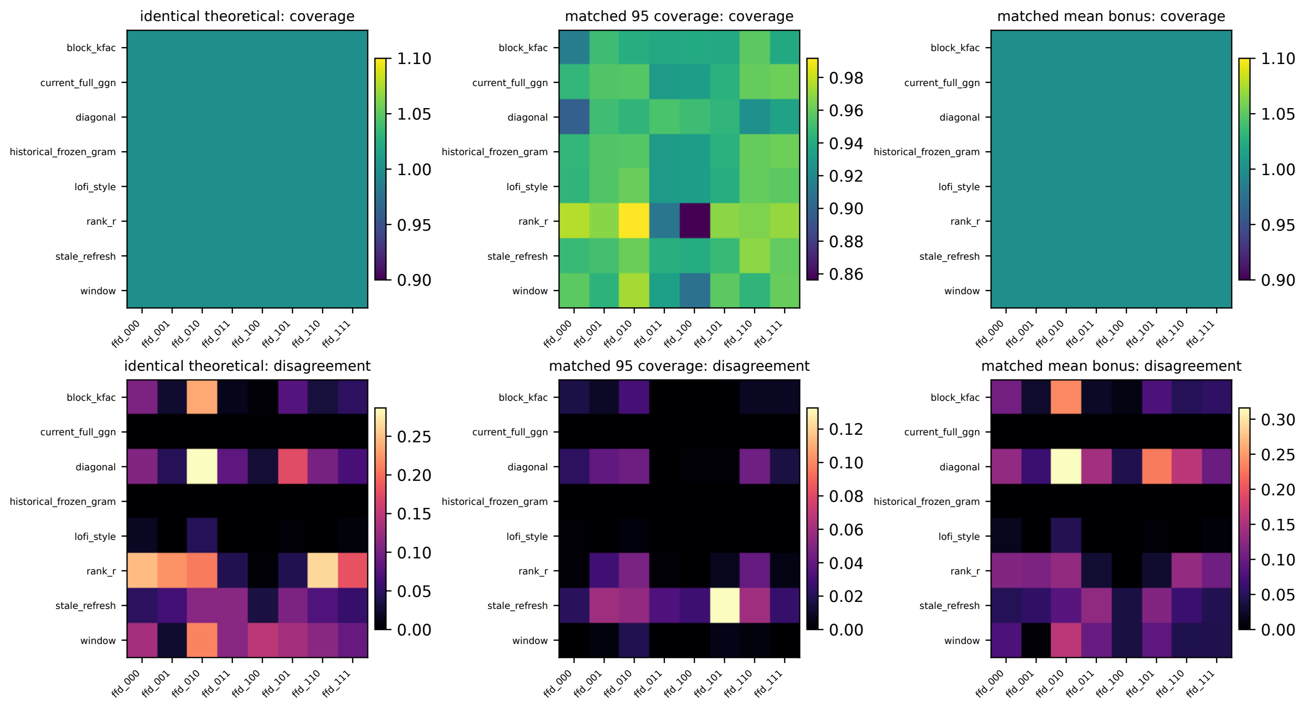

Figure 13: Full-grid evaluation coverage and top-action disagreement under all three frozen calibration protocols. These are independently executed policy diagnostics, not causal operator effects.

include wall time, host memory, operator calls, equivalent sample CVPs, per-action iterations and residuals, and width error. These are synthetic float64 operator timings on one CPU host and use different computational primitives. The timed operator stores an $n \times d$ random-feature matrix and applies an averaged Gram; it does not time model training or on-the-fly network JVP/VJP. Timings are descriptive resource accounting under the executed implementations, not matched-wall-clock comparisons; no trained-network or accelerator extrapolation is made. The Jacobi diagonal is fixed during each solve, so its recurrence is the symmetric transformed PCG of Lemma 27. We report its measured iterations and make no unproved claim that Jacobi improves the transformed condition number. A Buck-managed Conda toolchain runs a separate real-`torch.func` pipeline. All 360 predeclared evaluation cells complete for the 131,841- and 9,972,737-parameter MLPs. At $m = 512$, $K = 10$, and target residual 10^{-3}, mean batched-CG times are 0.0395 and 0.5674 seconds with peak allocated accelerator memory 0.126 and 4.219 GiB, respectively. The corresponding separate-CG times are 0.2101 and 0.7532 seconds. On the small model, the strict target's upper 95% aggregate bound on maximum relative squared-width error is 9.38×10^{-5}; one of the prespecified groups has top-action and complete rank agreement 0.9 because a near tie changes order. The large model has no explicit-Jacobian reference. Figure 14 and Table 26 report the full accuracy and resource audit.

E.1 Covertype real-data baseline audit

We use UCI Covertype DOI `10.24432/C50K5N` via scikit-learn 1.9.0; the cached dataset checksum is `eeab5db05bb3a9c00c62ced4c7d13112a00cd3487f59ac1943aaf5983d7c3ddf`. A PCG64 permutation with seed 20260717 creates non-stratified train/validation/test counts 348,607/116,202/116,203. Training means and scales standardize columns 0–9; binary columns 10–53 are unchanged. A width-four tanh network is initialized from seeded fan-in normals and optimized by one online SGD step (learning rate 0.01, ridge 1) using only selected-arm feedback. Full-GGN, frozen-Gram, diagonal, and greedy policies train all 255 parameters; last-layer policies train 35 head parameters. This one-sample update is not the full-history objective (3) and supplies no optimizer-residual certificate. Damping $\{1, 10\}$ and bonus $\{1, 2, 4\}$ are selected on validation seeds 50–59; greedy fixes bonus zero. Evaluation seeds 150–159 use test contexts only after selection. CG uses zero initialization, relative residual 10^{-3}, at most 80 iterations, and one fixed operator per solve. The selection artifact and its validation evidence are

κ	Method	Status	Regret slope	Work slope
10	Exact current	PASS	0.167	0.000
10	Full CG	PASS	0.167	2.025
10	Window $q = 1/2$	PASS	0.167	1.564
10	Window $q = 2/3$	PASS	0.167	1.678
10	Window $q = 1$	PASS	0.167	2.025
10	Frozen	PASS	0.167	0.000
10	Diagonal	control	0.167	0.000
10	Greedy	control	0.167	0.000
100	Exact current	PASS	0.227	0.000
100	Full CG	PASS	0.227	2.012
100	Window $q = 1/2$	PASS	0.227	1.621
100	Window $q = 2/3$	PASS	0.227	1.692
100	Window $q = 1$	PASS	0.227	2.012
100	Frozen	PASS	0.227	0.000
100	Diagonal	control	0.227	0.000
100	Greedy	control	0.227	0.000
1000	Exact current	PASS	0.236	0.000
1000	Full CG	PASS	0.236	2.022
1000	Window $q = 1/2$	PASS	0.236	1.640
1000	Window $q = 2/3$	PASS	0.236	1.745
1000	Window $q = 1$	PASS	0.236	2.022
1000	Frozen	PASS	0.236	0.000
1000	Diagonal	control	0.236	0.000
1000	Greedy	control	0.236	0.000

Table 17: Finite-grid log–log slope diagnostics for $d = 2048, r = 16$. Fits use the six prespecified nested horizons and are not asymptotic-rate estimates.

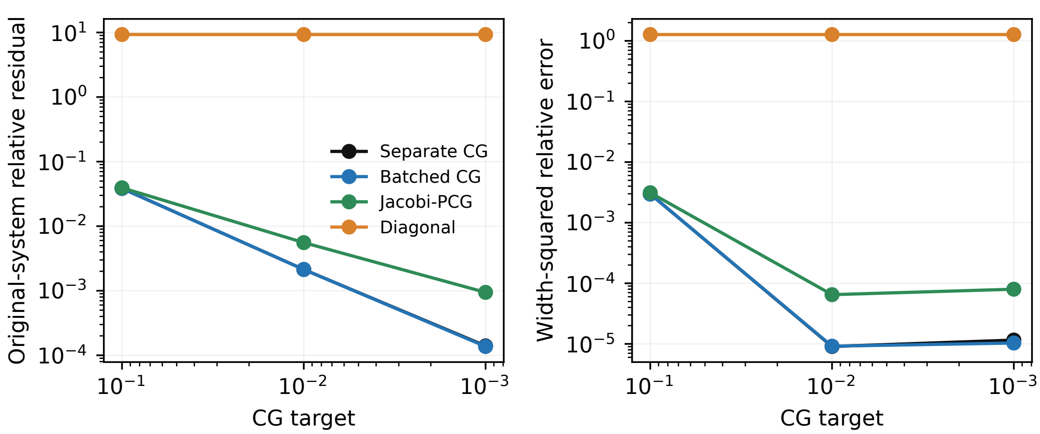

Figure 14: Accuracy diagnostics for the real-autodiff GGN study. Dense energy/width errors are available only for the 131,841-parameter model; no reference value is imputed for the larger model.

Regime	Method	Regret	γ_T	Γ_{tail}	Disagree
$p = 0$, head	Exact dense full	29.744	540.75	16832.75	0.000
$p = 0$, head	Residual-checked full CG	29.744	540.75	16832.75	0.000
$p = 0$, head	Rank truncation	0.156	86.73	86.75	0.000
$p = 0$, head	Diagonal	84.238	850.98	47246.75	0.940
$p = 0$, head	Block diagonal	69.721	787.07	39192.75	0.020
$p = 0$, head	Frequent Directions	1.517	113.33	868.75	0.000
$p = 0$, head	Greedy	0.148	86.73	86.75	0.000
$p = 0$, tail	Exact dense full	35.316	622.19	20012.75	0.000
$p = 0$, tail	Residual-checked full CG	35.316	622.19	20012.75	0.000
$p = 0$, tail	Rank truncation	0.516	95.20	314.75	0.000
$p = 0$, tail	Diagonal	73.999	800.94	41600.75	0.960
$p = 0$, tail	Block diagonal	65.089	748.83	36602.75	0.020
$p = 0$, tail	Frequent Directions	2.287	129.52	1332.75	0.000
$p = 0$, tail	Greedy	0.127	86.73	86.75	0.000
$p = 1$, head	Exact dense full	9.549	254.11	5400.75	0.000
$p = 1$, head	Residual-checked full CG	9.549	254.11	5400.75	0.000
$p = 1$, head	Rank truncation	0.120	85.32	86.75	0.000
$p = 1$, head	Diagonal	16.302	320.74	8856.75	0.940
$p = 1$, head	Block diagonal	18.303	353.75	9870.75	0.000
$p = 1$, head	Frequent Directions	1.372	110.03	822.75	0.000
$p = 1$, head	Greedy	0.049	85.33	86.75	0.000
$p = 1$, tail	Exact dense full	14.266	322.23	8084.75	0.000
$p = 1$, tail	Residual-checked full CG	14.266	322.23	8084.75	0.000
$p = 1$, tail	Rank truncation	7.456	134.62	4212.75	0.000
$p = 1$, tail	Diagonal	52.778	448.79	27800.75	0.940
$p = 1$, tail	Block diagonal	20.298	335.56	11126.75	0.000
$p = 1$, tail	Frequent Directions	2.630	126.31	1498.75	0.000
$p = 1$, tail	Greedy	0.127	86.67	86.75	0.000
$p = 2$, head	Exact dense full	0.714	87.87	330.75	0.000
$p = 2$, head	Residual-checked full CG	0.714	87.87	330.75	0.000
$p = 2$, head	Rank truncation	0.283	77.75	86.75	0.000
$p = 2$, head	Diagonal	1.319	93.71	448.75	0.420
$p = 2$, head	Block diagonal	1.937	102.82	738.75	0.000
$p = 2$, head	Frequent Directions	0.693	87.14	318.75	0.000
$p = 2$, head	Greedy	0.141	78.44	86.75	0.000
$p = 2$, tail	Exact dense full	1.771	106.68	1012.75	0.000
$p = 2$, tail	Residual-checked full CG	1.771	106.68	1012.75	0.000
$p = 2$, tail	Rank truncation	88.176	149.80	44330.75	0.000
$p = 2$, tail	Diagonal	5.699	123.75	3178.75	0.580
$p = 2$, tail	Block diagonal	3.079	127.41	1716.75	0.000
$p = 2$, tail	Frequent Directions	1.510	104.99	864.75	0.000
$p = 2$, tail	Greedy	0.120	86.73	86.75	0.000

Table 18: Complete displayed-rank ($r = 8$) spectral-tail summary at $T = 4096$. All six power/alignment regimes and all seven methods are retained. The disagreement column is the common-trajectory top-score diagnostic; online regret comes from independently executed paired policies.

Study	Policy	Regret [95% CI]	Measured run time (s)	Event fail.	Status
Tanh fixed	Original center	7.86 [7.66, 8.07]	0.094	0/50	Post-hoc verified
Tanh fixed	Corrected center	5.69 [5.46, 5.92]	0.095	0/50	Post-hoc verified
Balanced tuned	CC-UCB (full GGN-CG)	21.94 [20.64, 23.23]	0.966	–	Uncertified
Balanced tuned	Diagonal full network	16.11 [14.62, 17.61]	0.024	–	Uncertified
Balanced tuned	LinUCB	10.58 [9.82, 11.33]	0.015	–	Uncertified
Balanced tuned	LinearTS	12.67 [12.02, 13.31]	0.017	–	Uncertified
Balanced tuned	NeuralLinear	10.01 [9.33, 10.68]	0.014	–	Uncertified
Balanced tuned	NeuralUCB	23.04 [21.81, 24.27]	0.027	–	Uncertified
Balanced tuned	NeuralTS	25.07 [23.40, 26.75]	0.032	–	Uncertified
Balanced tuned	Frozen last layer	7.44 [6.90, 7.99]	0.012	–	Uncertified
Balanced tuned	Greedy	16.26 [14.44, 18.09]	0.017	–	Uncertified
Balanced tuned	UCB1	90.35 [87.95, 92.75]	0.003	–	Uncertified
Balanced tuned	Context-free TS	89.40 [86.53, 92.26]	0.004	–	Uncertified

Table 19: Supplementary executed-policy summary. Tanh rows use one fixed theoretical configuration; balanced rows are selected only on tuning seeds and rerun on disjoint evaluation seeds. Float64 event checks are not verified enclosures, and balanced rows are uncertified performance comparisons. Measured run times are resource accounting under each executed implementation, not a controlled speed or matched-wall-clock comparison.

Policy	R_{10}	R_{100}	R_{1000}	R_{10000}	Log–log slope
Exact full	0.90	0.90	0.90	0.90	0.000
Residual-checked full CG	0.90	0.90	0.90	0.90	0.000
Diagonal (raw)	9.00	90.00	900.00	9000.00	1.000
Diagonal (uniform transfer)	9.00	90.00	900.00	9000.00	1.000
Diagonal (actionwise reference)	0.90	0.90	0.90	0.90	0.000
Greedy	9.00	90.00	900.00	9000.00	1.000

Table 20: Executed analytic off-diagonal linear-Gram witness by horizon. Every value is generated from the provenance-validated derived artifact; fitted slopes are finite-horizon diagnostics.

SHA-256 committed inside every evaluation manifest.

The exact test-split label counts are $(42267, 56737, 7221, 536, 1902, 3444, 4096)$, giving a label-2 majority rate 0.488258. Uniform random has exact expected accuracy $1/7 = 0.142857$. The fixed majority arm is a whole-test-label oracle diagnostic, not a deployable policy. We add deployable context-free UCB1 (one forced pull per arm, then $\hat{\mu}_a + \sqrt{2 \log t / N_a}$) and independent Beta(1, 1)–Bernoulli Thompson sampling. All eight executed methods share evaluation seeds and context order.

Both non-contextual policies beat every contextual method at every horizon. At $T = 1500$, UCB1 and Thompson sampling have regret 895.6 and 802.6 (accuracy 0.403 and 0.465), while full GGN-CG has 1189.7 (0.207). Their paired method-minus-full differences are -294.1 $[-388.1, -200.1]$ and -387.1 $[-479.3, -294.9]$. Full and diagonal last-layer policies have regret 1273.1 and 1274.9 (accuracy 0.151 and 0.150). Mean complete-run times at $T = 1500$ are 5.665 seconds for full GGN-CG, 0.602 for diagonal full-network, 0.007 for UCB1, and 0.017 for Thompson sampling. No method is theorem-certified: binary rewards are routed through Gaussian squared-loss curvature. The full-GGN CG calls reach their relative-residual rule, but no energy-error certificate is supplied, so they are not called solver-certified. This study is a failed baseline check, not external validation of curvature-aware exploration.

Environment and artifacts. All reported new runs used CPython 3.14.0 on one arm64 host with 16 logical CPUs, 64 GiB physical memory, Darwin 25.5.0, and no accelerator. Core versions are NumPy 2.5.1, SciPy 1.18.0, matplotlib 3.11.0, psutil 7.2.2, and scikit-learn 1.9.0. Each run directory contains the resolved configuration, seed, UTC timestamp, Git revision and dirty flag, package and hardware record, raw per-round JSONL, and a summary. Derived JSON, CSV, table, and figure artifacts carry SHA-256 sidecars linking them to the aggregate and raw inputs.

Corollary 38 (Looser worst-case reduction)**.** *If $\alpha_t \le \alpha_{\max}$, $u_t(a_t) \le u_{\max}$, and $\omega_t \le \omega_{\max}$ for all $t \le T$, then $S_T \le \alpha_{\max}^2 u_{\max} \omega_{\max}^2 T$ and*

$$R_T \;\le\; 2\,\alpha_{\max}\,\omega_{\max}\,\sqrt{u_{\max}}\,\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\Lambda_T^{\mathcal{C}}} \;+\; 2E_T.$$

This $\sqrt{T\,\Lambda_T^{\mathcal{C}}}$ form recovers the familiar worst-case expression; it is weakly looser than (15) *(no tighter, with*

Method	$T = 250$	$T = 500$	$T = 1000$
full CG	375.7	438.1	479.3
full dense	361.5	422.4	461.7
diagonal	4120.8	7710.5	13456.0
Lanczos–Ritz	495.7	496.4	509.3
subsample 64	339.7	373.1	396.6
refresh 20	370.3	465.1	526.7
window 64	252.0	255.3	260.2

Table 21: Fixed-reference linear RHS-to-regret ratio RHS/R_T. The JSON artifact also records R_T, RHS, R_T/T, and RHS/T at every horizon for both fixed and tuned configurations. Every displayed RHS exceeds T times the exact maximum one-round pseudo-regret 0.8118 on the finite context support, so the reported RHS values are numerically vacuous. For the four affected policies they are not rigorously certified bounds because their spectral point estimates lack verified upper enclosures.

Operator	Identical	95% coverage	Mean bonus
block_kfac	1.000	0.222	1.137
current_full_ggn	1.000	0.208	1.000
diagonal	1.000	0.248	1.337
historical_frozen_gram	1.000	0.208	1.000
lofi_style	1.000	0.208	1.012
rank_r	1.000	0.154	0.421
stale_refresh	1.000	0.198	0.685
window	1.000	0.163	0.665

Table 22: Tuning-selected scalar multipliers for the coverage-matched operator study. Evaluation outcomes are not used to select these values.

equality when the per-round factors are constant) and is not *the main result.*

F Proof of Theorem 1

We recall the key objects: $\mathbf{g}_s(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}, a)$ (frozen features), $\bar{\mathbf{C}}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s<t} \mathbf{g}_s(\mathbf{x}_s, a_s)\mathbf{g}_s(\mathbf{x}_s, a_s)^\top$ (frozen-feature Gram matrix), $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}$, the algorithmic curvature $\mathcal{C}_t$ with width $s_{\mathcal{C},t}(a) = \sqrt{\mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a)}$, the linearized ridge estimator $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ (Eq. 107), the combined width factor $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, the per-round CG inflation $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, and the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$ (Eq. 79). Write $\widehat{\mu}_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.

Step 1: Confidence, centering, and prediction-error bound. By Lemma 29, with probability $\geq 1 - \delta$,

$$\left|\mathbf{g}_t(a)^\top (\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^*)\right| \;\leq\; \bar{s}_t(a)\, \beta_t, \qquad \forall\, t,\, a, \tag{108}$$

where β_t is as in (99). The self-normalized martingale inequality (Abbasi-Yadkori et al., 2011) is applied to the 1-sub-Gaussian sequence $\xi_s = \eta_s/\sigma$ with the predictable design $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$. Each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is $\mathcal{G}_s$-measurable (where $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ is the pre-reward sigma-algebra of (105)), and the sequential update (7) ensures the hypotheses of the self-normalized bound are satisfied. By Lemma 32 (Assumption 4), $|\mathbf{g}_t(a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \leq \bar{\psi}_t\, \bar{s}_t(a)$. Combined with $|\rho_t| \leq \varepsilon_{\text{lin}}(t)$ and $\bar{\beta}_t \geq \beta_t$, the triangle inequality gives the prediction-error bound (100)

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \widehat{\mu}_t(a)| \;\leq\; (\beta_t + \bar{\psi}_t)\, \bar{s}_t(a) + \varepsilon_{\text{lin}}(t) \;\leq\; \omega_t\, \bar{s}_t(a) + \varepsilon_{\text{lin}}(t), \qquad \forall\, a. \tag{109}$$

(The corrected-center variant of Corollary 10 reaches (109) with $\omega_t = \bar{\beta}_t$ and without Assumption 4.)

Step 2: Bonus sandwiching. By Lemma 28 (which uses Lemma 26 with $\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t$ and the one-sided transfer (60)), for all a,

$$\omega_t\, \bar{s}_t(a) \;\leq\; w_t(a) \;\leq\; \alpha_t\, \omega_t\, \sqrt{u_t(a)}\, s_{\mathcal{C},t}(a). \tag{110}$$

Cell	Surrogate	Mean diff.	Raw p	Holm p	Classification
ffd_000	block_kfac	-0.828	2.17e-14	2.99e-12	surrogate lower
ffd_000	diagonal	-1.044	1.26e-17	1.82e-15	surrogate lower
ffd_000	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_000	lofi_style	-0.045	2.55e-01	1.00e+00	unresolved
ffd_000	rank_r	3.087	1.11e-29	1.75e-27	full lower
ffd_000	stale_refresh	0.369	7.72e-07	7.87e-05	full lower
ffd_000	window	1.728	3.29e-22	4.97e-20	full lower
ffd_001	block_kfac	-0.510	3.55e-04	3.16e-02	surrogate lower
ffd_001	diagonal	-1.380	4.19e-13	5.57e-11	surrogate lower
ffd_001	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	rank_r	9.930	1.86e-25	2.87e-23	full lower
ffd_001	stale_refresh	1.380	4.36e-06	4.31e-04	full lower
ffd_001	window	1.110	7.40e-09	8.65e-07	full lower
ffd_010	block_kfac	-0.981	9.20e-17	1.32e-14	surrogate lower
ffd_010	diagonal	-1.134	2.72e-20	4.02e-18	surrogate lower
ffd_010	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_010	lofi_style	0.117	2.65e-02	1.00e+00	unresolved
ffd_010	rank_r	2.187	5.12e-22	7.68e-20	full lower
ffd_010	stale_refresh	1.242	6.81e-15	9.47e-13	full lower
ffd_010	window	2.700	1.13e-30	1.81e-28	full lower
ffd_011	block_kfac	-0.270	1.92e-03	1.65e-01	unresolved
ffd_011	diagonal	-0.780	2.40e-09	2.90e-07	surrogate lower
ffd_011	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	rank_r	1.650	3.63e-10	4.50e-08	full lower
ffd_011	stale_refresh	4.740	5.53e-41	9.23e-39	full lower
ffd_011	window	4.950	3.96e-35	6.42e-33	full lower
ffd_100	block_kfac	0.009	5.69e-01	1.00e+00	unresolved
ffd_100	diagonal	-0.207	2.70e-06	2.72e-04	surrogate lower
ffd_100	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	rank_r	0.027	8.32e-02	1.00e+00	unresolved
ffd_100	stale_refresh	0.387	9.36e-09	1.09e-06	full lower
ffd_100	window	1.989	2.19e-33	3.52e-31	full lower
ffd_101	block_kfac	-0.930	5.03e-10	6.14e-08	surrogate lower
ffd_101	diagonal	-1.620	1.10e-20	1.63e-18	surrogate lower
ffd_101	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	lofi_style	-0.060	3.22e-01	1.00e+00	unresolved
ffd_101	rank_r	1.740	1.78e-13	2.42e-11	full lower
ffd_101	stale_refresh	4.560	1.11e-25	1.72e-23	full lower
ffd_101	window	5.940	3.77e-36	6.14e-34	full lower
ffd_110	block_kfac	-0.117	3.53e-03	2.93e-01	unresolved
ffd_110	diagonal	-0.729	3.78e-11	4.84e-09	surrogate lower
ffd_110	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_110	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_110	rank_r	3.447	3.31e-30	5.26e-28	full lower
ffd_110	stale_refresh	0.225	2.38e-02	1.00e+00	unresolved
ffd_110	window	1.503	2.25e-25	3.45e-23	full lower
ffd_111	block_kfac	-0.990	9.24e-08	1.03e-05	surrogate lower
ffd_111	diagonal	-1.650	3.22e-13	4.32e-11	surrogate lower
ffd_111	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_111	lofi_style	-0.180	3.24e-02	1.00e+00	unresolved
ffd_111	rank_r	7.950	7.01e-26	1.09e-23	full lower
ffd_111	stale_refresh	1.680	3.12e-08	3.55e-06	full lower
ffd_111	window	3.960	1.88e-26	2.95e-24	full lower

Table 23: Paired terminal-regret inference for the Identical theoretical coefficient protocol. Differences are surrogate minus current full GGN; Holm adjustment uses the complete 168-test family.

Cell	Surrogate	Mean diff.	Raw p	Holm p	Classification
ffd_000	block_kfac	-0.162	1.31e-05	1.26e-03	surrogate lower
ffd_000	diagonal	-0.198	5.56e-07	5.79e-05	surrogate lower
ffd_000	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_000	lofi_style	-0.018	1.59e-01	1.00e+00	unresolved
ffd_000	rank_r	-0.126	6.55e-05	6.22e-03	surrogate lower
ffd_000	stale_refresh	0.135	4.24e-03	3.43e-01	unresolved
ffd_000	window	-0.099	5.17e-04	4.55e-02	surrogate lower
ffd_001	block_kfac	-0.180	1.28e-02	9.96e-01	unresolved
ffd_001	diagonal	-0.720	1.77e-08	2.03e-06	surrogate lower
ffd_001	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	rank_r	1.110	4.44e-07	4.66e-05	full lower
ffd_001	stale_refresh	2.130	4.95e-13	6.49e-11	full lower
ffd_001	window	-0.000	1.00e+00	1.00e+00	unresolved
ffd_010	block_kfac	-0.090	1.68e-02	1.00e+00	unresolved
ffd_010	diagonal	-0.153	1.79e-04	1.61e-02	surrogate lower
ffd_010	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_010	lofi_style	-0.009	5.69e-01	1.00e+00	unresolved
ffd_010	rank_r	0.531	4.40e-09	5.23e-07	full lower
ffd_010	stale_refresh	0.612	9.72e-11	1.23e-08	full lower
ffd_010	window	0.207	2.37e-07	2.59e-05	full lower
ffd_011	block_kfac	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	diagonal	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	rank_r	0.090	8.32e-02	1.00e+00	unresolved
ffd_011	stale_refresh	1.470	5.04e-09	5.95e-07	full lower
ffd_011	window	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	block_kfac	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	diagonal	0.009	3.22e-01	1.00e+00	unresolved
ffd_100	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	rank_r	-0.009	3.22e-01	1.00e+00	unresolved
ffd_100	stale_refresh	0.369	3.38e-07	3.58e-05	full lower
ffd_100	window	-0.009	3.22e-01	1.00e+00	unresolved
ffd_101	block_kfac	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	diagonal	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	rank_r	0.330	5.17e-04	4.55e-02	full lower
ffd_101	stale_refresh	5.970	6.42e-73	1.08e-70	full lower
ffd_101	window	0.270	1.92e-03	1.65e-01	unresolved
ffd_110	block_kfac	-0.063	3.34e-02	1.00e+00	unresolved
ffd_110	diagonal	-0.405	5.71e-11	7.25e-09	surrogate lower
ffd_110	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_110	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_110	rank_r	0.468	6.98e-08	7.82e-06	full lower
ffd_110	stale_refresh	0.333	7.40e-05	6.88e-03	full lower
ffd_110	window	-0.018	5.69e-01	1.00e+00	unresolved
ffd_111	block_kfac	-0.120	4.43e-02	1.00e+00	unresolved
ffd_111	diagonal	-0.420	6.55e-05	6.22e-03	surrogate lower
ffd_111	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_111	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_111	rank_r	0.090	3.71e-01	1.00e+00	unresolved
ffd_111	stale_refresh	1.050	3.35e-07	3.58e-05	full lower
ffd_111	window	-0.120	4.43e-02	1.00e+00	unresolved

Table 24: Paired terminal-regret inference for the Tuning-matched 95% coverage protocol. Differences are surrogate minus current full GGN; Holm adjustment uses the complete 168-test family.

Cell	Surrogate	Mean diff.	Raw p	Holm p	Classification
ffd_000	block_kfac	-0.765	2.98e-13	4.02e-11	surrogate lower
ffd_000	diagonal	-0.918	4.71e-15	6.59e-13	surrogate lower
ffd_000	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_000	lofi_style	-0.036	3.51e-01	1.00e+00	unresolved
ffd_000	rank_r	0.765	5.51e-08	6.22e-06	full lower
ffd_000	stale_refresh	0.135	3.43e-02	1.00e+00	unresolved
ffd_000	window	0.720	3.34e-10	4.17e-08	full lower
ffd_001	block_kfac	-0.360	6.03e-03	4.76e-01	unresolved
ffd_001	diagonal	-1.200	4.36e-10	5.37e-08	surrogate lower
ffd_001	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_001	rank_r	4.620	1.29e-15	1.81e-13	full lower
ffd_001	stale_refresh	1.110	7.40e-05	6.88e-03	full lower
ffd_001	window	-0.180	1.59e-01	1.00e+00	unresolved
ffd_010	block_kfac	-0.747	8.30e-12	1.08e-09	surrogate lower
ffd_010	diagonal	-0.720	1.42e-11	1.83e-09	surrogate lower
ffd_010	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_010	lofi_style	0.126	2.16e-02	1.00e+00	unresolved
ffd_010	rank_r	0.513	1.19e-05	1.15e-03	full lower
ffd_010	stale_refresh	0.387	3.47e-03	2.91e-01	unresolved
ffd_010	window	1.431	1.09e-19	1.60e-17	full lower
ffd_011	block_kfac	0.030	7.43e-01	1.00e+00	unresolved
ffd_011	diagonal	-0.210	3.34e-02	1.00e+00	unresolved
ffd_011	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_011	lofi_style	0.060	1.59e-01	1.00e+00	unresolved
ffd_011	rank_r	0.120	3.76e-01	1.00e+00	unresolved
ffd_011	stale_refresh	4.740	5.53e-41	9.23e-39	full lower
ffd_011	window	3.450	2.08e-36	3.41e-34	full lower
ffd_100	block_kfac	0.189	2.73e-07	2.95e-05	full lower
ffd_100	diagonal	0.288	3.62e-09	4.34e-07	full lower
ffd_100	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	lofi_style	0.000	1.00e+00	1.00e+00	unresolved
ffd_100	rank_r	-0.684	7.19e-19	1.04e-16	surrogate lower
ffd_100	stale_refresh	0.081	1.72e-01	1.00e+00	unresolved
ffd_100	window	0.108	1.29e-01	1.00e+00	unresolved
ffd_101	block_kfac	-0.660	5.56e-07	5.79e-05	surrogate lower
ffd_101	diagonal	-1.230	4.86e-13	6.41e-11	surrogate lower
ffd_101	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_101	lofi_style	-0.060	3.22e-01	1.00e+00	unresolved
ffd_101	rank_r	-0.330	7.01e-02	1.00e+00	unresolved
ffd_101	stale_refresh	4.290	1.52e-36	2.51e-34	full lower
ffd_101	window	3.180	5.17e-24	7.87e-22	full lower
ffd_110	block_kfac	0.054	2.43e-01	1.00e+00	unresolved
ffd_110	diagonal	-0.504	5.96e-06	5.84e-04	surrogate lower
ffd_110	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_110	lofi_style	0.072	3.63e-03	2.98e-01	unresolved
ffd_110	rank_r	1.332	7.08e-19	1.03e-16	full lower
ffd_110	stale_refresh	-0.045	6.46e-01	1.00e+00	unresolved
ffd_110	window	0.306	1.20e-04	1.10e-02	full lower
ffd_111	block_kfac	-0.930	2.00e-07	2.20e-05	surrogate lower
ffd_111	diagonal	-1.620	3.31e-14	4.54e-12	surrogate lower
ffd_111	historical_frozen_gram	0.000	1.00e+00	1.00e+00	unresolved
ffd_111	lofi_style	-0.120	1.03e-01	1.00e+00	unresolved
ffd_111	rank_r	3.330	1.01e-16	1.44e-14	full lower
ffd_111	stale_refresh	0.600	4.42e-03	3.53e-01	unresolved
ffd_111	window	1.230	3.08e-06	3.08e-04	full lower

Table 25: Paired terminal-regret inference for the Tuning-matched mean bonus protocol. Differences are surrogate minus current full GGN; Holm adjustment uses the complete 168-test family.

Table 26: Real-autodiff resource summary at $m = 512$, $K = 10$, and target original-system residual 10^{-3}. Values are evaluation-seed means of the timed primitive; the explicit reference is intentionally skipped for the larger model.

Model	Method	Time (s)	Peak GiB	Sample-CVPs	Max residual
mlp_100k	Separate CG	0.2101	0.108	25600	1.40e-04
mlp_100k	Batched CG	0.0395	0.126	25600	1.37e-04
mlp_100k	Jacobi-PCG	0.05416	0.141	28570	9.44e-04
mlp_100k	Diagonal	0.4428	0.086	0	9.30e+00
mlp_100k	Last layer	0.002994	0.097	0	–
mlp_100k	Explicit dense reference	0.02338	1.477	0	0.00e+00
mlp_10m	Separate CG	0.7532	2.844	61440	6.17e-04
mlp_10m	Batched CG	0.5674	4.219	61440	6.60e-04
mlp_10m	Jacobi-PCG	0.5825	5.370	61440	7.65e-04
mlp_10m	Diagonal	0.4889	1.839	0	2.76e+02
mlp_10m	Last layer	0.002785	2.387	0	–
mlp_10m	Explicit dense reference	skipped	–	–	–

Policy	$T = 200$	$T = 500$	$T = 1000$	$T = 1500$
full GGN-CG	163.3 / 0.183	399.0 / 0.202	794.3 / 0.206	1189.7 / 0.207
frozen Gram	164.0 / 0.180	401.1 / 0.198	797.5 / 0.202	1193.0 / 0.205
diagonal full	166.4 / 0.168	407.7 / 0.185	807.2 / 0.193	1204.4 / 0.197
last-layer full	170.7 / 0.146	420.9 / 0.158	849.4 / 0.151	1273.1 / 0.151
last-layer diagonal	171.1 / 0.144	426.0 / 0.148	851.6 / 0.148	1274.9 / 0.150
greedy	162.3 / 0.189	393.9 / 0.212	781.9 / 0.218	1166.2 / 0.223
UCB1	145.4 / 0.273	332.5 / 0.335	616.2 / 0.384	895.6 / 0.403
Beta–Bernoulli TS	124.7 / 0.376	284.9 / 0.430	544.0 / 0.456	802.6 / 0.465

Table 27: Covertype evaluation-seed means, shown as cumulative pseudo-regret / accuracy. Contextual hyperparameters are selected only on validation seeds; UCB1 and Thompson sampling are parameter-free. Per-horizon marginal and paired 95% Student-t intervals, runtimes, policy types, and certification status are in `results/derived/covertype_horizon_results.json`.

The lower bound and (109) give $\mu^*(\mathbf{x}_t, a) \le \widehat{\mu}_t(a) + w_t(a) + \varepsilon_{\rm lin}(t)$ for all a: the UCB covers μ^* up to the additive $\varepsilon_{\rm lin}(t)$ model-mismatch term.

Step 3: Instantaneous regret. From the UCB rule $\widehat{\mu}_t(a_t) + w_t(a_t) \ge \widehat{\mu}_t(a_t^*) + w_t(a_t^*)$, and using $\mu^*(a_t^*) \le \widehat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\rm lin}(t)$ (Step 2 at a_t^*, valid since $a_t^* \in \mathcal{A}_t$),

$$\begin{aligned}
\mu^*(a_t^*) - \mu^*(a_t) \;&\le\; [\widehat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\rm lin}(t)] - \mu^*(a_t) \\
&\le\; [\widehat{\mu}_t(a_t) + w_t(a_t) + \varepsilon_{\rm lin}(t)] - \mu^*(a_t) \quad \text{(UCB selection)} \\
&=\; w_t(a_t) + [\widehat{\mu}_t(a_t) - \mu^*(a_t)] + \varepsilon_{\rm lin}(t) \\
&\le\; w_t(a_t) + \omega_t\, \bar{s}_t(a_t) + 2\varepsilon_{\rm lin}(t) \quad \text{((109) at } a_t) \\
&\le\; 2\, w_t(a_t) + 2\varepsilon_{\rm lin}(t) \quad \text{(lower bound in (110)).}
\end{aligned}$$

Step 4: Upper bonus bound. The upper bound in (110) gives

$$\mu^*(a_t^*) - \mu^*(a_t) \;\le\; 2\,\alpha_t\,\omega_t\,\sqrt{u_t(a_t)}\, s_{\mathcal{C},t}(a_t) \;+\; 2\varepsilon_{\rm lin}(t). \tag{111}$$

Step 5: Cauchy–Schwarz after summing. Summing (111) over $t = 1, \ldots, T$ and applying Cauchy–Schwarz to the sum (*not* per round),

$$R_T \;\le\; 2\sum_{t=1}^{T} \alpha_t\omega_t\sqrt{u_t(a_t)}\, s_{\mathcal{C},t}(a_t) + 2E_T \;\le\; 2\sqrt{\sum_{t=1}^{T}\alpha_t^2\omega_t^2 u_t(a_t)}\,\sqrt{\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t)} + 2E_T.$$

Step 6: Dynamic potential and collection. Lemma 23 gives $\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \le (\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}$. Substituting,

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}\,\sum_{t=1}^{T}\alpha_t^2\, u_t(a_t)\,\omega_t^2} \;+\; 2E_T,$$

which is (15). The interpretable and observable versions (35) follow from $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\rm dyn}$ (Lemma 23) and $\Lambda_T^{\mathcal{C}} \le \widehat{\Lambda^{\mathcal{C}}}_T$ (Corollary 25). The Cauchy–Schwarz step is applied only after summing, so the time-varying factors $\alpha_t, u_t(a_t), \omega_t$ enter through a sum of squares; no horizon maximum is taken before Cauchy–Schwarz, and the usual $\sqrt{T}$ scaling is recovered when the per-round factors are uniformly bounded (Corollary 38). The stated probability follows from a union bound over the self-normalized confidence event $\mathcal{E}_{\rm conf}$ ($\ge 1-\delta$) and the certificate event $\mathcal{E}_{\rm cert}$ ($\ge 1-\delta_{\rm cert}$), giving $\ge 1-\delta-\delta_{\rm cert}$. $\square$

G CG Convergence Analysis

Fix a round t and an action a with $\mathbf{g} := \mathbf{g}_t(a) \neq \mathbf{0}$. After $I_{t,a}$ iterations of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ (starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$), the standard energy-norm bound gives

$$\varepsilon_{t,a} \;=\; \frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\leq\; 2\left(\frac{\sqrt{\kappa_t}-1}{\sqrt{\kappa_t}+1}\right)^{I_{t,a}},$$

where $\kappa_t = \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number. The bound is right-hand-side independent, so a common iteration budget bounds $\max_a \varepsilon_{t,a}$. This zero-start form is what the fixed-budget certificate (112) uses. For a *warm start* $\tilde{\mathbf{u}}_0 \neq \mathbf{0}$, energy-norm minimization over the shifted Krylov space instead gives

$$\frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\leq\; 2\,q^{\,I_{t,a}}\,\frac{\|\tilde{\mathbf{u}}_0 - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}, \qquad q = \frac{\sqrt{\kappa_t}-1}{\sqrt{\kappa_t}+1},$$

i.e. the extra factor is the initial relative energy error $\|e_0\|_{\mathcal{C}_t}/\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}$ (equal to one at $\tilde{\mathbf{u}}_0 = \mathbf{0}$, recovering the zero-start bound). Under the same certified condition-number bound, adaptive residual stopping (line 5) remains valid with a warm start because it certifies the realized residual directly; but the fixed-budget count (112) is valid *as written* only for zero initialization, and a fixed-budget warm-start certificate additionally requires a bound on that initial relative energy error (the zero-start iteration count does not by itself certify a warm-started solve). Any certified upper bound $\bar{\kappa}_t \geq \kappa_t$ may replace κ_t; for a nonnegative weighted outer-product operator, $\lambda_{\min}(\mathcal{C}_t) \geq \lambda$ and $\lambda_{\max}(\mathcal{C}_t) \leq \lambda + W_t G^2/\sigma^2$ (where $W_t = \sum_{s\in\mathcal{S}_t} w_s$) give

$$\kappa_t \leq \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \leq 1 + \frac{W_t G^2}{\lambda\sigma^2} =: \bar{\kappa}_t,$$

whereas a general SPD operator requires a separately certified $\bar{\kappa}_t$.

Two cases at value one must be distinguished. The true condition number $\kappa_t = 1$ forces all eigenvalues equal, i.e. $\mathcal{C}_t = c_t\mathbf{I}$ for some scalar $c_t > 0$, and one CG iteration from $\tilde{\mathbf{u}}_0 = \mathbf{0}$ is *exact* ($\varepsilon_{t,a} = 0$) for every $\mathbf{g} \neq \mathbf{0}$; this does *not* by itself imply $c_t = \lambda$. With the specific certificate used here, $\bar{\kappa}_t = 1$ additionally forces $\lambda_{\max}(\mathcal{C}_t) = \lambda$, which together with $\mathcal{C}_t \succeq \lambda\mathbf{I}$ gives $\mathcal{C}_t = \lambda\mathbf{I}$ exactly. In either reading the single-step solve is exact, so we keep $\bar{\kappa}_t = 1$ as a separate branch and never evaluate the logarithmic budget below at its zero denominator. For $\bar{\kappa}_t > 1$, setting the right-hand side of the rate equal to $\bar{\varepsilon}_{\mathrm{CG}}$ and requiring $\bar{\varepsilon}_{\mathrm{CG}} < 1$ (where $\bar{\varepsilon}_{\mathrm{CG}}$ denotes the round-t target tolerance $\bar{\varepsilon}_t$) gives the per-solve budget

$$I_{\min}(\bar{\kappa}_t, \bar{\varepsilon}_{\mathrm{CG}}) \;=\; \left\lceil \frac{\log(2/\bar{\varepsilon}_{\mathrm{CG}})}{\log\big((\sqrt{\bar{\kappa}_t}+1)/(\sqrt{\bar{\kappa}_t}-1)\big)} \right\rceil \;=\; O\big(\sqrt{\bar{\kappa}_t}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}})\big), \tag{112}$$

which, applied to every scored action, ensures $\max_a \varepsilon_{t,a} \leq \bar{\varepsilon}_{\mathrm{CG}} = \bar{\varepsilon}_t$. For $W_t = n$ (unrescaled fixed buffer size), $\bar{\kappa}_t \leq 1 + nG^2/(\lambda\sigma^2)$ is bounded independently of T, so $I_{\min}$ is constant. For full history, and for rescaled uniform subsampling with $w_s = (t-1)/n$, we have $W_t = t-1$, so the condition-number *bound* $\bar{\kappa}_t = O(tG^2/(\lambda\sigma^2))$ grows with t, and the worst-case certificate is then obtained with $I_{\min} = \Theta\big(G\sqrt{t/(\lambda\sigma^2)}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}})\big)$ iterations per solve—i.e. $I = O(\sqrt{t}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}}))$ suffices under this bound (this is an upper-bound certificate, not a lower bound on the number of iterations CG actually needs). CG may converge faster than this worst-case certificate suggests under favorable spectral clustering of $\mathcal{C}_t$; the realized iteration counts would be measured empirically.

H Two-Sided Spectral-Distortion Corollary

The main-text Theorem 34 uses only a one-sided upper transfer factor. For comparison we record the earlier two-sided result, which instead assumes both an upper *and* a lower spectral factor and yields the familiar $\sqrt{\rho_+/\rho_-}$ inflation. The two results trade different assumptions—the corollary below needs a lower spectral bound; the dynamic Theorem 1 needs the dynamic width complexity $\Lambda_T^{\mathcal{C}}$ instead—and neither numerically dominates the other in general.

Corollary 39 (Two-sided spectral-distortion inflation)**.** *Adopt the hypotheses of Theorem 1 with $\mathcal{C}_t = \hat{\mathbf{C}}_t$, retaining the* original nonlinear center *(so $\omega_t = \bar{\beta}_t + \bar{\psi}_t$; in the convex last-layer regime under exact optimization,*

$\bar{\psi}_t = 0$ and $\omega_t = \bar{\beta}_t$), and a uniform CG tolerance $\bar{\varepsilon}_t \le \bar{\varepsilon}$ so that $\alpha_t \le \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$. Suppose predictable factors $\rho_{-,t}, \rho_{+,t}$ are certified before action selection and the two-sided *sandwich $\rho_{-,t}\bar{\mathbf{C}}_t \preceq \hat{\mathbf{C}}_t \preceq \rho_{+,t}\bar{\mathbf{C}}_t$ holds for all $t \le T$ with $0 < \rho_{-,t} \le \rho_{+,t} < \infty$; set $\rho_- = \min_t \rho_{-,t}$, $\rho_+ = \max_t \rho_{+,t}$, and (to satisfy the convention $u_t \ge 1$ of Assumption 5) take the transfer factor $u_t := \max\{1, \rho_{+,t}\} \le \max\{1, \rho_+\}$. Write $\omega_{\max,T} := \max_{t\le T}(\bar{\beta}_t + \bar{\psi}_t)$ and $\rho_+^\star := \max\{1, \rho_+\}$. Then, on the intersection of the confidence event $\mathcal{E}_{\mathrm{conf}}$ ($\ge 1-\delta$) and the certificate event $\mathcal{E}_{\mathrm{cert}}$ (which now also contains the two-sided sandwich event; $\ge 1 - \delta_{\mathrm{cert}}$),*

$$R_T \;\le\; 2\,\alpha_I \sqrt{\rho_+^\star/\rho_-}\;\omega_{\max,T}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} \;+\; 2E_T$$

with probability at least $1-\delta-\delta_{\mathrm{cert}}$, where γ_T is the frozen-feature information gain (31). *Under the standard normalization $\rho_+ \ge 1$ (achievable by rescaling the sandwich so its upper factor is at least one), $\rho_+^\star = \rho_+$ and the inflation is the familiar $\sqrt{\rho_+/\rho_-}$.*

Proof. This is a *direct* argument through the monotone frozen Gram $\bar{\mathbf{C}}_t$; it does *not* substitute $\hat{s}_t$ for the dynamic width in Corollary 38. *Step 1 (variance sandwich).* Inverting the two-sided sandwich gives $\rho_{+,t}^{-1}\bar{\mathbf{C}}_t^{-1} \preceq \hat{\mathbf{C}}_t^{-1} \preceq \rho_{-,t}^{-1}\bar{\mathbf{C}}_t^{-1}$, so, evaluating the quadratic form at $\mathbf{g}_t(a)$, $\rho_{+,t}^{-1/2}\,\bar{s}_t(a) \le \hat{s}_t(a) \le \rho_{-,t}^{-1/2}\,\bar{s}_t(a)$. The lower half gives $\bar{s}_t^2(a) \le \rho_{+,t}\hat{s}_t^2(a) \le u_t\hat{s}_t^2(a)$, so Assumption 5 holds with $u_t = \max\{1, \rho_{+,t}\}$. *Step 2 (per-round regret).* With $s_{\mathcal{C},t} = \hat{s}_t$, Theorem 1 Steps 1–4 give $\mathrm{regret}_t \le 2\,\alpha_t\,\omega_t\,\sqrt{u_t}\,\hat{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t)$; using $\hat{s}_t(a_t) \le \rho_{-,t}^{-1/2}\bar{s}_t(a_t)$, $u_t = \max\{1, \rho_{+,t}\} \le \rho_+^\star$, $\alpha_t \le \alpha_I$, and $\omega_t \le \omega_{\max,T}$,

$$\mathrm{regret}_t \;\le\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_{-,t}}\;\bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t) \;\le\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\;\bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

Step 3 (frozen-feature elliptic potential). Sum over t and apply Cauchy–Schwarz, then the classical elliptic-potential identity for the *monotone* Gram $\bar{\mathbf{C}}_t$ ((7)), $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\sum_t \log(1 + \sigma^{-2}\bar{s}_t^2(a_t)) = (\sigma^2 + G^2/\lambda)\,\gamma_T$:

$$R_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{T\sum_t \bar{s}_t^2(a_t)} + 2E_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} + 2E_T.$$

The probability statement is the union bound over $\mathcal{E}_{\mathrm{conf}}$ and $\mathcal{E}_{\mathrm{cert}}$. □

For Lemma 22, allocate δ_{cert}/T per round and use the worst-case $m = T-1$ in (78) to obtain the required time-uniform sandwich.

I GGN Approximation Error

Lemma 40 (GGN approximation error)**.** *Suppose $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a)$ is twice differentiable in $\boldsymbol{\theta}$. Suppose $|r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)| \le \epsilon_r$ and $\|\nabla^2\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\|_{\mathrm{op}} \le H$ for all $s < t$. Define $\zeta := (t-1)\,\epsilon_r\,H/(\sigma^2\lambda)$. Then:*

(i) $\|\mathbf{H}_t - \mathbf{C}_t\|_{\mathrm{op}} \le (t-1)\,\epsilon_r\,H/\sigma^2 = \zeta\lambda$.

(ii) If $\zeta < 1$, then $(1-\zeta)\mathbf{C}_t \preceq \mathbf{H}_t \preceq (1+\zeta)\mathbf{C}_t$.

Proof. For $D_t = \sigma^{-2}\sum_{s<t}(r_s - \mu_{\boldsymbol{\theta}_t})\nabla^2\mu_{\boldsymbol{\theta}_t}$, the triangle inequality gives $\|D_t\|_{\mathrm{op}} \le \zeta\lambda$, proving (i). Combining $-\|D_t\|\mathbf{I} \preceq -D_t \preceq \|D_t\|\mathbf{I}$ with $\mathbf{C}_t \succeq \lambda\mathbf{I}$ proves (ii). □

J Exponential-Family Curvature Definitions

For a regular exponential-family negative log-likelihood with natural parameter $z_{\boldsymbol{\theta}}(\mathbf{x}, a) \in \mathbb{R}^k$ (scalar for Bernoulli, k-dimensional for categorical) and log-partition function A, the GGN/Fisher curvature matrices are

$$\mathbf{C}_t^{\exp} \;=\; \lambda\,\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_t(\mathbf{x}_s, a_s)^\top\,\nabla^2 A\big(z_t(\mathbf{x}_s, a_s)\big)\,J_t(\mathbf{x}_s, a_s), \tag{113}$$

$$\bar{\mathbf{C}}_t^{\exp} \;=\; \lambda\,\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_s(\mathbf{x}_s, a_s)^\top\,\nabla^2 A\big(z_s(\mathbf{x}_s, a_s)\big)\,J_s(\mathbf{x}_s, a_s), \tag{114}$$

where $J_t(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \in \mathbb{R}^{k \times d}$ is the Jacobian of the natural parameter and $\nabla^2 A(z)$ is the output-space Hessian of the log-partition function. For Bernoulli rewards $\nabla^2 A(z) = p(1-p)$ (scalar); for categorical rewards $\nabla^2 A(z) = \mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$). Setting $k = 1$ and $\nabla^2 A = \sigma^{-2}$ recovers the Gaussian curvature (5)–(6).

K Broader Impact

Exploration can expose users to under-vetted actions and amplify feedback loops. Deployment requires safety constraints, action filters, offline evaluation, and online monitoring; the regret certificates here do not establish safety.

L Data and Code Availability

The repository contains the Buck pipeline, tests, artifact scripts, validated legacy derived results, the retained $d = 128, r = 4$ slice, and deterministic reproduction entry points for the spectral-tail, premise-clean scaling, real-autodiff, scaled-tanh, and positive-gap studies. Those full grids have run; their validated derived aggregates, generated figures, tables, hashes, and provenance records are in the reviewed tree, while their seed-level raw directories remain outside Git. The scaled-tanh artifacts record the failed premise conjunction rather than a theorem-instantiation claim. The Wheel full grid has not run; its smoke profile is not paper evidence. The 2.1 GB legacy scaling tree and historical inputs outside Git are absent, so not every legacy raw-derived chain is reproducible. Public dataset caches are optional local accelerators rather than release requirements; the builder packages loaders/configurations and sanitizes identities and hashes/provenance without depending on a specific scikit-learn cache layout. Covertype is CC BY 4.0 (DOI `10.24432/C50K5N`) (Blackard, 1998). The autodiff study status is `full_evaluation`: 360 cells over ten fixed evaluation seeds completed with no skipped cell under the 24-accelerator-hour cap. The balanced-MNIST study is `full_evaluation`: canonical archives pass their published checksums, both online pools are exactly class-balanced, and all 480 tuning plus 240 evaluation policies complete. The ignored local cache is not a clean-checkout fixture; the public download path is retained. No current bundle or identifying URL is claimed.

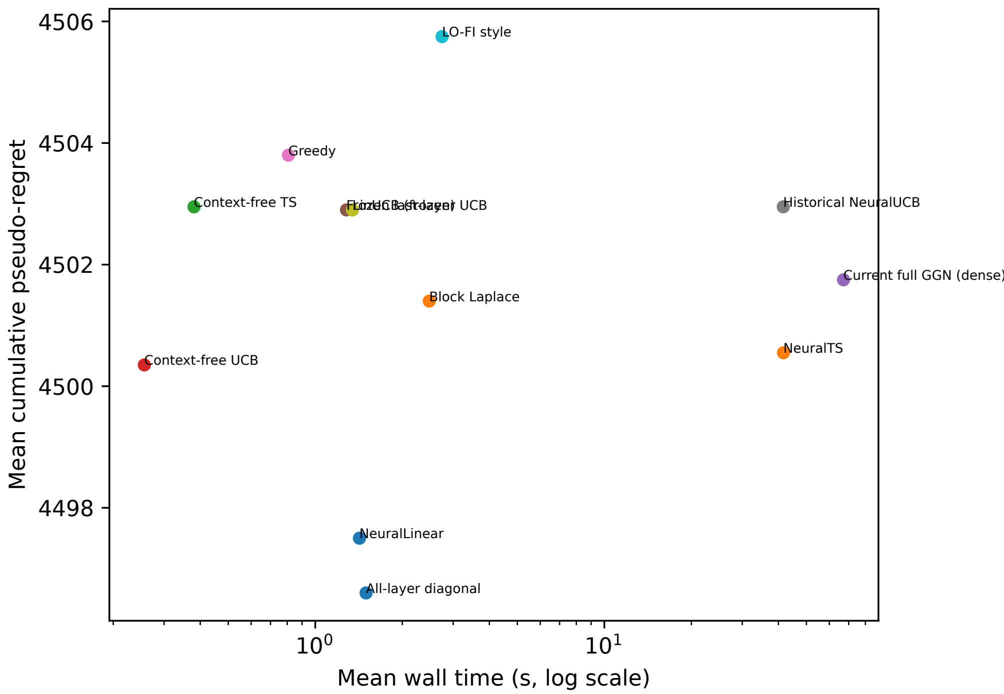

Figure 15: Balanced-MNIST terminal regret versus contention-affected per-policy wall time. The online model has 85 parameters and dense full-Gram solves, so this figure is not a matrix-free scalability comparison.